Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

Pablo Paramio Pérez
CEMFI Master in Economics and Finance

June 2026

Contents

Glossary of acronyms **3**

1 Introduction **4**

2 Institutional setting: sequential markets, three reforms, and a blackout **4**
- 2.1 Spanish wholesale architecture: sequential auctions and balancing 4
- 2.2 15-minute granularity transition 6

3 Theory: a sequential-market model of bid ladders with granularity selectors **7**
- 3.1 Setup: three sequential stages, three agents, and the within-hour state 7
- 3.2 Bid ladders within a product: step offers along the monopoly line 10
- 3.3 Arbitrage regimes and the pre-reform baseline 14
- 3.4 Spot reform $(1, Q)$: contractibility arrives 14
- 3.5 Forward reform (Q, Q): relocation and limited per-product arbitrage 15
- 3.6 Comparative statics and welfare across the reform path 16

4 Data and variables of interest **17**
- 4.1 Bid offers, reform path, and notation 18
- 4.2 In-band region and bid-shape outcomes 18
- 4.3 Aggregate outcomes 20
- 4.4 Critical / flat hour-class partition 21

5 Empirical strategy **23**
- 5.1 Level variables: long pre-windows, renewable controls, and a counterfactual 23
- 5.2 Bid-shape outcomes: a within-day difference-in-differences 24

6 Results **25**
- 6.1 Prices: a robust drop at ID15, a null at DA15, and matching cross-market effects . 25
- 6.2 The margin channel: residual demand becomes more price-elastic and margins compress 27
- 6.3 Within-hour wedge dispersion turns on at ID15 and does not collapse at DA15 . . 28
- 6.4 Bid curves disperse and gain tranches in critical hours on each reformed market . . 30

6.5 Imbalance: the price of a deviation falls at ID15, the volume does not 32
6.6 Ancillary-services costs are dominated by the reinforced-operation regime, not granularity 33

7 Conclusion **34**

References **36**

A Theoretical derivations **38**

B Robustness **54**

C Measurement details: bid curves, residual demand, and the margin proxy **64**

D Additional descriptive findings **68**

Glossary of acronyms

OMIE	Operador del Mercado Ibérico de Energía	PDBC	Programa Diario Base de Casación
REE	Red Eléctrica de España	PDBF	PDBC plus bilateral contracts
CNMC	Comisión Nacional de los Mercados y la Competencia	PIBCI	Programa Intradiario Base de Casación Incremental
ENTSO-E	European Network of TSOs for Electricity	TR	Tiempo Real (real-time technical restrictions)
ESIOS	Sistema de Información del Operador del Sistema	Fase I/II	pre-/post-IDA technical-restriction redispatch
EU	European Union	PO	Procedimientos de Operación
DA	day-ahead auction	aFRR	automatic Frequency Restoration Reserve
IDA	intraday auctions	mFRR	manual Frequency Restoration Reserve
SIDC	Single Intraday Coupling	RR	Replacement Reserve
XBID	cross-border intraday continuous market	BRP	Balance Responsible Party
EPEX	European Power Exchange	BSP	Balance Service Provider
MTU	market time unit (MTU60 hourly; MTU15 15-min)	CCGT	combined-cycle gas turbine
ISP15	imbalance settlement at 15 min (2024-12-11)	RES	renewable energy sources
ID15	intraday at 15 min (2025-03-19)	PV	photovoltaic
DA15	day-ahead at 15 min (2025-10-01)	MIC	minimum-income condition (offer type)
DA60	pre-DA15 hourly day-ahead	HHI_{tr}	tranche-HHI: MW-weighted concentration of in-band bid tranches
MCP	market clearing price	MW / MWh / GWh	megawatt / megawatt-hour / gigawatt-hour

1 Introduction

[To be written. Concepts to cover, mirroring the "Question and headline" slide:]

- **Question.** Does finer time resolution reshape strategic bidding? Does market power go up or down? What is the effect on prices, on the day-ahead vs intraday wedge, and on imbalance penalties?
- **Headline answer.** Granularity made the market more competitive without moving aggregate quantities: prices fell substantially at ID15 and were unchanged at DA15; the residual demand each dominant firm faces became more price-elastic at every reform; dispatchables grade their critical-hour ladders more finely while renewables compress; imbalance penalty per MWh dropped by a third at ID15 with aggregate imbalance volume essentially unchanged.
- **Identification challenge.** An April 2025 blackout triggered a separate "operación reforzada" regulatory regime that inflated ancillary-services costs. Granularity and that regime are calendar-overlapping but mechanistically distinct; the empirical windows are constructed to keep them separable.
- **Contribution.** Bid-level identification on the new MTU15 sequence; a sequential-markets model in which the optimal step offer is solved exactly — the bid ladder places equal-MW tranches on an even price grid along the bidder's monopoly line, so the measured bid-shape statistics $(\sigma_p, \mathrm{HHI}_{\mathrm{tr}}, \hat{\beta})$ have closed-form model counterparts; per-firm Ito-Reguant residual-demand recovery for Spanish data; convex-renewable robustness on the price effect; a strategic-markup proxy that nets out infra-marginal own production from the Cournot first-order condition.
- **Roadmap.** Institutional setting (§2); theory (§3); data (§4); empirical strategy (§5); results (§6); conclusion (§7).

2 Institutional setting: sequential markets, three reforms, and a blackout

2.1 Spanish wholesale architecture: sequential auctions and balancing

Spanish wholesale electricity markets are run in sequential order by the Iberian wholesale market operator (OMIE[1]), which organizes auctions for both Spain and Portugal jointly. The product of each auction is the delivery of electricity, in MW[2], over a fixed short time slot: 60 minutes before the reform, 15 minutes after. Even though the product is homogeneous, a mix of technologies competes to supply it. The most important ones are nuclear, combined-cycle gas turbines (CCGT), hydro (run-of-river and pumped-storage), wind and solar photovoltaic (PV), with smaller contributions from cogeneration, biomass and a fast-growing battery-storage segment. These technologies differ sharply in marginal cost and in how flexibly they can move output within an hour, and it is precisely this heterogeneity that makes the granularity of the delivery slot economically relevant: nuclear runs as near-inflexible baseload, CCGT and hydro are the usual price-setters at the margin, and wind and solar are near-zero-marginal-cost but variable within the hour.

The auctions[3] are the following:

[1] *Operador del Mercado Ibérico de Energía*

[2] Megawatts are Joules per second, so it is a rate, not a stock. The stock is MWh, which is the MW that are delivered in 1 hour. Key distinction for the granularity reform.

[3] Note that these auctions are separate from bilateral contracts between the agents, which are over-the-counter

- Day-ahead (DA) auction or forward market, which is the biggest market and clears the day before delivery.
- Intraday auctions (IDA) or spot markets, to adjust positions of the day-ahead. Since June 2024 there are 3 European auctions, previously they were 6 Iberian auctions (MIBEL)[4].
- Continuous intraday markets, for adjusting positions near delivery. Not much volume in this market.

Both DA and IDA auctions are **uniform-price multiunit auctions**, in which bidders submit step-function[5] supply or demand curves (not both). A complex algorithm called EUPHEMIA considers all these bids and selects, for each auction, the supply and demand offers that maximize total welfare of the market at European level, considering cross-border restrictions[6]. The price is therefore not a mechanical crossing of two fixed aggregate curves: because some offers are indivisible blocks or carry the complex conditions above, and because adjacent zones are coupled through limited interconnection capacity, accepting one offer changes which others can clear — so the algorithm must *select* which offers enter the clearing, jointly across all coupled zones, and only once that selection is fixed does a marginal accepted offer set the price. This algorithm gives different prices for each bidding zone, so in practice we have one price for Spain and another one for Portugal.

On top of these markets, in Spain there is a public[7] system operator, *Red Eléctrica de España* (REE), which operates the very-high-voltage transmission grid and is responsible for keeping generation and demand in balance in real time.

To fulfil this task, REE runs a sequence of **balancing and redispatch markets** that sit on top of the OMIE auctions and clear progressively closer to delivery. After the day-ahead auction it resolves grid-security and congestion constraints through a pay-as-bid redispatch (the *technical-restrictions* market); it then procures upward and downward frequency reserves (secondary and tertiary regulation) through separate auctions; and, close to delivery, it settles the residual gap between each agent's scheduled and delivered energy in the imbalance (deviation) market.

These ancillary markets are economically large and very important for welfare analysis as recently documented by Brown and Reguant (2026) and Daví-Arderius and Graf (2026), but they are not the main focus of this master thesis.

(OTC).

[4]The difference between an "Iberian" and a "European" auction is the scope of the clearing: the European sessions are coupled with the rest of the continent, so cross-border interconnection capacity and the surplus of adjacent bidding zones enter the clearing through the common EUPHEMIA algorithm, whereas the old MIBEL sessions cleared the Iberian zone in isolation (NEMO Committee, 2025).

[5]Offers fall into two families, documented in OMIE's public file specification (OMIE, 2025a): *simple* offers (price–quantity steps for a single period) and *complex* offers, which add cross-period conditions — chiefly the minimum-income condition (MIC), requiring that a unit's day-ahead revenue cover its quasi-fixed and start-up costs, together with load-gradient and indivisibility conditions. The minimum-income condition was removed on the day-ahead market at the 2025 fifteen-minute reform. Block offers are a separate multi-period fill-or-kill format.

[6]EUPHEMIA was developed from the Price Coupling of Regions (PCR) in 2009-2012 and it is used in the DA since 2014, via Single Day-ahead Coupling (SDAC) across countries. For the intraday auctions in Spain EUPHEMIA started to set prices in June 2024, when the 6 MIBEL auctions were changed to 3 European ones, which cleared via Single Intraday Coupling (SIDC), previously existing for the continuous market since June 2018 (CNMC, 2024b). For more details on the algorithm, see NEMO Committee (2025).

[7]Strictly it is a listed private company, but the state holding company SEPI is the largest single shareholder of its parent, Redeia, with a 20% stake (the rest is free float), and by statute must hold no less than 10%; the system operator is therefore kept under public control (Redeia, 2025).

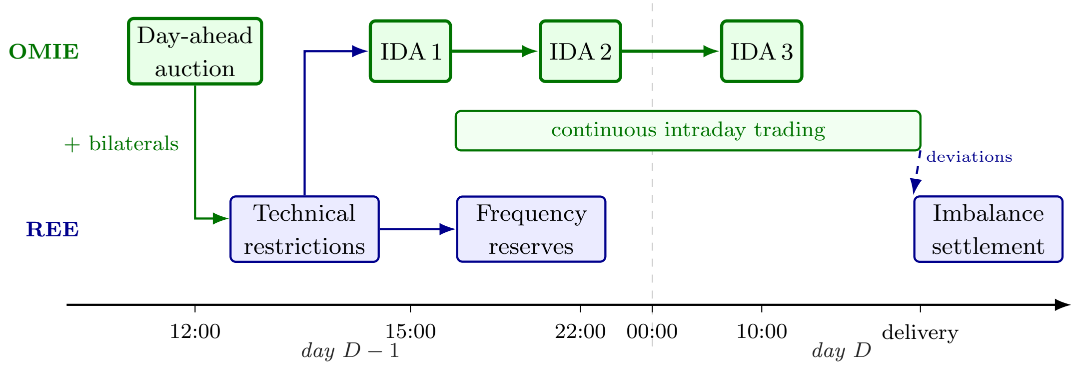


Figure 1: The sequence of Spanish wholesale markets for one delivery day D. OMIE energy markets in green, REE processes in blue. The day-ahead schedule — auction results plus bilateral contracts — passes to REE for the resolution of technical restrictions before intraday trading begins; deviations from final schedules are settled ex post. *Sources:* market sessions and gate-closure times per the CNMC market rules (CNMC, 2024b, 2025); REE processes per Red Eléctrica de España (2024).

Figure 1 sketches a stylised summary of the full sequence. In this master thesis we concentrate on the **day-ahead and intraday auctions**, because they are where the main granularity reforms actually land, but they were not the only markets affected by the reforms.

2.2 15-minute granularity transition

The move from hourly to quarter-hourly products was a European mandate, and it arrived in steps over almost a decade. The quarter-hour first entered European law through the Electricity Balancing Guideline of 2017, which harmonised the period over which imbalances are settled at 15 minutes across Member States (European Commission, 2017). The stated rationale was the integration of variable renewables: as wind and solar vary within the hour, a quarter-hourly reference makes schedules and imbalance prices track the sub-hourly variation that hourly settlement averages away, sharpening the incentive to forecast and balance close to real time.

The recast Electricity Regulation of 2019 then made the timetable binding: every scheduling area had to settle imbalances at 15 minutes by January 2021, with national derogations allowed only until the end of 2024, and the day-ahead and intraday markets had to offer products at least as short as the settlement period (European Union, 2019). The endpoint was therefore fixed years in advance. What remained open was how each country would phase it in.

Spain, which had taken the derogation, phased it in during 2024 from the balancing layer inward. In the first half of 2024 the ancillary-service products and provider qualification moved to a 15-minute reference[8]. On **14 June 2024** the six regional MIBEL intraday auctions were consolidated into three European auctions cleared under the common Single Intraday Coupling (CNMC, 2024b); this reorganised the session structure but left the traded product hourly. On **11 December 2024** the imbalance settlement period itself moved from 60 to 15 minutes (**ISP15**), just ahead of the European deadline, so that deviations from schedule began to settle at quarter-hourly resolution while the OMIE auctions still traded hourly blocks. By the time the auctions were reformed, every clearing already settled against a quarter-hourly reference.

The reforms this thesis studies are the final two steps, in which the traded product itself became

[8]This adaptation followed the European Electricity Balancing Guideline (Regulation (EU) 2017/2195) and the connection of the Spanish reserves to the European balancing platforms. The automatic secondary-reserve (aFRR) energy market was split into separate upward and downward clearings, each with its own marginal price, and linked to the European PICASSO platform; the manual tertiary reserve (mFRR) joined the European MARI platform at the end of 2024. Both standard products are defined on a 15-minute reference (CNMC, 2024a).

quarter-hourly (CNMC, 2025). On **19 March 2025** the intraday auctions and the continuous intraday market switched from 60- to 15-minute delivery slots, so that each clock-hour of intraday trading now hosts four products where it previously hosted one; we label this cutover **ID15**. On **1 October 2025** the much larger day-ahead auction followed the same logic and completed the rollout; we label it **DA15**. These are the two blue markers in Figure 2, and they are the cutovers around which the whole empirical strategy is built.

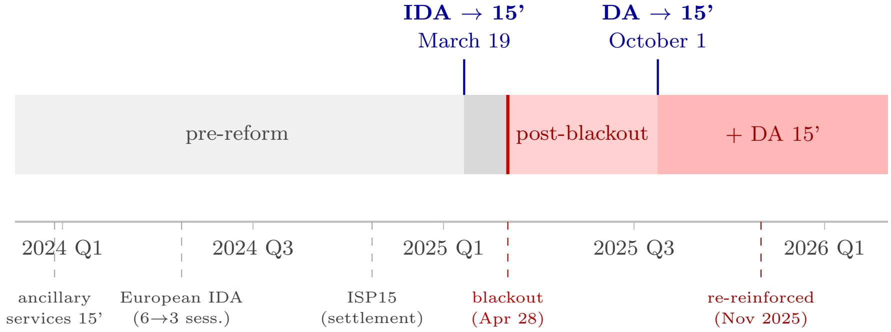


Figure 2: The Spanish reform sequence, 2024–2026.

Cutting across this sequence, the **28 April 2025 blackout** fell between ID15 and DA15 and triggered a separate, continuously operating regime called reinforced operation[9], which modified the resulting quantities from the auctions in each period. This regime still persists and since November 2025 it has moved these restrictions even more from real time to the post-day-ahead technical-restriction redispatch — the *Fase I* phase, run after the day-ahead market and before the intraday auctions, with its later unwinding back into the schedule called *Fase II* — to increase predictability.[10] The Spanish balancing layer thus has redispatch at several stages: the Fase I/II technical restrictions before and around the intraday auctions, the real-time technical restrictions between gate closure and delivery, and the frequency reserves (aFRR, mFRR, RR) and imbalance settlement at delivery — the concepts §6 returns to when it reaches this layer.

As was mentioned, the Commission's case for finer granularity was a balancing one: better incentives to forecast and balance variable renewables. This thesis asks a different question — what finer granularity does to *strategic conduct* in the auctions themselves: to prices, to cleared quantities, and to the shape of firms' bids in the day-ahead and intraday markets. To isolate the effects of ID15 and DA15 on these outcomes, I build the following sequential-market model.

3 Theory: a sequential-market model of bid ladders with granularity selectors

3.1 Setup: three sequential stages, three agents, and the within-hour state

The model is a sequential-market framework in the tradition of Ito and Reguant (2016), with three agents: a strategic bidder, a renewable aggregator that schedules on forecasts, and an arbitrageur with bounded capacity. I add two ingredients. First, each market carries a **granularity selector** $Q^m \in \{1, Q\}$: it clears either once per clock-hour or once per subperiod, the margin the reforms moved. Second, an explicit **within-hour state** — a deterministic ramp plus stochastic redistributive news — that an hourly product averages away but a quarter-hourly product can

[9] *Operación reforzada*

[10] What Colón (2025) calls the *re-reinforced* mode (*modo re-reforzado*).

price. I build the model in reading order — the stages and the reform path, the three agents, the within-hour state and what is known about it when, and how the bidder responds — and solve it from §3.2 on. Full derivations are in Appendix A.

A representative clock-hour contains Q equal-length subperiods, and three stages operate in order: a **forward** auction (Stage 1, $m = F$), a **spot** auction (Stage 2, $m = S$), and ex-post **balancing** settlement (Stage 3, $m = B$). Each stage has its own granularity Q^m, so the profile (Q^F, Q^S, Q^B) defines a $2^3 = 8$-state policy space. The main analysis fixes the auction-side path $(Q^F, Q^S) = (1, 1) \to (1, Q) \to (Q, Q)$: the forward corresponds to the day-ahead auction and the spot to the intraday auctions, so the first step of the path is ID15 and the second is DA15. The agents, stages, and reform path are diagrammed in Figure 3.

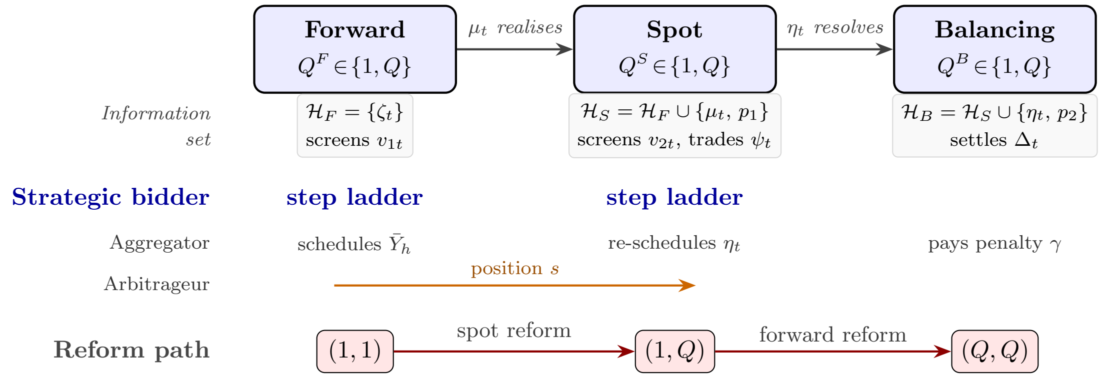


Figure 3: Sequential markets, agents, and information sets (§3.1). The filtration $\mathcal{H}_F \subset \mathcal{H}_S \subset \mathcal{H}_B$ grows as each shock is revealed: the deterministic ramp ζ_t is known throughout, the news μ_t realises between forward and spot, and the aggregator's forecast error η_t resolves by balancing. Each committed offer *screens* its clearing state v_{mt} — revealed only at clearing — so $v_{mt} \notin \mathcal{H}_m$ when the stage-m offer is submitted; p_1, p_2 are the realised forward and spot prices.

The **strategic bidder** is a residual monopolist with constant marginal cost c. It delivers exactly what it sells: once the auction clears its assigned quantity is fixed and, being dispatchable, it produces precisely that[11] — it carries no production shock, and so does not internalise the imbalance penalty γ. Its offer for each delivery product is a non-decreasing step function with corner points $\{(p_k, q_k)\}$ — the unit offers cumulative quantity q_k once the price reaches p_k — submitted before the product's clearing state is known and cleared at uniform price (§3.2). The **operational aggregator**[12] is a price-taker with stochastic production $y_t = \bar{y}_t + \eta_t$, where the forecast error η_t has range Δ_η. It allocates its total hourly energy $\bar{Y}_h$ across forward and spot, and it *does* pay the imbalance penalty, at the rate $\gamma(Q^B)$ set by the balancing granularity. The **arbitrageur** carries a net position s from forward to spot, in one of the four regimes of Ito and Reguant (§3.3); arbitrage transmits price impact but not imbalance risk.

What granularity acts on is the **within-hour state**. Write the bidder's residual demand for delivery quarter t in market m as $q_{mt} = A_{mt} - b_{mt}\, p_{mt}$, net of arbitrage and aggregator supply (Appendix A.1 gives the full specification), with intercept

$$A_{mt} \;=\; \bar{A} \;+\; \zeta_t \;+\; v_{mt}.$$

The ramp ζ_t is the *deterministic* within-hour profile: known to all at every stage, summing to zero over the hour, with dispersion $\sigma_\zeta^2 = Q^{-1} \sum_t \zeta_t^2$ large in steep-ramp hours and near zero when the hour is internally flat. The *clearing state* v_{mt} is what the bidder screens: unknown when

[11] Realistic assumption if we think on CCGT.

[12] Renewable aggregator analogue.

the offer is committed, realised at clearing. It splits into an hour-level draw and a redistributive piece, $v_{mt} = \theta_m + \mu_{mt}$ with $\sum_t \mu_{mt} = 0$, so an hourly product — clearing against the within-hour average $Q^{-1} \sum_t v_{mt} = \theta_m$ — sees the redistributive part cancel, while a quarter-hourly product clears against its own v_{mt} and is exposed to μ as well. (The spot ends up trading whatever the forward leaves uncontracted, its *update* ψ_t, equal to $\zeta_t + \mu_t$ when the forward is hourly; I take it up at the reforms.)

The news μ_{mt} is *stochastic* — mean-zero, independent of θ_m, reallocating MW between quarters without touching the hour-level average. The two granularity assumptions — $\theta_m \sim U[-\Delta_m, \Delta_m]$ for an hourly product and $v_{mt} \sim U[-\rho\Delta_m, +\rho\Delta_m]$ for a quarter-hourly one — are then not independent stipulations but one. The tiling Lemma 1 (Appendix A.1) shows the uniform family is *closed* under the aggregation $v_{mt} = \theta_m + \mu_{mt}$: it exhibits a discrete symmetric μ_{mt} — atoms at $\pm(2j-1)\Delta_m$, $j = 1, \dots, J$ — for which the convolution is *exactly* uniform on $[-\rho\Delta_m, \rho\Delta_m]$ with $\rho = 2J$. The amplification $\rho \geq 1$ is exactly the spread of the news: $\rho = 1$ when $\mu_{mt} \equiv 0$ (an internally flat hour), larger as its dispersion grows (I carry a general ρ). Granularity's whole effect on a single product is therefore to replace its screened width Δ_m with $\rho\Delta_m$, nothing else about the state changing. Figure 4 shows the decomposition over two different-ramp hours.

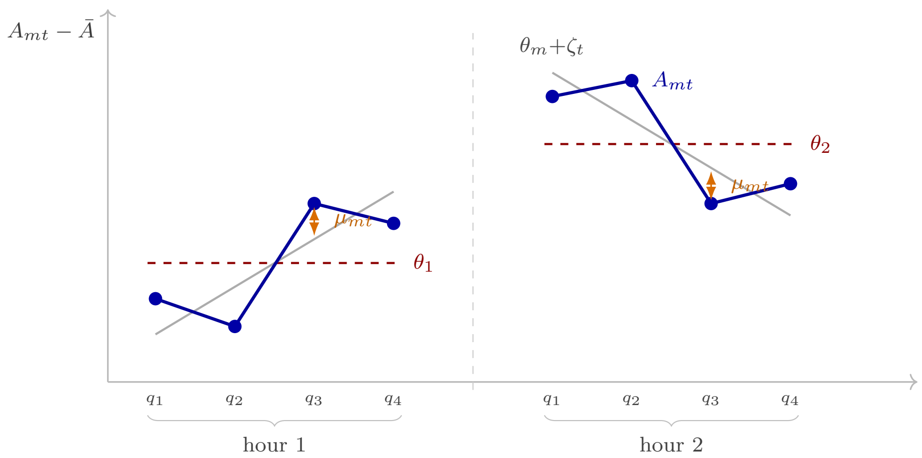


Figure 4: The state the bidder faces, $A_{mt} = \bar{A} + \zeta_t + \theta_m + \mu_{mt}$, over two hours (§3.1). Dots: the realised per-quarter state A_{mt}. Dashed: the hour level θ_m (a single uniform draw, fixed within the hour). Grey: the backbone $\theta_m + \zeta_t$ — that level plus the deterministic ramp ζ_t — with A_{mt} scattered around it by the news μ_{mt}. The ramp rises in hour 1 and falls in hour 2.

What is known when completes the setup. The ramp ζ_t is known throughout. The forward clears first; only afterwards does the news μ realise, and the aggregator's forecast error η_t resolves later still, by balancing. So the information a stage-m offer is committed against grows down the sequence, $\mathcal{H}_F \subset \mathcal{H}_S \subset \mathcal{H}_B$ (Figure 3): the forward conditions on ζ alone, the spot adds the forward price and the realised news, balancing adds the forecast error. Each offer is committed *before* its own clearing state is realised — the bidder screens v_{mt}, it does not know it — and the news μ enters only at the spot, never at the forward.

This pins down the strategic bidder's problem. Having market power but no production shock, its only uncertainty is the clearing state, which the auction resolves before delivery: it is free to supply whatever quantity it is assigned, so its whole task is to commit, before the state is known, an offer that clears well across every state it might face. Because a price-setter's optimal clearing point moves with the state, no single quantity will do, but a committed schedule will — and the step ladder is exactly that schedule, the instrument that lets one offer *screen* across states. It is the absence of a production constraint, not its presence, that lets the bidder bid a clean curve at all; the aggregator is the mirror image, with a production shock but no market power, and faces a scheduling-and-hedging problem instead. Granularity reaches the two through separate channels — bid shape for the ladder, imbalance for the schedule — that never mix.

The ladder does two things at once, and it helps to separate them. *Where* the offer clears in each state is governed by the bidder's **monopoly line**, the price–quantity point a residual monopolist would pick for each realised demand, and this sets the price *level*; *how* the finite ladder approximates that line sets the bid *shape*. I solve the game by backward induction, and the organising principle is that the sequential coupling runs entirely through the line: backward induction sets each stage's monopoly line — its level, and through the spot continuation its slope — while the ladder simply approximates whichever line it is handed, over the width of the state it screens. The forward and spot are linked because the forward price moves the spot *line*, not because either ladder reacts to the other. Granularity therefore touches a product's bid shape twice: directly, by widening the state it screens on its own market, and indirectly, by tilting — through the continuation slope — the line its ladder tracks. The model makes three deliberate abstractions — a single residual monopolist per product, no supply-function interaction across large firms in the sense of Klemperer and Meyer (1989), and a passive balancing stage — discussed, together with the model's relation to the implicit competitive fringe of Ito and Reguant, in Appendix A.1. §3.2 solves the single-product ladder; the reforms then follow.

3.2 Bid ladders within a product: step offers along the monopoly line

I now solve the bid-shape layer: how the finite step ladder approximates the monopoly line within one delivery product. With linear residual demand and one-dimensional clearing-state uncertainty, a continuous supply schedule would sit exactly on the line and replicate the full-information monopoly outcome state by state, so the only friction is the finite number of tranches.[13] The optimal n-tranche ladder has a closed form for every bid-shape metric, it leaves the mean clearing price untouched (so the pricing layer is unaffected), and it interacts with granularity in a single line (Corollary 2).

Fix one delivery product with residual demand $q = A - b\,p$, where $A = \bar{a} + v$, $v \sim U[-\Delta, \Delta]$, and marginal cost is a constant c; this is the generic stage-m product, with the mt subscripts of §3.1 suppressed here and restored per stage when the reforms apply the result — for a spot product the centre $\bar{a}$ carries the realised forward price. The bidder submits a non-decreasing step offer before v realises, and the auction clears at the crossing of the realised residual demand with the offer. When the crossing lands on a horizontal segment the marginal tranche sets the price; when it lands on a vertical riser the realised demand does. Both cases occur in equilibrium.

Here v is the screened state of §3.1: an hourly product screens it with width $\Delta = \Delta_m$, a quarter-hourly product with $\Delta = \rho\Delta_m$, and the deterministic ramp ζ never enters the shape, so the bid-shape problem reduces to (Δ, b). Expectations are conditional on the information available when the offer is committed (Figure 3): a stage-m offer conditions on $\mathcal{H}_m$, and since the only product-level randomness left given $\mathcal{H}_m$ is the screened state, $\mathbb{E}[\,\cdot \mid \mathcal{H}_m]$ integrates over $v \sim U[-\Delta, \Delta]$. I keep the stage explicit — $\mathcal{H}_F$ for forward (stage 1) objects, $\mathcal{H}_S$ for spot (stage 2) — so that the forward's continuation value, for instance, is $\mathbb{E}[\text{spot profit} \mid \mathcal{H}_F]$, an expectation over the news μ and the spot state that stage 1 has not yet seen.

[13] This friction is institutional, not merely a modelling convenience: a simple offer is capped at 25 price–quantity steps per period in the day-ahead auction (OMIE, 2025a) and at 5 in the intraday auctions (OMIE, 2025b), so the number of tranches is genuinely bounded — and far more tightly on the intraday side, where the bid-shape margin is institutionally narrower regardless of any strategic choice.

The frictionless benchmark is a continuous supply schedule: $S^*(p) = b\,(p-c)$ clears at the full-information monopoly point $p^m(A) = (A+bc)/(2b)$, $q^m(A) = (A-bc)/2$ in *every* state and earns the monopoly value $\mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m]/(4b)$[14] (Lemma 2, Appendix A.3). With one-dimensional demand uncertainty a single firm's optimal supply function is the locus of its profit-maximizing (monopoly) points as demand varies (Klemperer and Meyer, 1989, p. 1258), so it reproduces the full-information monopoly outcome in every state; Lemma 2 is its linear-demand case. The bidder's instrument, however, is a step offer with n price levels, not a continuum. The relevant problem is therefore the best n-step approximation of the monopoly line, and it has an exact solution.

By a "best-response price" $p^m(A)$ I mean the point on the monopoly line for state A, the outcome the bidder would reach if it could condition on A. It cannot — the offer is committed before the state realises — so the ladder *implements* the line with finitely many steps: a continuum would be exact in every state, but n tranches are exact only at $2n+1$ states and approximate between them. That approximation error is the sole friction the next proposition characterises.

Proposition 1 (Optimal n-tranche ladder)**.** *Let $w = 2\Delta/(2n+1)$ and let the monopoly quantity be interior in every state by a quarter-cell margin ($q^m(\bar{a}-\Delta) \geq \Delta/(2(2n+1))$, Assumption 1 in Appendix A.3). The optimal offer with n competing price levels consists of an inframarginal base of $q^m(\bar{a}-\Delta+w/2)$ MW priced at the floor, plus n competing tranches of exactly w MW each at prices $p_k = p^m\big(\bar{a}-\Delta+(2k-\frac{1}{2})\,w\big)$, evenly spaced with grid step $p_{k+1}-p_k = w/b$ and spanning $(n-1)\,w/b$: the ladder splits the state range into $2n+1$ equal intervals, each served at its midpoint monopoly point, and expected profit is*

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] \;=\; \frac{\mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m]}{4b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2}.$$

(Proof: Appendix A.3.)

Figure 5 draws the result for two residual-demand slopes: each ladder weaves across its own monopoly line, crossing it at the midpoint of every horizontal segment and every vertical riser, and a flatter line (larger b) is tracked by a more compressed ladder.

[14]In closed form $\mathbb{E}[(A-bc)^2 \mid \mathcal{H}_m] = (\bar{a}-bc)^2 + \sigma_v^2$, with $\sigma_v^2 = \Delta^2/3$ for $v \sim U[-\Delta,\Delta]$. The level term $(\bar{a}-bc)^2$ drops out of every bid-shape statistic below — all of which depend on (Δ, b) alone — so I carry the expectation as a symbol.

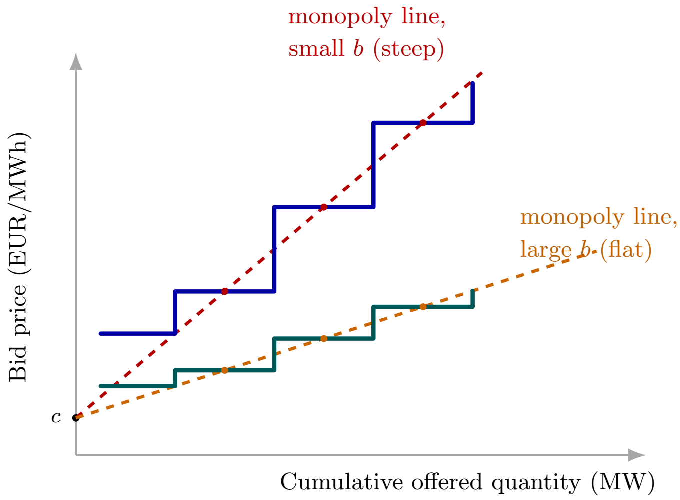


Figure 5: Different residual-demand slopes, different ladders. Each monopoly line $p^m = c + q/b$ (common cost c, different slope $1/b$) is tracked by its own optimal step ladder, weaving across it at every segment midpoint (dots mark the implemented points). Competing tranches carry the same width w, so the price grid step w/b — and hence the dispersion of tranche prices — shrinks as residual demand flattens: a steep line (small b) yields a tall, widely-spaced ladder; a flat line (large b) a compressed one. This is the bid-shape comparative static the reforms move (§3.2).

Corollary 1 (Geometry of the competing tier)**.** *Computed over the n competing tranches — netting out the inframarginal base, which never competes — the optimal tier has equal MW shares, so its MW-Herfindahl index for the tranches is $1/n$ and any weighting of its tranches is immaterial; the dispersion of its tranche prices is*

$$\sigma(\Delta;\, b) \;=\; \frac{2\Delta}{(2n+1)\,b}\sqrt{\frac{n^2-1}{12}} \;\xrightarrow[n\to\infty]{}\; \frac{\sigma_v}{2b},$$

with $\sigma_v = \Delta/\sqrt{3}$ the SD of the screened state; and the ladder climbs at the constant gradient $1/b$ — the inverse of the residual-demand slope — with zero curvature, since tranche prices and cumulative quantities are both evenly spaced.

Ladder geometry is also *level-neutral*: $\mathbb{E}[\text{clearing price} \mid \mathcal{H}_m] = p^m(\bar{a})$ for every n, because within each interval of states the realised price averages to the monopoly price at the interval's midpoint (Remark 1, Appendix A.3). Bid-shape changes and price-level changes are therefore separate channels: a reform can change the spacing and dispersion of a bid curve's tranche prices strongly while leaving its price level untouched.

What *does* move the level is the markup. At the monopoly point the margin over cost is the net sold quantity over the slope of residual demand,

$$p_t - c \;=\; \frac{q_m}{b_{mt}}, \tag{1}$$

the Cournot identity (the first-order condition of $\max_q \big(p(q) - c\big)\, q$ against $p(q) = (A - q)/b_{mt}$). The level moves only through the quantity a firm clears and the slope it faces, never through ladder geometry: a more price-elastic residual demand (larger b_{mt}) compresses the margin, a less elastic one raises it ($\partial(p_t - c)/\partial b_{mt} = -q_m/b_{mt}^2 < 0$). Granularity can push b_{mt} either way — finer products make residual demand more elastic if rivals and re-scheduled volume adjust per quarter, less elastic if a fixed pool of responsive supply is split across more products — so the model leaves the direction free and reads its sign off the data. A slope change is in any case common across hours, so on the bid-shape side it cancels when an amplified hour is differenced against an internally flat one, leaving the shape response to within-hour variation alone.

The ladder stays finite because price steps are not free. Each competing tranche is a distinct price–quantity pair the bidder must design, submit, monitor, and risk-manage — across every delivery product and session — so finer discrimination carries a *bidding-complexity cost*: a menu, or attention, cost paid per competing tranche. It is the smooth counterpart of the hard institutional cap on the number of steps an offer may carry (twenty-five in the day-ahead, five in the intraday); indeed dispatchables use far fewer steps than the day-ahead cap allows (§6.4), so it is this cost, not the cap, that pins their depth there. With a linear such cost $\kappa > 0$ per competing tranche the optimal depth satisfies $2n^* + 1 = \big(\Delta^2/(3b\kappa)\big)^{1/3}$ (Proposition 2, Appendix A.3); without any such cost the bidder would use a continuum of tranches and track the monopoly line exactly, leaving no finite tranche count to respond to anything. Ladders therefore gain tranches as the screened range widens, and lose them as residual demand becomes more price-elastic or tranches more costly. A wider screened range raises the tranche-price dispersion and the number of tranches *together* — a joint signature, not two independent movements.

Corollary 2 (Granularity $\times$ within-hour variation, in closed form)**.** *Let a market switch from hourly to per-quarter products, so that in an hour with amplification ρ the screened range moves from Δ to $\rho\Delta$ and the per-product residual-demand slope from b to b'; write $\sigma(\Delta;\, b)$ for the fixed-n dispersion of Corollary 1. Then, exactly: (i) the conditional monopoly-price range moves from Δ/b to $\rho\Delta/b'$, and with it the dispersion (σ multiplies by $\rho\, b/b'$ at fixed depth) and the tranche-price span; (ii) with the depth endogenous, $2n^*+1$ multiplies by $\rho^{2/3}\,(b/b')^{1/3}$, so the number of tranches rises — the Herfindahl $1/n^*$ falls — exactly when the tranche-price dispersion rises; and (iii) the post-minus-pre double difference between that hour and an internally flat one ($\rho = 1$) nets out the slope change, which is common to both: at fixed depth,*

$$\underbrace{\big[\sigma(\rho\Delta;\, b') - \sigma(\Delta;\, b)\big]}_{\textit{hour with amplification } \rho} \;-\; \underbrace{\big[\sigma(\Delta;\, b') - \sigma(\Delta;\, b)\big]}_{\textit{internally flat hour}} \;=\; (\rho - 1)\,\sigma(\Delta;\, b'),$$

zero when the hour is internally flat and strictly increasing in the within-hour variation that ρ encodes (Lemma 1). (Proof: Appendix A.3.)

Read the corollary as two things happening to one product. Granularity widens the state the ladder must cover, $\Delta \to \rho\Delta$, by exposing the within-hour news — the bid-shape channel, alive only where the hour's profile varies. Separately the residual-demand slope moves, $b \to b'$, as the market re-clears with finer products; but that slope is an equilibrium object, handed to the ladder by the backward-induction solution of the pricing layer (§3.4), and it shifts every hour alike. Part (iii) differences the two: the common slope change cancels and the hour-specific width channel $(\rho - 1)\,\sigma$ remains. This is the mechanism of the thesis in one display — the reform changes bid shape exactly where, and exactly as much as, within-hour variation amplifies the state a product screens; anything common to the two hours (the slope change, complexity costs, price-level seasonality) drops out by construction.

Corollary 3 (Continuation slope and the forward monopoly line)**.** *Under the stage-1 second-order condition $B_2 \equiv \sum_t b_{2t} < 2b_1$ (Assumption 2, Appendix A.3), the slope of the stage-1 monopoly line is $dp_1^*/dv = 1/(2b_1 - B_2/2)$, the forward ladder's gradient is $1/(b_1 - B_2/2)$, and the stage-1 clearing-price range over the screened states is exactly $2\Delta_1/(2b_1 - B_2/2)$. As the spot market becomes per-quarter, B_2 jumps from a single hourly slope to the sum of Q per-quarter slopes; the forward monopoly line, the ladder gradient, and the price range all steepen and widen correspondingly — the cross-market anticipation channel of §3.4.*

A price-taking unit, by contrast, simply posts its marginal-cost schedule. For a zero-marginal-cost renewable this is a single block at the floor — a one-tranche tier with zero price dispersion, out of band at any granularity (Remark 2, Appendix A.3). The bid-shape margin is therefore specific to dispatchable technologies; the renewable margin of the reform is the quantity re-scheduling of §3.4.

3.3 Arbitrage regimes and the pre-reform baseline

Between the two auctions sits the arbitrageur. Following Ito and Reguant (2016, §I.A and Table 1), arbitrage profit per matched product is $\pi^A(s) = s\,(p_1 - p_2)$, and four regimes pin the position. Under *no arbitrage*, $s = 0$ and a positive forward premium $p_1 > p_2 > c$ survives. Under *full arbitrage*, s closes the spread and the strategic bidder withholds more in response. Under *limited arbitrage*, a capacity ceiling $s \leq K_a$ binds and sustains a wedge that depends on K_a. Under *strategic arbitrage*, finally, the arbitrageur internalises its own price impact and stops short of parity at the closed-form benchmark

$$\boxed{p_1 - p_2 \;=\; \frac{p_1 - c}{4} \;>\; 0.} \tag{2}$$

Rather than carry all four through the analysis, I **focus on the limited-arbitrage regime** as the maintained case, for one empirical reason: an arbitrageur with a bounded balance sheet and price impact does not close the spread, and full arbitrage would eliminate the forward premium that sequential-market models exist to explain (Ito and Reguant, 2016). The choice is about the empirically relevant magnitude, not the direction — the comparative-statics signs agree across the no-arbitrage, limited, and strategic regimes, and Appendix A records each at every state for completeness. Under limited arbitrage, once granularity makes the wedge per-product at (Q, Q) the capacity constraint binds at K_a/Q per product — the reason, in the model, why within-hour wedge dispersion need not collapse even when both auctions are granular.

Pre-reform $(1, 1)$**.** Before the reforms both auctions are hourly. The Ito and Reguant scarcity wedge $p_1 > p_2 > c$ obtains under any regime short of full arbitrage (Appendix A.4). Neither auction can carry the within-hour profile or the redistributive news, so whatever the contracting grid cannot carry loads onto the balancing residual: the balancing discrepancy is $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$. All within-hour dispersion objects — across-quarter price dispersion, the within-hour wedge SD — are mechanically zero, and the competing tiers screen the hour-level ranges Δ_1 and Δ_2: the pre-reform baseline of the bid-shape statistics.

3.4 Spot reform $(1, Q)$: contractibility arrives

At ID15 the spot becomes quarter-hourly while the forward stays hourly. The first granular market makes the within-hour state contractible for the first time, and four mechanisms activate.

The aggregator gains from granular spot scheduling. A granular spot lets the aggregator wait until its forecast update η_t resolves and re-schedule per period, driving its per-period imbalance to zero. An hourly spot, in contrast, forces one schedule before η_t is known, so the update loads onto the imbalance bill. Under uniform shocks the expected hourly hedging gain is the avoided cost

$$\boxed{G(\Delta_\eta) \;=\; \frac{\gamma Q\,\Delta_\eta}{2}, \qquad \frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0,} \tag{3}$$

which is strictly increasing in the forecast-error range — largest for wind and solar. The individual split is a corner solution, so the migration is an aggregate equilibrium statement: $G > 0$ is a participation premium that shifts aggregator volume toward the granular spot (Appendix A.2).

Price levels: market re-clearing and cross-market transmission. This is the pricing layer: the spot best response per quarter is the monopoly line evaluated at the realised spot state, $p^*_{2t} = (p_1 + c)/2 + (\psi_t + s_t - S^{op}_{2t})/(2b_{2t})$ (Appendix A.5). The spot level therefore falls with migrated aggregator supply, and with any rise in the per-product residual-demand slope through the markup identity (1). On the forward leg two forces compete: arbitrage transmits the spot discount upstream, while the strengthened continuation value pushes toward more withholding. The net forward effect is ambiguous in the model.

Block parity reveals within-hour spot dispersion. The hourly forward can be arbitraged only against the *average* of the Q spot prices — block parity — so arbitrage disciplines the mean wedge but not the within-hour dispersion of the granular leg. The spot prices spread across quarters with the update:

$$\boxed{\mathrm{sd}_t(p_{2t}) \;=\; \frac{\sigma_\psi}{2\,b_2}, \qquad \sigma^2_\psi = \sigma^2_\zeta + \sigma^2_\mu,} \tag{4}$$

Since p_1 is one number per clock-hour, (4) is also the within-hour wedge SD. It was mechanically zero before the reform, and it is largest in ramp hours, where σ_ζ is large (corrections for asymmetric per-quarter slopes in Appendix A.5).

Bid ladders: own-market exposure and cross-market anticipation. Two exact ladder results follow. *(a) Spot tier, own market.* Each quarter product screens the ρ-amplified state, so by Corollary 2 the spot tranche-price dispersion and the number of tranches rise in proportion to $\rho - 1$ — strongly where the within-hour state varies, not at all where it is flat — while slope shifts common to all hours net out in the double difference. *(b) Forward tier, cross-market.* The stage-1 problem internalises the spot continuation value, and the slope of the stage-1 monopoly line is

$$\boxed{\frac{dp^*_1}{dv} \;=\; \frac{1}{2b_1 - B_2/2}, \qquad B_2 \equiv \sum_t b_{2t},} \tag{5}$$

with B_2 the total slope of the spot products hanging off the forward product (I maintain $B_2 < 2b_1$, Assumption 2). At the spot reform B_2 jumps from the single hourly slope b_2 to the sum of Q per-quarter slopes, so whenever the per-quarter slopes are comparable to the hourly one the jump is large and the forward line steepens: the stage-1 clearing-price range is exactly $2\Delta_1/(2b_1 - B_2/2)$ and the ladder gradient becomes $1/(b_1 - B_2/2)$ (Corollary 3). Through this anticipation channel the spot reform reshapes the *forward* bid curve as well — its tranche-price dispersion and its slope both rise — even though the forward product itself is unchanged. The tranche-count response on the forward side is not signed by this channel alone, and I do not claim it.

On the balancing side, the quarter-hourly spot absorbs $\zeta_t + \mu_t$ and the aggregator hedges η_t, so the balancing discrepancy is $\Delta_t = \varepsilon_t$ already at this state (Appendix A.8). Contractibility arrives with the *first* granular market.

3.5 Forward reform (Q, Q): relocation and limited per-product arbitrage

The forward reform brings no new contractibility: it *relocates* the already-contractible within-hour objects from the spot to the forward. But only the ramp can relocate. The ramp ζ_t is known at every stage, so once the forward is granular it prices it; the news μ_t realises only after the forward gate — it is never in the forward's information set $\mathcal{H}_F$ (Figure 3) — so no forward product, however fine, can contract it. The news therefore stays at the spot, and that retained within-hour object is why the spot keeps trading inside the hour and the wedge does not collapse.

The ramp becomes forward-contractible. Per-quarter forward products price the ramp at the day-ahead margin: $\mathbb{E}[p_{1t} \mid \mathcal{H}_F] = p_1^*(\bar{A}+\zeta_t)$, with pass-through $\lambda_1 \equiv (2b_1 - B_2/2)^{-1}$ per unit of ramp. The across-quarter dispersion of forward prices therefore turns on at $\mathrm{sd}_t(\mathbb{E}[p_{1t} \mid \mathcal{H}_F]) = \lambda_1\sigma_\zeta$: a dispersion of forward price *levels* across the quarters of the hour, where before the reform there was a single price. The spot's update demand correspondingly shrinks from $\zeta_t + \mu_t$ to μ_t.

Per-product limited arbitrage and arbitrageur market power. The arbitrageur now earns $s_t\,(p_{1t} - p_{2t})$ per matched quarter. Full arbitrage would give $p_{1t} = p_{2t}$ and a wedge SD of zero; unlimited strategic arbitrage would give the per-product wedge (2) and a wedge SD of $\lambda_1\sigma_\zeta/4$. Neither obtains once the per-product capacity K_a/Q binds. The symmetric state therefore sits in the **limited-arbitrage regime** (Ito and Reguant Result 3) applied per product: the per-product wedge becomes $(p_{1t} - c)/2$ rather than $(p_{1t} - c)/4$, doubling the wedge SD to

$$\boxed{\mathrm{sd}_h(p_{1t} - p_{2t}) \;=\; \frac{\lambda_1\,\sigma_\zeta}{2}} \tag{6}$$

(Appendix A.6.3). The wedge SD does *not* collapse at the symmetric state: bounded, market-power-conscious arbitrage sustains it.

Bid ladders relocate, not duplicate. The forward screen widens from Δ_1 to $\rho\Delta_1$, because the redistributive component becomes contractible at the forward, so by Corollary 2 the forward tranche-price dispersion and tranche count rise in high-ρ hours exactly as the spot's did at the spot reform — the same contrast, now on the forward. Symmetrically, the spot screen loses its ramp-timing component. Writing $\rho^\mu \le \rho$ for the news-only amplification (Lemma 1 on the smaller support; equal to one when the hour is internally flat), the spot screen falls from $\rho\Delta_2$ to $\rho^\mu\Delta_2$ and the spot bid curve *concentrates* — fewer tranches on a narrower price range — by the same closed forms run in reverse.

Price level. As at the spot reform, the forward price level moves only through the markup channel (1): it falls where the per-quarter re-clearing makes residual demand more price-elastic (a larger b_{1t}) and rises where it makes it less so, never through ladder geometry (Remark 1). Which way it goes is an empirical question, with exact comparative statics for both branches in Appendix A.9.4.

Balancing unchanged. Nothing new loads onto the balancing residual: $\Delta_t = \varepsilon_t$ at both $(1, Q)$ and (Q, Q), so no further imbalance gain arrives at the forward reform.

3.6 Comparative statics and welfare across the reform path

Taken together, the model tells a simple story. When the spot becomes quarter-hourly, the within-hour state becomes contractible for the first time, and everything else follows from that. The aggregator no longer needs the balancing layer to absorb its forecast errors: it re-schedules per quarter and shifts volume toward the granular market, the more so the larger its uncertainty. The granular-side price falls through market re-clearing; the slope of each firm's residual demand then steers the level further — prices and markups fall if finer products make it more price-elastic, rise if they make it less so — but never through ladder geometry, which is level-neutral (Remark 1). The wedge between the two auctions starts to vary within the hour, most where the ramp is steep. And the bid curves change shape: in the spot, tranche-price dispersion rises and the quantity splits into more tranches where within-hour variation is large (Corollary 2); in the forward, the slope of the bid curve increases in anticipation, through the continuation channel (Corollary 3).

When the forward follows, nothing new becomes contractible; the within-hour objects relocate. The ramp moves to the forward margin, so the forward bid curve disperses and gains tranches exactly as the spot's did at the first reform, while the spot bid curve — left with the news alone — concentrates back into fewer, more closely priced tranches. The within-hour wedge dispersion does not collapse, because the arbitrage that would close it is carried by capacity-bounded position-takers whose per-product capacity shrinks with the number of products.

This asymmetry organises the whole reform path: new contractibility arrives only at the *spot* reform, the *forward* reform relocates it, and whatever the balancing layer gains, it gains at the first step. Table 8 (Appendix A.7) collects the state-by-state closed forms behind this narrative.

Welfare. Three channels raise welfare and one is distributional. The imbalance saving $\gamma Q(\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2)$ arrives entirely at the spot reform, and it is net-positive whenever $\gamma > 1/(8b_2)$ after the small allocative cost of per-quarter monopoly pass-through. Markup compression operates wherever the residual demand a firm faces becomes more price-elastic (b_f rises), through (1); by Remark 1 it is the only channel that moves mean prices. The discreteness loss $\Delta^2/(12b(2n^*+1)^2)$ shrinks where ladders gain tranches. Bid-ladder rent, finally, is mostly a transfer from consumers to the strategic bidder. Per-channel derivations are in Appendix A.9.

4 Data and variables of interest

The model in §3 speaks in objects a researcher rarely observes directly: the screen each unit faces, its monopoly line, the within-hour state. What is observable is the offer that screens that state — the step ladder of §3.2 — together with the prices it clears against and the balancing residual it leaves behind. The three data sources below are chosen to recover exactly these: the bid ladder and the auction prices from OMIE, the balancing layer from REE, and the within-hour state from the weather that drives it.

The primary source is the public file archive of the market operator (OMIE). It carries the clearing prices of every period of the day-ahead and the three intraday auctions,[15] the continuous-market trades, and — the object the bid-shape analysis is built on — the accepted offer curve of each unit in the day-ahead and in each intraday session. The quantity side is a cascade of "programs": each auction produces a schedule, before and after bilateral contracts and before and after technical restrictions, so cleared MW is jointly determined across many stages rather than read off a single market. This is why the strategic content of the model lives in the *bid curve*, observed before clearing, rather than in cleared quantities, which the cascade makes endogenous (a point I return to in §4.3).

The balancing layer — the model's residual Δ_t and its quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ — comes from the system operator (REE) through the ESIOS API: imbalance settlement prices, ancillary-service energy costs, and the redispatch ledger that the reinforced-operation regime of §6.6 moves. The within-hour state itself is not a market object; I proxy it with the residual demand

[15]The third intraday session is observed only from 13:00 of the delivery day onward, since it clears at 10:00 (Figure 1).

that dispatchables must cover — load net of zero-marginal-cost wind and solar — built from the European Network of Transmission System Operators (ENTSO-E), which also supplies cross-border flows and the renewable forecasts whose error η_t the aggregator hedges.

The panel runs, for most series, from January 2018 to the present, spanning the three granularity reforms that repeatedly reshape the file formats.[16]

4.1 Bid offers, reform path, and notation

The object I observe, and the one the analysis is built on, is a single offer curve: one generating unit u, on date d, bidding into one delivery period p — a clock-hour before that market's switch to fifteen minutes, a quarter-hour after it. Each sell offer arrives in the tranche format of the OMIE files, a step ladder $\{(p_k, \Delta q_k)\}_{k=1}^{K_{u,d,p}}$ sorted by ascending price, where Δq_k is the extra megawatts the unit adds once the price reaches p_k; cumulating the increments, $q_k = \sum_{j \le k} \Delta q_j$, recovers the corner points of the model's step offer. Offers come in three forms — a bare price–quantity ladder, one carrying a minimum-income condition, or an all-or-nothing block spanning periods — and most technologies bid the bare form, the combined-cycle plants being the exception, bidding mostly income-conditioned or block. Against each offer I line up the auction's clearing price $\text{MCP}_{d,p}$ and its cleared quantity $Q^\star_{d,p}$. Throughout, $h \in \{1, \ldots, 24\}$ indexes the clock-hour and $q \in \{1, \ldots, 4\}$ its quarters (defined only after the switch), s the three intraday sessions, t a technology, $m \in \{\text{DA}, \text{IDA}\}$ the market, k the rungs of a ladder, and T_r the cutover date of a given reform.

Three reforms move the granularity of this object in under a year, and the whole empirical design turns on keeping them apart. The settlement clock went quarter-hourly first (ISP15, 11 December 2024), then the intraday auctions and the continuous market (ID15, 19 March 2025), and finally the day-ahead (DA15, 1 October 2025); the late-April 2025 blackout opened the reinforced-operation regime that then runs alongside the last cutover. Because that regime overlaps the day-ahead reform on the calendar but works through a different mechanism, every comparison window below is cut inside a span where the regime is constant, so a reform effect and a regime effect are never confounded.

4.2 In-band region and bid-shape outcomes

The model gives the bid curve a shape only where the curve actually competes. The optimal ladder of §3.2 lives in a neighbourhood of the realised monopoly point; the tranches that the screen reaches are the ones that price the state, and everything outside that neighbourhood is, in the model, irrelevant to the firm's problem. Real offer ladders, by contrast, carry a long *parked tier* of tranches priced far from any plausible clearing level — a near-zero floor that guarantees must-run dispatch, a near-cap ceiling that insures against scarcity — which never compete and which would dominate any dispersion or concentration statistic computed over the whole curve. To recover the competing tier the model cares about, I keep only the tranches within a bandwidth δ of the contemporaneous clearing price.

This bandwidth has an empirical job the model does not pin down. The interior-solution assumption (Assumption 2, which keeps the stage-1 problem concave and the ladder centred on the monopoly line) tells us *that* the active tranches sit near clearing, but it does not deliver a number: the screen half-width Δ_m is a latent object, not an observable price. So δ is the

[16] Daylight-saving transitions give a 23-hour and a 25-hour day each year; I keep both at their true length and index periods within the local clock day, so the within-hour quarter structure of §4.1 is never misaligned across the change.

empirical analogue of "the part of the ladder the screen reaches," not the screen itself — related to the interior-solution restriction, but set from the data rather than from Δ_m. Two choices define the band, and it helps to separate them. Its *centre* is never fixed: for every (unit, date, period) the band is re-centred on that observation's *own* realised clearing price, so each curve is read against the price it actually faced, not against a window average. Its *half-width* is a single number per window, $\delta = p_{90} - p_{50}$ of the clearing-price distribution in the analysis window, market by market — the typical distance from the median price up to its upper decile, a measure of how wide the contested margin runs in that regime. Keeping the half-width common while letting the centre float means the band tracks the price level without letting a high-price window mechanically admit more tranches than a low-price one: a period that clears at 120 and one that clears at 40 EUR/MWh are each judged over the same-width neighbourhood around their own clearing price. The eight values (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are in Table 1; the bid-shape results are robust to widening δ (Appendix B.2). CCGT's explicit two-tier offer structure (Appendix C.1) makes the competing-versus-parked distinction visible in the raw data.

Per in-band curve I compute the five sample counterparts of the competing-tier geometry of Corollary 1: the MW-weighted price dispersion σ_p (the tier dispersion $\sigma(\Delta; b)$), the bid-stack concentration $\mathrm{HHI}_{\mathrm{tr}}$ (the inverse depth $1/n$), the slope $\hat{\beta}$ (the inverse residual-demand slope $1/b_{mt}$), the curvature $\hat{\gamma}$ (zero on the even grid), and the level $\hat{\alpha}$ (granularity-invariant by Remark 1). Two of these are robust price-side anchors and three are mechanically fragile; I make the distinction precise after defining them.

Window	δ in DA (EUR/MWh)	δ in IDA (EUR/MWh)
ID15 real	50	62
ID15 placebo (2024)	45	46
DA15 real	50	58
DA15 placebo (2024)	45	49

Table 1: Bandwidths δ used to define the in-band region. δ is set to $p_{90} - p_{50}$ of the MCP distribution in each (window, market). Window-and-market-specific by construction. Bandwidth robustness in Appendix B.2.

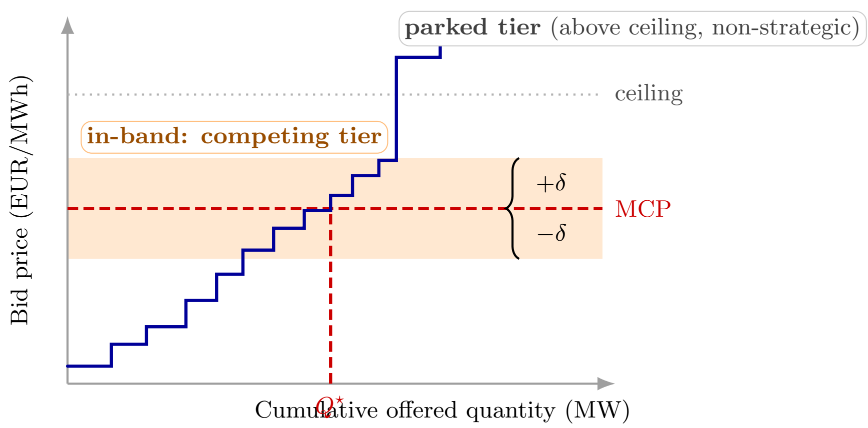


Figure 6: Bandwidth δ and the in-band region. The blue step function is the sorted sell-bid ladder for one quarter. Only tranches falling within $\pm\delta$ of the clearing price (orange band) enter the bid-shape outcomes; the parked tier above the clearing-price ceiling is excluded (Appendix C.1).

For a given offer curve, then, the in-band set collects the rungs within δ of clearing, with megawatt shares and a share-weighted mean price,

$$\mathcal{B}_{u,d,p} \;=\; \big\{k:\; |p_k - \mathrm{MCP}_{d,p}| \le \delta\big\}, \qquad w_k = \frac{\Delta q_k}{\sum_{j\in\mathcal{B}_{u,d,p}} \Delta q_j}, \qquad \bar{p}_{u,d,p} = \sum_{k\in\mathcal{B}} w_k\, p_k,$$

and on that set I summarise the ladder with five numbers, the sample images of the competing-tier geometry the theory delivers. (What these look like for each technology, dispatchable ladders against single renewable blocks, is in Appendix C.1.) Writing $Q_k = \sum_{j\in\mathcal{B},\, j\le k} \Delta q_j$ for the cumulative in-band megawatts up to the kth rung, the one I lean on is the share-weighted price dispersion,

$$\sigma_{p,\,u,d,p} \;=\; \Big[\sum_{k\in\mathcal{B}} w_k\,(p_k - \bar{p}_{u,d,p})^2\Big]^{1/2}$$

— how spread out the in-band rungs are, weighted by megawatts because profit risk scales with tranche size and an unweighted spread would let a handful of micro-rungs dominate a statistic meant to capture where the offered megawatts actually sit. Equally robust is the bid-stack concentration, the megawatt-weighted Herfindahl of those rungs,

$$\mathrm{HHI}_{\mathrm{tr},\,u,d,p} \;=\; \sum_{k\in\mathcal{B}} w_k^2 \;\in\; \big[1/|\mathcal{B}|,\,1\big],$$

which falls as the ladder splits into more, more even steps — the empirical face of the ladder's depth. The other three come from fitting the rungs: an unweighted regression of price on cumulative in-band megawatts,

$$p_k \;=\; \hat{\alpha}_{u,d,p} + \hat{\beta}_{u,d,p}\, Q_k + e_k,$$

returns a level $\hat{\alpha}$ and a slope $\hat{\beta}$, the latter the empirical counterpart of the inverse residual-demand slope; and a separate quadratic fit,

$$p_k \;=\; \tilde{\alpha}_{u,d,p} + \tilde{\beta}_{u,d,p}\, Q_k + \hat{\gamma}_{u,d,p}\, Q_k^2 + \tilde{e}_k,$$

adds a curvature $\hat{\gamma}$ — positive when cheap megawatts are packed at the bottom and the marginal ones priced steeply, negative when concave, and meaningful only on curves with at least three rungs. Reading all five against the contemporaneous clearing price through δ is what makes them immune to price-level seasonality: a January curve and a July curve are compared by shape, not by the level the season puts them at.

The five are not on an equal footing, and the difference governs which I lead with. The dispersion and the concentration are defined on every curve — even the single- and few-rung ones where a fitted slope is pure noise — and they shrug off the mechanical changes in tranche count and price-grid spacing that the 2024 session reform and ID15 introduced. The slope and curvature inherit those mechanical shifts, so their pre-reform trends drift (Appendix B), and the level is useful only as a check that a reform moved a curve's shape without moving where it sits. I therefore lead with the dispersion and the concentration and keep the other three in support.

4.3 Aggregate outcomes

Four system-level outcomes are built off the bid stack and the cleared prices, each the empirical counterpart of one model object: the day-ahead/intraday wedge and its within-day dispersion (the arbitrage object of §3.3); the intraday-auction in-band sell-share (the aggregator-migration mechanism of §3.4); the system adjustment cost (the balancing layer); and five post-MTU15 within-hour dispersion outcomes that read within-hour discrimination off the data directly.

The model also has a clean quantity-side prediction — more capacity should compete near clearing once the state is contractible — but I do not build an outcome for it. A per-curve in-band offered-energy series would have to be summed across the program cascade, where cleared and offered megawatts are jointly determined over many stages, and it would inherit the bandwidth δ, which itself moves with the price level; the resulting quantity is too endogenous to read as a treatment response. I keep the quantity-side prediction as an open, in-reach test (§6.3) rather than force it through a noisy proxy.

The first is the wedge between the two auctions, the day-ahead price minus the intraday for the same clock-hour, together with its within-day spread,

$$\Delta_{d,h} \;=\; \mathrm{MCP}^{\mathrm{DA}}_{d,h} - \mathrm{MCP}^{\mathrm{IDA}}_{d,h}, \qquad \Delta^{\mathrm{SD}}_{d} \;=\; \mathrm{sd}_h(\Delta_{d,h}).$$

The level of Δ is the forward premium the two auctions carry; its within-day standard deviation is the dispersion granularity unlocks. The second places each technology on the intraday by its in-band sell-share,

$$\mathrm{SS}_{t,d} \;=\; \frac{\sum_{q,k\in\mathcal{B}} q^{\mathrm{sell}}_{t,d,q,k}}{\sum_{q,k\in\mathcal{B}}\big(q^{\mathrm{sell}}_{t,d,q,k} + q^{\mathrm{buy}}_{t,d,q,k}\big)},$$

the share of its in-band megawatts offered to sell rather than to buy — which separates the rent-capturing sellers on the granular leg from the forecast-driven rebalancers, the footprint the aggregator's migration would leave. The third is the system adjustment cost, the daily bill REE runs to keep the system feasible, summed from the ESIOS settlement series across its up- and down-direction channels (the Fase I and Fase II technical restrictions, the real-time restrictions, and the secondary, tertiary and replacement-reserve energy); it enters as complementary evidence on the balancing layer, not as a primary object. The last set exists only after the fifteen-minute switch and reads within-hour discrimination straight off the data: for each unit, date and clock-hour I take the across-quarter standard deviation of, in turn, the curve's price level, its offered volume (as a coefficient of variation), the system clearing price, the ladder's spread, and its tranche concentration,

$$D^{\bar p}_{d,h} = \mathrm{sd}_q(\bar p_q), \quad D^{m}_{d,h} = \mathrm{sd}_q(m_q)/\bar m, \quad D^{\mathrm{MCP}}_{d,h} = \mathrm{sd}_q(\mathrm{MCP}_q), \quad D^{\sigma_p}_{d,h} = \mathrm{sd}_q(\sigma_{p,q}), \quad D^{\mathrm{HHI}}_{d,h} = \mathrm{sd}_q(\mathrm{HHI}_{\mathrm{tr},q}).$$

Each is mechanically zero under hourly products — a single hourly figure cannot differ from itself across quarters — so any positive value after the switch is a unit deliberately treating its four quarters differently.

4.4 Critical / flat hour-class partition

Granular pricing has economic content only where the within-hour state varies. Recall the decomposition of §3.1: the state a unit faces in quarter t of an hour is $A_{mt} = \bar A + \zeta_t + \theta_m + \mu_{mt}$. Of these four terms, the level $\bar A$ and the hour draw θ_m are constant across the four quarters of a clock-hour; only the deterministic ramp ζ_t and the redistributive news μ_{mt} move within the hour. The whole granularity mechanism — the screen amplification ρ of Corollary 2, the wedge dispersion, the extra tranches — is driven by the dispersion of $\zeta_t + \mu_{mt}$, and is mechanically absent in an hour where that dispersion is near zero. So the partition I need is one that sorts clock-hours by how much $\zeta_t + \mu_{mt}$ varies inside them.

The within-hour state is not observed, but its physical driver is. The state the dispatchable bidder screens is its residual demand — system load net of the zero-marginal-cost wind and solar that clear infra-marginally and never sit on the dispatchable's monopoly line.[17] The across-quarter standard deviation of residual demand inside a clock-hour is therefore the empirical counterpart of the within-hour dispersion of $\zeta_t + \mu_{mt}$: a high-ramp evening hour, with load climbing and solar collapsing across its four quarters, has a large one; a flat overnight hour has a small one. I measure it two ways on 15-minute ENTSO-E data (A65 load; A75 codes B16, B18, B19 for wind and solar) — residual demand rather than raw load precisely because renewables enter the dispatchable's problem only through what they leave behind. The *level* measure is the across-quarter standard deviation itself, $\sigma_{\text{within}}(h,d) = \text{sd}_q[\text{load} - \text{wind} - \text{solar}]_{q,h,d}$, in MW — the volume a firm could in principle reprice between the four quarters. Its *scale-free* counterpart is the within-hour coefficient of variation, computed per hour-day and then averaged over days, $\text{CV}(h) = \mathbb{E}_d\big[\sigma_{\text{within}}(h,d)/\overline{\text{RD}}(h,d)\big]$, which divides out the level so that two hours with the same proportional swing score alike regardless of their load.[18] Partitioning clock-hours by their median σ_{within} over the post-2024-06-14 regime gives three classes, and the level and scale-free measures rank them alike where it matters:

- **Critical** $\mathcal{C} = \mathcal{C}_M \cup \mathcal{C}_E$ with $\mathcal{C}_M = \{5,\dots,8\}$ (morning ramp) and $\mathcal{C}_E = \{16,\dots,22\}$ (evening ramp); median $\sigma_{\text{within}} \approx 830$ MW (per-hour medians 356–1355), CV $\approx 6.3\%$.
- **Flat** $\mathcal{F} = \{1,2,3\}$: overnight; median $\sigma_{\text{within}} \approx 200$ MW (162–242), CV $\approx 1.9\%$.
- **Midday** $\mathcal{M} = \{11,\dots,14\}$ (solar-marginal): median $\sigma_{\text{within}} \approx 250$ MW (212–292), close to flat in level, *yet* CV $\approx 6.8\%$, the highest of all classes. Excluded from main specs; kept for falsification (Appendix B.3).

In the language of the model, critical hours are the large-ρ hours and flat hours the $\rho \approx 1$ benchmark: critical residual demand swings about four times as much within the hour as flat residual demand in level terms (σ_{within} near 830 versus 200 MW), and roughly three times as much scale-free (CV near 6% versus 2%), so the within-hour state these hours hand the bidder is genuinely different on either reading. Midday is the instructive exception, and it is why I partition on the level rather than the ratio: its coefficient of variation is critical-sized, but only because the solar-suppressed denominator is small — in absolute MW, the volume a firm could reprice across the four midday quarters is no larger than at flat overnight hours. The granularity mechanism acts on that volume, not on its ratio to a shrinking base, so σ_{within} is the economically right sorter; midday is held out precisely because the two measures disagree there, which makes it a clean placebo. The critical-minus-flat, post-minus-pre double difference is then the empirical counterpart of Corollary 2(iii), and the bid-shape DiD of §5.2 uses this contrast as its within-day identification device. The partition earns its keep: the descriptive gap below shows that wherever the model says discrimination can happen — the critical hours — the within-hour dispersion outcomes light up, and in flat hours they stay near zero.

To see this before any reform comparison, take each of the five within-hour dispersion outcomes on the post-switch sample and regress it on nothing more than a critical-hour indicator with date fixed effects,

$$D^{\bullet}_{d,h} \;=\; \eta_d \;+\; \theta\, \mathbf{1}\{h \in \mathcal{C}\} \;+\; \varepsilon_{d,h},$$

so that θ is the bare critical-minus-flat gap — a description, not yet a causal estimate. The pattern is the one the partition promises: every dispersion outcome runs several times larger in

[17] Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

[18] The mean-of-ratios here, rather than the ratio-of-means $\mathbb{E}_d[\sigma_{\text{within}}]/\mathbb{E}_d[\overline{\text{RD}}]$, keeps the statistic a genuine per-day object; the two agree to within a percentage point in every class, so nothing below turns on the choice.

critical hours than in flat ones, and the gap is sharpest exactly where the model expects the most discrimination.

Outcome	Subsample	ID15	DA15	crit/flat ratio (IDA, DA)
$D^{\bar{p}}$ (EUR/MWh)	per unit (pooled)	**0.39***** (0.05)	**1.35***** (0.11)	4.1×, **7.4**×
D^{m} (CV)	per unit (pooled)	**0.024***** (0.003)	**0.033***** (0.002)	4.3×, 4.9×
D^{MCP} (EUR/MWh)	system	**4.55***** (0.40)	**3.97***** (0.37)	2.8×, 2.6×
D^{σ_p} (EUR/MWh)	CCGT	−0.19 (0.36)	**0.61***** (0.12)	0.8×, **6.5**×
	Hydro	**0.48***** (0.08)	**1.32***** (0.17)	2.2×, 3.3×
	Hydro pump	**0.14***** (0.03)	**0.70***** (0.17)	3.3×, 10.9×
	Wind	0.00 (0.01)	0.03** (0.01)	0.9×, 1.9×
$D^{1/\mathrm{HHI}}$	CCGT	0.17 (0.16)	**0.17***** (0.02)	1.1×, **14.3**×
	Hydro	**0.05***** (0.01)	**0.19***** (0.03)	2.0×, 2.8×
	Hydro pump	**0.03***** (0.01)	**0.13***** (0.03)	2.6×, 4.3×
	Wind	0.00 (0.00)	0.01 (0.00)	1.0×, 1.3×

Table 2: Post-MTU15 critical−flat within-hour dispersion gap θ, by outcome. Post-only cross-sectional gap. Windows: ID15 post 2025-03-19 to 2025-04-27; DA15 post 2025-10-01 to 2025-12-31. SE clustered by date; $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$ (applies to all subsequent tables). Bid-level rows ($D^{\bar{p}}$, D^{m}, D^{MCP}) are bandwidth-free; the per-tech D^{σ_p} and $D^{1/\mathrm{HHI}}$ rows use the window-specific p_{90} bandwidth (§4.2): ID15 column δ=62 (post-ID15 IDA); DA15 column δ=50 (post-DA15 DA). $D^{1/\mathrm{HHI}}$ uses the effective-tranche-count framing ($N_{\mathrm{eff}} = 1/\mathrm{HHI}_{\mathrm{tr}}$) because the within-hour SD of HHI is mechanically tiny on a $[0, 1]$ scale; the inverse-HHI quantity carries the same content on a more readable scale.

5 Empirical strategy

The empirical strategy pairs two specifications with two economic objects. For **level outcomes** (prices, wedge SD, imbalance penalty, cleared MWh) we run hourly OLS and a BSTS counterfactual on long pre-windows (2022 onwards) with year-by-year renewable interactions, and check robustness to convex (quadratic and cubic) renewable terms. For **bid-shape outcomes** (σ_p, $\hat{\alpha}$, $\hat{\beta}$, $\hat{\gamma}$, $\mathrm{HHI}_{\mathrm{tr}}$) we run a bid-curve-level DiD with flat hours as the within-day control for critical hours.

5.1 Level variables: long pre-windows, renewable controls, and a counterfactual

For the level outcomes the threat is not seasonality within a regime but the renewable build-out across regimes. Spanish solar capacity grew about forty per cent between January 2024 and January 2025, so a spring-2025 price that merely clears flat is, net of solar's downward push on the merit order, already a fall; telling the reform's effect from the fleet's is the whole identification problem, and it dictates every choice that follows. The pre-window runs from January 2022 — long enough for the renewable coefficients to be pinned down across several years of the build-out before they are extrapolated to the post-reform months, where a short window could not separate granularity from solar. The post-windows are kept short and regime-clean: forty days for the spot reform, ending before the April 2025 blackout so that the reinforced-operation regime touches neither side; ninety-two days for the forward reform, with its pre-window started on the blackout date itself, so reinforced operation is held fixed across the cutover rather than switched on by it.

The headline regression is an hourly ordinary-least-squares fit of the outcome on a post-reform indicator, wind, solar and gas, each renewable interacted with a post-2024 indicator so its merit-order coefficient can shift as the fleet grew, and hour-of-day, month and day-of-week fixed effects, with Newey–West standard errors at a weekly bandwidth. The year-by-renewable interactions are load-bearing because the build-out changed the *slope* of the price–renewable relationship, not merely its level; that the merit-order curvature is fully absorbed I confirm by replacing the linear renewable terms with quadratic and then cubic ones, where the quadratic does the work and the cubic adds nothing. Because any single regression can be questioned on functional form, each level result is shadowed by a Bayesian structural time-series counterfactual (Brodersen et al., 2015; Märkle-Hußet al., 2020): a state-space trend, a weekly seasonal and the same renewable regressors, fit under a spike-and-slab prior on the pre-window, with the effect read as the gap between the realised series and its projected counterfactual over the post-window. A result is retained only when regression and counterfactual land in the same region, and only when a same-calendar placebo one year before the true cutover comes back null or opposite-signed — the guard against anything that simply recurs each spring.

The same level machinery delivers the supply-side reading of §6.2. Following Ito and Reguant (2016, §III.A), I rebuild each dominant firm's residual demand from the offer book — system demand net of rivals' aggregate supply — and read its slope b_f at the clearing price from a local quadratic fit within ±50 EUR/MWh of it. A descriptive margin index, the firm's cleared megawatts net of its own near-zero-price renewable output divided by that slope, then moves as the markup channel would without being a fitted first-order condition; I read it as suggestive supply-side evidence on the competitive margin, not as an estimated conduct parameter.

5.2 Bid-shape outcomes: a within-day difference-in-differences

The bid-shape outcomes call for a different design, because their threat is different. Each is built within a band re-centred on the contemporaneous clearing price, so price-*level* seasonality is already differenced out of the curve. What a before-after comparison cannot remove is a common change in bid *shape*: a shift in which technologies are marginal — the low-gas, renewable-heavy spring of 2025 that pushed gas off the merit order at the spot reform, the reinforced-operation regime at the forward reform — compresses or stretches the whole in-band ladder regardless of granularity. These shape shocks are large enough to dominate: in most technology cells the simple before-after in critical hours points the opposite way to the design below, because dispersion falls in critical hours but falls further in flat ones, so the change that granularity actually caused is the gap between them, not either fall on its own. The design that isolates that gap is a difference-in-differences across hours of the same day.

I summarise each unit's in-band ladder on each date and period by the quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_{\mathrm{tr}})$ and compare its post-minus-pre change in critical hours against the same change in flat hours,

$$Y_{u,d,p[,s]} = \alpha_u + \beta\,\mathbf{1}\{d \geq T_r\} + \xi\,\mathbf{1}\{h \in \mathcal{C}\} + \theta\, D_{d,h} + \varepsilon, \qquad \text{standard errors clustered by date,}$$

with $D_{d,h}$ the post-by-critical interaction and θ the object of interest. That contrast is not a nuisance to net out but the quantity itself: differentiated bidding pays off only where residual demand is steep enough to reward it, which is precisely the critical hours, while in flat hours the curve is shallow and the extra quarter-prices are wasted optionality. The critical-minus-flat differential therefore *is* the granularity-enabled discrimination of Corollary 2(iii), and the flat partition supplies a within-day control that needs no time-series extrapolation.

Identification rests on parallel critical-and-flat trends absent the reform. Price dispersion σ_p and tranche concentration $\mathrm{HHI}_{\mathrm{tr}}$ satisfy it cleanly for the price-setting technologies and carry the headline; the fitted slope $\hat{\beta}$ and curvature $\hat{\gamma}$ drift in the pre-period, mechanically sensitive to tranche count and price-grid density, both of which moved at the 2024 and 2025 reforms, so I read those two as supporting rather than decisive. Each reform is a clean two-by-two with no staggered-treatment complications (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3), and the parallel paths that license the comparison are shown before the coefficients in §6.4.

6 Results

> Finer granularity made the wholesale market more competitive without moving aggregate quantities, and it did so *asymmetrically across the two reforms*, exactly as §3.6 predicts. New contractibility arrives only at the spot reform (ID15): prices fall on the granular leg and transmit cross-market, the within-hour wedge starts to disperse, the imbalance penalty drops, and the aggregator's footprint shifts toward the granular auction. The forward reform (DA15) *relocates* the already-contractible objects rather than creating new ones: the bid curves move — the forward ladder disperses and the spot ladder concentrates — while prices and the balancing layer do not. Underneath both, the residual demand each firm faces becomes more price-elastic — suggestive, on the supply side, of a compressing markup.

I read the results through the model in its own causal order. The price level comes first (§6.1), since market re-clearing is the headline outcome; then the margin channel (§6.2), where the measured residual-demand slope b_f selects which branch of the comparative static the data sit on; then the within-hour wedge (§6.3), the sharp test of block parity and its limits under bounded per-product arbitrage; then the bid curves (§6.4), where exposure at ID15, relocation at DA15, and level-neutrality appear together; and finally the balancing layer — the imbalance penalty (§6.5) and the ancillary-cost evidence (§6.6), where the reform's effect is swamped by the post-blackout reinforced-operation regime the model does not speak to. Each subsection closes with the model predictions in that domain that are within empirical reach but that I have not yet tested; where a result does not fit the model, I say so and offer a reading.

6.1 Prices: a robust drop at ID15, a null at DA15, and matching cross-market effects

The spot reform cut the granular-leg clearing price, the cut carried over to the day-ahead market that ID15 never touched, and the forward reform moved neither price. That is the whole level story, and Table 3 shows it holding across a deliberately demanding ladder of specifications.

What makes the ID15 drop credible is not its size in any single regression but its refusal to shrink as the renewable controls grow more flexible. Spain's solar fleet expanded sharply between the comparison years, so part of any raw price fall is simply more zero-marginal-cost supply pushing down the merit order; the year-by-renewable interactions, and then the quadratic and cubic renewable terms, exist to absorb exactly that confound. This is also why the price regressions take a long pre-window, from 2022 onward: the renewable coefficients have to be identified across several years of the build-out before they can be applied to the post-reform months — a short window could not tell granularity from solar. (The within-hour and slope outcomes later use short, regime-constant windows instead, for the opposite reason: there the offer-book regime, not the multi-year renewable trend, is the thing that must be held fixed.) As they do, the spot estimate settles from the mid-forties into the low-twenties of EUR/MWh — it narrows, but it never travels toward zero, and the independent BSTS counterfactual lands in the same region. The granularity-attributable cut is whatever survives once the enlarged solar fleet is controlled for, and a cut of around twenty EUR/MWh is what survives. It helps to anchor that magnitude, because the raw before-after fall is far larger — from the high nineties in the weeks before the switch to the mid-twenties after — and almost all of that gap is the winter-to-spring renewable swing the controls remove, not the reform. The estimate that matters is the controlled one, measured against the pre-reform price level it is identified relative to: day-ahead and intraday both averaged near a hundred EUR/MWh in the months before ID15, so a cut of around twenty is about a fifth of the prevailing price. That is a moderate re-clearing, not a dramatic one — the reform shaves a fifth off the granular leg, it does not halve the market. The day-ahead, left at hourly resolution by ID15, moves almost one-for-one with the spot: this is the model's cross-market transmission, the forward bidder pricing in a cheaper granular continuation through backward induction rather than anything changing in its own product.

DA15 is the mirror image. Every specification leaves both its own forward market and the cross-market spot wandering insignificantly around zero, the sign of the point estimate flipping with the controls precisely because there is no effect to pin down. The asymmetry — **a clean level cut at the spot reform, nothing at the forward reform** — is the first and sharpest fingerprint of the model's central claim that new contractibility, and with it market re-clearing, arrives only when the spot becomes granular. The supply side tells the same story through the slope of the residual demand each firm faces, which the next subsection takes up; window-length, hour-class, per-session, same-calendar-placebo, and cross-border-flow-control robustness are in Appendix B and B.6.

	ID15 (spot reform)		**DA15** (forward reform)	
Specification	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS, daily averages				
base	-31.7^{***}	-32.8^{***}	-8.3	-8.4
year × renewable (per year)	-45.8^{***}	-46.7^{***}	-8.6	-10.0
year × renewable (pooled)	$\mathbf{-38.7}^{***}$	$\mathbf{-38.7}^{***}$	-4.6	-6.0
quadratic renewable × year	-22.7^{***}	-23.2^{***}	-2.8	-3.4
OLS, hourly				
base	-33.6^{***}	-33.6^{***}	-10.3^{**}	-10.2^{**}
year × renewable (per year)	-40.4^{***}	-39.0^{***}	-3.0	-4.0
year × renewable (pooled)	-30.1^{***}	-28.7^{***}	2.9	1.7
quadratic renewable × year	-24.3^{***}	-23.6^{***}	-2.3	-2.5
BSTS counterfactual				
daily, year × renewable	-34.3^{*}	-33.9^{*}	7.8	1.0
daily, quadratic renewable	-32.8	-32.2	8.5	1.6

Table 3: Clearing-price effect by specification and reform-side leg (EUR/MWh). Common pre-window from 2022-01-01; post-windows 2025-03-19 to 2025-04-27 (ID15, 40 days, pre-blackout) and the 90 days from 2025-10-01 (DA15). Base controls (all rows): wind, solar, gas, month FE, day-of-week FE, with hour-of-day FE added in the hourly rows; the renewable terms named are added on top. The quadratic row adds wind^2 and solar^2 with year-by-renewable interactions on the linear and quadratic terms; the cubic spec gives -23.5 / -22.9 ($p < 0.001$). Newey–West HAC; BSTS adds a state-space level and 7-day seasonality, its * marking the posterior tail. Headline pooled-renewable ID15 spec in **bold**. The only significant DA15 cell is the hourly base, and it vanishes the moment renewable controls enter — a confound, not an effect.

> ***Within reach, not yet tested.*** In the model the level moves through one channel — the markup, as residual demand becomes more or less price-elastic — and never through ladder

geometry, which is level-neutral (Remark 1). The evidence is consistent: at ID15 the spot level falls while the slope of residual demand rises (§6.2). But I do not split the realised cut into how much is margin compression and how much is a contemporaneous shift in marginal cost and the merit order; a descriptive level decomposition against fuel and carbon prices (ESIOS gas and CO_2) is within reach and would sharpen the attribution. I read the slope as suggestive evidence for the mechanism, not as an estimate of a structural pass-through.

6.2 The margin channel: residual demand becomes more price-elastic and margins compress

The residual demand each dominant firm faces became **more price-elastic at every reform-side leg** (Table 4). I recover the slope of that residual demand at the clearing price for each firm in the spirit of Ito and Reguant (2016, §III.A), and I read it as suggestive evidence on the competitive margin — not as an estimate of a structural conduct parameter, which this design does not attempt. The reading the model invites is direct: a steeper residual demand leaves a firm less room to hold price above cost, so the slope increase the supply side shows should map into the level cuts the previous subsection found, on the same legs. The rise is corroborated by two estimators that disagree about almost everything else — hourly OLS and a daily BSTS counterfactual agree that the cross-market day-ahead slope roughly doubles at ID15, and agree in sign on the DA15 day-ahead. The thickening is, moreover, a *local* phenomenon: measured over the whole residual-demand curve rather than in a tight band around the clearing price it shrinks several-fold (the last slope column of Table 4), because the reform makes residual demand more elastic right where the market clears — exactly the margin a strategic bidder prices against — and much less so far from it. The windows differ in kind from the price regressions, and deliberately so. The price level is read from a long horizon, because the renewable build-out that dominates its variation plays out over years; the slope is instead read off the offer book, which the post-blackout reinforced operation reshapes by moving dispatchables into the technical-restriction redispatch (the Fase I phase of §2), so I hold that regime fixed across each cutover — the ID15 windows sit entirely before the April 2025 blackout, the DA15 windows entirely after it (the row groups of Table 4) — rather than let a window straddle it. A long-window robustness on the slope, where the rise shrinks but stays positive, is in Appendix B.8.

A coarse margin index points the same way without pretending to more. Taking each firm's cleared megawatts net of its own renewable output — which clears at essentially zero price and carries no margin — and scaling by the slope gives a quantity-over-elasticity index that should move as the markup channel would; it is a descriptive indicator, not a fitted first-order condition. At the DA15 day-ahead it falls for every dominant firm, the margin-compression direction, exactly where the model places the strategic bidder. At ID15 its sign is mixed, but for a mundane reason rather than a model failure: the forty pre-blackout days have nuclear offline and gas called hard, so the composition of who is marginal is unstable. The one genuinely ambiguous cell is the DA15 cross-market spot, where the two estimators disagree in sign; the model is comfortable with either, because relocating within-hour pricing to the forward can leave the spot residual demand flatter or steeper depending on which screen component departs, and the negative BSTS branch coheres with the spot bid curve concentrating at DA15 (§6.4). The two reforms thus light up the model's two agents in turn — the aggregator, a price-taker hedging its forecast error, at the spot reform; the strategic bidder, a residual monopolist, at the forward reform.

Reform-side leg	Δb_f in-band (± 50) OLS	BSTS	Δb_f full curve	Margin index
Around ID15 (reforzada-absent, 2024-06-14 to 2025-04-27)				
IDA (granular spot)	16 to 25%	10 to 16%	−2 to 3%	mixed
DA (cross-market)	**109** to **141%**	**99** to **124%**	14 to 19%	mixed
Around DA15 (reforzada-present, 2025-04-28 to 2025-12-31)				
DA (granular forward)	19 to 42%	12 to 13%	9 to 13%	−**12** to −**27%**
IDA (cross-market)	13 to 20%	−11 to −8%	3 to 8%	—

Table 4: Residual-demand slope change with and without the bandwidth, and a descriptive margin index. Δb_f: percentage change (post vs pre, range across the four dominant firms) in the slope of the residual demand each faces, following Ito and Reguant (2016, §III.A). *In-band* fits the slope locally, within ±50 EUR/MWh of the clearing price, under controlled hourly OLS on $\log b_f$ and a daily BSTS counterfactual; *full curve* is the raw post/pre change in the slope fit over the entire residual-demand curve. The thickening is concentrated at the clearing price — several times larger in-band than over the full curve, where strategic bidders operate (the contrast survives like-for-like: the in-band *raw* change is 93 to 116% on ID15 DA against 14 to 19% full curve). Margin index: pre/post change in cleared MW net of own renewable clearance, divided by the slope; descriptive, not a fitted first-order condition. Reforzada-constant windows; per-firm detail in Appendix C.3.

Within reach, not yet tested. The slope is suggestive evidence, and I stop short of a structural conduct estimate by design; the natural next steps are descriptive, not structural. The margin index moves with the markup channel, but I do not benchmark it against a direct price-cost margin from heat-rate-implied marginal cost (ESIOS gas and CO_2), which would say whether the two agree in magnitude and not only in sign. And the slope can steepen for a reason the model holds fixed — renewables crowding the merit order — so the granularity-driven part of the change is not separated from the renewable-driven part. Both are within reach with data already in hand.

6.3 Within-hour wedge dispersion turns on at ID15 and does not collapse at DA15

The within-day dispersion of the day-ahead – intraday wedge is the model's sharpest sign test, and it **turns on exactly where the model says it should**. The dispersion was mechanically zero before ID15 — a single hourly day-ahead price cannot differ from itself within the hour — and at ID15 it appears, concentrated in the critical ramp hours and absent overnight (the SD columns of Table 5, sharpest in the morning ramp). This is block parity at work. The hourly forward can be arbitraged only against the *average* of the four spot quarter-prices, so arbitrage disciplines the average wedge far more than its within-hour spread, and that spread is largest precisely where the ramp is steepest. The one off-sign cell, midday, is the solar-marginal block I exclude from the partition: there the granular prices *converge* as midday solar flattens the within-hour profile, so a narrowing wedge is the same mechanism running the other way, not a counterexample. The comparison windows are chosen to fit the object: they start no earlier than the June 2024 European-IDA reform — before that the intraday ran six sessions rather than three, so the day-ahead – intraday wedge is a different object — and, as for the slope, they hold the reinforced-operation regime fixed across each cutover rather than spanning a long horizon. A long window is right for the price level, whose variation is dominated by the multi-year renewable build-out; it would only inject regime changes into a within-hour object like this one. The average wedge *level* moves too, but more modestly and through the separate forward-premium margin —

at ID15 it rises in the flat hours rather than the ramps, the mirror image of where the dispersion lives.

At DA15 the dispersion does not change in any hour class, and that null is itself a prediction borne out. With both auctions granular, full arbitrage would drive the within-hour wedge SD to zero; the model instead keeps it alive, because the position-takers who would close it hold a per-product capacity that shrinks as the number of products grows (§3.5). The dispersion that switched on at the spot reform stays on through the forward reform — bounded, price-impact-aware arbitrage sustains it rather than competing it away. The per-session wedge *levels* at ID15 are consistent with the same story: the earliest intraday session widens robustly across estimators, while an apparent narrowing of the latest session appears only in the daily specification and does not survive the hourly or BSTS estimators; the placebo cutover is null, and DA15 leaves the session wedges unchanged.

The non-collapse has a mechanism, and the continuous market lets me look at it directly rather than infer it. If the arbitrage that would close the wedge were unbounded it would scale with the number of delivery products; the model instead fixes each position-taker's capacity, so spreading it over more products thins it per product. The continuous (XBID) market shows precisely this. When it went quarter-hourly its daily product count quadrupled — from twenty-four to ninety-six — yet total traded volume did not rise to match but *fell*, from about 25 to 16 GWh a day, so liquidity *per product* collapsed to roughly a sixth of its former level. And per-product liquidity is what compresses the wedge: regressing the within-day wedge SD on per-product continuous-market liquidity, with the net cross-border flow that bounds feasible arbitrage held fixed, the liquidity coefficient is negative and significant ($p < 0.01$), while a post-DA15 indicator adds nothing once liquidity and cross-border flow are in. Bounded arbitrage that thins as products multiply is, then, not merely the model's assumed reason the wedge survives — it is visible in the one market built to do the arbitraging.

	ID15		**DA15**		
Hour class	Mean	SD	Mean	SD	**plac. SD**
Overall	1.0	**1.1****	-1.9^{*}	-0.1	-1.0^{*}
Critical (ramps)	1.1	**2.1*****	0.2	-0.5	-1.1^{*}
Morning ramp		**2.3*****		0.5	-0.5
Evening ramp		1.2^{*}		-1.4	-0.9^{*}
Midday	1.3	-1.6^{***}	-2.5	-0.7^{*}	-0.7^{*}
Flat	**4.7*****	-0.0	-2.0^{*}	-0.5	-0.9^{*}

Table 5: Mean and within-day SD effect on the day-ahead – intraday wedge, by hour class (EUR/MWh, BSTS). *Mean* is the average wedge-level effect, *SD* the within-day dispersion effect; the last column is the SD effect at the same-calendar 2024 placebo cutover (ID15). Windows hold the IDA session structure and the reinforced-operation regime fixed: the pre-window starts at the June 2024 European-IDA reform, post for ID15 is 2025-03-19 to 2025-04-27 (pre-blackout), post for DA15 is 2025-10-01 to 2025-12-31. Critical pools the morning (5–8) and evening (16–22) ramps; midday (11–14) is the excluded solar-marginal block; flat is overnight (1–3). Level series are published at the pooled-class level, so the ramp split is shown for the SD only. The ID15 SD jump concentrates in the morning ramp and is *opposite*-signed under the placebo, so it is not a recurring seasonal pattern; the level shifts more modestly and through a separate margin.

Within reach, not yet tested. These results establish the *directions*; two of the model's sharper claims here are still open. It attaches specific *magnitudes* to the wedge SD at each state — $\sigma_\psi/2b_2$ at the spot reform, $\lambda_1\sigma_\varsigma/2$ at the forward reform — which I do not take to the data. And its cleanest forward-leg prediction is descriptive and untested: across-quarter dispersion of *forward* prices should be absent at ID15 and switch on only at DA15, which deserves its own pre/post comparison rather than the market-pooled, post-only gap of §4.4. The dropped quantity-side prediction sits here too: the aggregator should shift volume to the granular leg in proportion to its forecast-error range, largest for wind and solar; the intraday in-band sell-share does rise for the forecast-error technologies at ID15 with aggregate cleared quantities unchanged, but a clean migration test is blocked by the program-cascade endogeneity of §4.3.

6.4 Bid curves disperse and gain tranches in critical hours on each reformed market

Both reformed markets **reshape the ladder of the dispatchable price-setter**, each in the direction the model assigns to it, while the price-takers move the opposite way. I read the reshaping off the two robust anchors — the price dispersion σ_p and the tranche-Herfindahl $\mathrm{HHI}_{\mathrm{tr}}$ — and leave the slope and curvature to the appendix, where their pre-trends drift (Table 6). CCGT, the marginal-cost unit that tends to set price, widens its day-ahead in-band ladder and splits it into more, more finely-priced steps at *both* reforms. That is one unit responding to two different mechanisms: at the spot reform the cross-market anticipation channel disperses the forward ladder ahead of the granular continuation (Corollary 3), and at the forward reform the forward screen itself widens as the ramp becomes contractible there (Corollary 2). The relocation runs in reverse wherever a market loses its within-hour pricing job: the same statistics shrink, the ladder concentrating into fewer, fatter tranches — visible in the cross-market CCGT response and in pump-storage, which coarsens its day-ahead ladder once it can discriminate in the now-granular intraday instead. The strategic renewables compress on the reformed leg, the price-takers again moving against the dispatchables. The morning-ramp-only double difference reproduces every headline sign, and the critical-versus-flat pre-trends — the identifying restriction — are shown for CCGT in Figure 7, with the full set of technologies in Appendix B.

Because the design identifies off parallel critical-versus-flat trends, those paths deserve to be seen before the coefficients, and they carry three cautions the table cannot. First, the DA15 day-ahead cell is the cleanest: the pre-reform lines move together and the critical dispersion lifts away only after the cutover. Second, the sharp fall in both DA15 lines at the end of the window is a regime artefact, not a reform reversal — once the post-blackout reinforced operation moved the dispatchables' business into the Fase I technical-restriction redispatch (the post-day-ahead phase of §2), the day-ahead curve stopped being where CCGT competes and its dispersion fell accordingly. Third, the ID15 day-ahead cell is thin: its post-window is only the forty pre-blackout days, and that spring was a low-gas, renewable-heavy period that pushed CCGT off the merit order for long stretches, so the ID15 day-ahead CCGT result is suggestive rather than decisive. The intraday cells need one further caution of their own: before the reform CCGT's flat-hour dispersion *exceeds* its critical-hour dispersion in the intraday auctions — the reverse of the day-ahead ordering — so the intraday baseline starts from a different place; and the third intraday session covers only the afternoon, with no overnight flat hours at all, so it cannot enter a critical-versus-flat comparison. The intraday bid-shape evidence therefore rests on the first two sessions (Appendix D.5), and I read the pooled intraday columns of Table 6 as the weakest in the set.

That this tranche-count margin is institutionally real, and not a free parameter, shows in how often it binds. In the intraday auctions — where a simple offer may carry at most five price steps — CCGT units submit the full five on 48% of their offers, against under 4% for wind and essentially

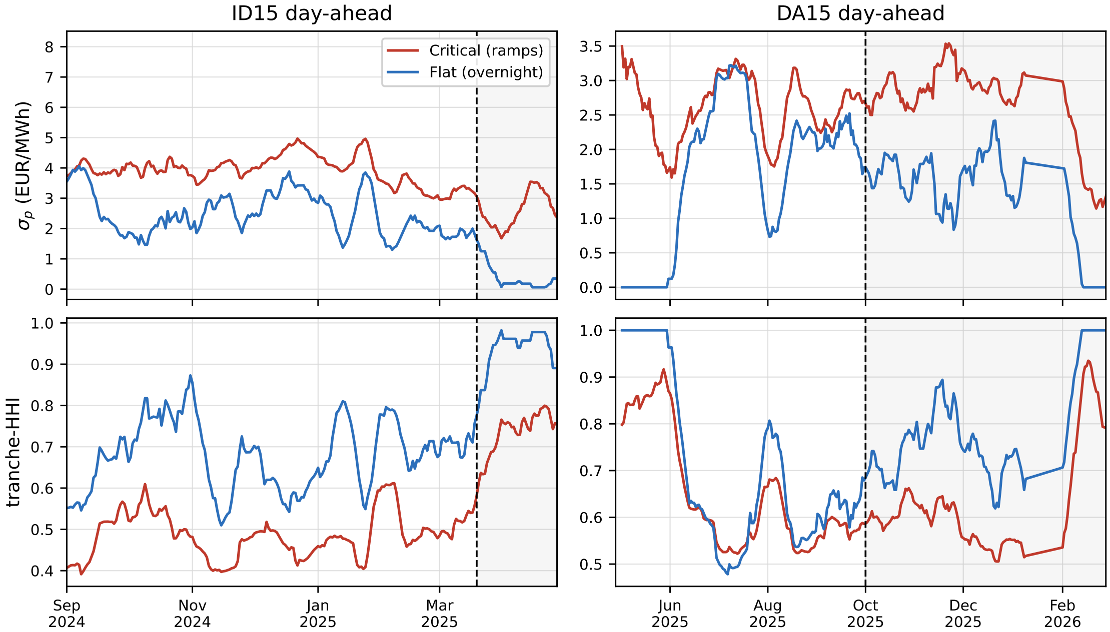


Figure 7: Critical-versus-flat pre-trends for CCGT on the two day-ahead cells (14-day rolling means; dashed line = cutover). Critical pools the morning and evening ramps; flat is overnight. At DA15 the critical-hour dispersion lifts above flat only after the cutover and the pre-reform paths run together; the collapse of both lines at the right edge of that window is the reinforced-operation regime moving CCGT into the Fase I technical-restriction redispatch (run before the intraday, §2), not the reform. The ID15 post-window is only the 40 pre-blackout days.

never for nuclear, and not one of the roughly 203 million sample offers exceeds the cap. In the day-ahead the looser ceiling of twenty-five steps is slack on the supply side, with CCGT ladders averaging five steps, so there the binding margin is the per-step cost of Proposition 2 rather than the cap itself. Either way it is the dispatchable price-setter, not the price-takers, that runs up against the finite-tranche friction the model places at its centre.

The cleanest single test in this section is on the bid-curve *level*. The model is emphatic that ladder geometry is level-neutral (Remark 1): a reform should reshape the curve without moving where it sits, and any level change has to arrive through the separate price-level channels instead. The MW-weighted mean in-band price bears this out. In the intraday cell at ID15 — where the exposure mechanism works purely on shape — the level does not move for CCGT, hydro or pump-storage even as their dispersion and tranche count clearly do: the shape changed and the level did not. And where the level *does* move, it moves in a price-level direction rather than a ladder direction — up at ID15 on the day-ahead, the continuation-value withholding of §3.4, and down everywhere at DA15, the margin compression of §3.5 — while the dispersion rises in those very same cells. Shape and level travel through different channels, with different signs, exactly as the model keeps them apart.

Outcome (DiD θ)	Tech	ID15		DA15	
		DA	IDA	DA	IDA
σ_p (EUR/MWh)	CCGT	$\mathbf{4.45}^{***}$	0.24	$\mathbf{1.05}^{***}$	$\mathbf{-1.01}^{***}$
	Hydro	-0.35^{**}	0.24	0.85^{***}	0.44^{***}
	Pump-storage	$\mathbf{-1.48}^{***}$	0.03	0.13	-0.02
$\mathrm{HHI}_{\mathrm{tr}}$	CCGT	$\mathbf{-0.18}^{***}$	-0.04	$\mathbf{-0.17}^{***}$	-0.00
	Hydro	0.010^{*}	-0.034^{***}	-0.037^{***}	-0.01
	Pump-storage	0.00	-0.02	0.00	0.01

Table 6: Bid-curve-level DiD on headline bid-shape outcomes. Critical-flat DiD on price dispersion σ_p and bid-stack concentration $\mathrm{HHI}_{\mathrm{tr}}$. Tight pre/post windows (reforzada held constant): ID15 pre 2024-12-19 to 2025-03-18, post 2025-03-19 to 2025-04-27 (40 days, pre-blackout); DA15 pre 2025-07-01 to 2025-09-30, post 2025-10-01 to 2025-12-31. Bandwidth $\delta = p_{90} - p_{50}$ window-specific. SEs clustered by date. A negative HHI DiD means the bid stack *deconcentrates* in critical hours relative to flat. The slope $\hat{\beta}$ and curvature $\hat{\gamma}$ — mechanically sensitive to tranche-count and grid density — are supporting; their full decomposition, additional technologies (Cogen, Hybrid, Wind, Biomass), morning-ramp-only robustness, and parallel-trends figures are in Appendix B.

Within reach, not yet tested. The bid-shape evidence confirms the *directions* of the ladder comparative statics but not the two quantitative magnitudes the model also implies. The tranche budget predicts a ladder depth $2n^* + 1 = (\Delta^2/3b\kappa)^{1/3}$ (Proposition 2) — a cube-root in the screen amplification — so the number of in-band tranches should scale roughly as $\sigma_{\text{within}}^{2/3}$ across hours; I show tranche counts rise in critical hours (the sign) but never check this scaling, a purely descriptive exercise. And the ID15 cross-market anticipation predicts the forward ladder steepens more the more the granular spot multiplies the products hanging off it (Corollary 3); I treat the slope as supporting because of its pre-trend drift and so do not pursue that magnitude. Level-neutrality, finally, is verified on the ID15 intraday cell; the sharp claim is that *no* shape change moves the level in *any* cell, and a full cell-by-cell level comparison across reforms, markets and technologies is in reach.

6.5 Imbalance: the price of a deviation falls at ID15, the volume does not

The balancing-layer evidence comes from REE's settlement, published through ESIOS, and it earns a word of caution before any number. The series are the quarter-hourly imbalance settlement prices — the per-megawatt-hour terms a balance-responsible party faces on an up- and on a down-deviation of its residual imbalance (metered delivery minus final schedule) — together with a net penalty per megawatt-hour and the aggregate imbalance volume, all averaged to the day. The settlement is ex post and price-harmonised: a single price per period by default, splitting into separate up- and down-deviation prices only when both balancing directions are activated heavily, and it is a distinct object from the deviation *procurement* market REE runs before delivery. Crucially, the price is not a black box — regulation builds it from the marginal prices of the secondary (aFRR), tertiary (mFRR), and replacement-reserve energy REE activated to cover the residual, so the cost of a deviation is the cost of the reserves it forced the system to call (Appendix C.4 gives the rule and its sources). I lead with the two directional prices because they read that cost most cleanly, and treat the net penalty as a one-number summary. To avoid resting on a single estimator I report both a daily OLS and the BSTS counterfactual (Table 7), and they agree — which matters, because a result visible under only one method would be the kind of thing worth distrusting.

The split between price and volume is exactly the model's. **The cost of a deviation falls at**

ID15 — by roughly a third on the net penalty and by some forty to fifty EUR/MWh on each directional price, significant under both estimators — while the megawatt-hours of imbalance do not move. Forecast errors are physical (wind, solar and demand stochasticity), so granularity cannot shrink the residual imbalance, only let it be covered with cheaper reserves scheduled earlier: the aggregator's hedging gain $G(\Delta_\eta) = \gamma Q \Delta_\eta/2$, a granular spot revealing positions before the forecast update η_t resolves, so REE pre-schedules cheaper reserves instead of dipping into expensive ones at the margin. The same residual imbalance is cleared with a cheaper reserve stack. And the timing is the model's signature: the saving arrives entirely at the spot reform, where the balancing residual collapses to $\Delta_t = \varepsilon_t$, with nothing left for the forward reform — the null DA15 column under both estimators. One caveat on the spec: the ISP15 settlement series only begins in December 2024, so the imbalance pre-window is a single short winter; I therefore drop month fixed effects from the OLS, which on so short a window would be collinear with the post period and would spuriously soak up the treatment.

		ID15		DA15	
Outcome	**Pre mean**	OLS	BSTS	OLS	BSTS
Settlement prices (EUR/MWh)					
Penalty per MWh of |dev|	3.45	**−1.3*****	**−1.1*****	−0.2	−0.1
Up-deviation price	69.4	**−48.5*****	**−44.6*****	−4.2	10.3
Down-deviation price	100.7	**−51.3*****	**−49.6*****	4.2	5.1
Volume (MWh/day)					
Aggregate |imbalance|	1,676	−167	−2	−287	−67

Table 7: Imbalance settlement prices and volume: OLS and BSTS agree. Outcomes are ESIOS quarter-hourly settlement series, daily means (Appendix C.4). Windows: ID15 pre 2024-12-01 (ISP15 settlement start) to 2025-03-18, post 2025-03-19 to 2025-04-27; DA15 pre 2025-04-28 to 2025-09-30, post 2025-10-01 to 2025-12-31. OLS: post + wind + solar + gas + day-of-week, Newey–West HAC lag 7, *no* month FE (the short ISP15 pre-window makes them collinear with the post); BSTS: state-space level, 7-day seasonality, same covariates. The penalty per MWh is the net settlement price per MWh of net imbalance; the up- and down-deviation prices are the gross prices to be short or long. Both estimators cut the ID15 prices, leave DA15 null, and leave volume unchanged. Stars: OLS HAC p; BSTS posterior tail.

Within reach, not yet tested. The hedging gain $G(\Delta_\eta) = \gamma Q \Delta_\eta/2$ is *increasing in the forecast-error range*, so the penalty saving should be concentrated in — and scale with — the technologies whose Δ_η is largest, namely wind and solar. I measure only the system-aggregate −33%; the cross-sectional prediction that the saving grades with per-technology forecast-error magnitude is in reach with the per-balance-party imbalance settlement detail (ESIOS *liquicierre*). One result also under-fits the model and is worth flagging plainly: the model sends the priceable imbalance cost to zero at the spot reform, yet the penalty falls only a third rather than to its floor. The gap is consistent with the irreducible production error ε_t (which no auction prices) plus incomplete hedging by smaller parties, but I do not decompose the residual penalty into these, and doing so would sharpen the claim.

6.6 Ancillary-services costs are dominated by the reinforced-operation regime, not granularity

REE's adjustment-cost lines balloon post-blackout, almost entirely on Fase I up-redispatch and almost entirely on CCGT. This is **a reforzada effect, mechanistically distinct from MTU15**.

Pre-blackout (blue zone in Figure 8), real-time channels (TR-up, aFRR-up) shrink slightly across ISP15-win and MTU15-IDA, but our identification is too tight to pin a granularity-attributable component on the EUR cost aggregates.

The cost composition makes the displacement concrete: post-blackout, the system operator pays CCGT to run (expensive) through the pre-intraday Fase I redispatch while curtailing solar (cheap), and the Fase I up-redispatch line alone balloons by roughly EUR 5 M/day, almost entirely CCGT voltage support. Whether this is granularity helping or merely procurement timing shifting cannot be settled on these aggregates. The real-time channels (technical restrictions and aFRR) shrink at the same time the pre-real-time Fase I rises, so the efficiency reading — a granular auction absorbing intra-hour forecast errors before they reach real time — and the pure displacement reading — the same procurement moved one stage upstream — are observationally equivalent in EUR, and I do not separate them. The independent estimates of Brown and Reguant (2026) and Daví-Arderius and Graf (2026) corroborate the size of the reinforced-operation cost rise.

This is also where the model is deliberately blind, and it is worth saying so. The model prices energy and the imbalance it leaves behind; it abstracts entirely from the reactive-power and voltage-support margin on which the largest cost changes of the whole sample actually land. The reinforced-operation regime is a grid-security response, not a granularity response, and the two only happen to overlap on the calendar. The reform's footprint on the ancillary layer, if any, is an order of magnitude smaller than the regime shift sitting on top of it — which is exactly why I keep this evidence complementary and make no granularity claim on the EUR aggregates.

Read together, the results trace the asymmetry of activation that §3.6 predicts. Everything that requires the within-hour state to become *newly* contractible happens at the spot reform and only there: the granular-leg price falls and transmits cross-market (§6.1), the residual demand each firm faces steepens and the markup compresses (§6.2), the within-hour wedge dispersion switches on in the ramp hours (§6.3), and the imbalance penalty drops as the balancing residual collapses to its irreducible core (§6.5). The forward reform adds no new contractibility, and the data show it adding none: prices and the balancing layer do not move, while the bid curves *relocate* — the forward ladder disperses and gains tranches exactly as the spot's did at the first reform, and the spot ladder concentrates back into fewer, more closely-priced steps (§6.4). The within-hour wedge dispersion that switched on at ID15 does not collapse at DA15, the fingerprint of bounded per-product arbitrage. Underneath all of it, the level moves only through the markup channel and never through ladder geometry, with the level-neutrality of the intraday bid curve at ID15 the cleanest single confirmation. The one place the model stays silent — the voltage-support layer where the reinforced-operation regime dominates the cost ledger — is also the place the largest numbers live, and I leave it labelled as such rather than forcing it into the granularity story.

7 Conclusion

[To be written. Concepts to cover:]

- **Aggregate effects of DA15 are small; ID15 delivered the headline gains.** Significant changes in bidding behaviour at both reforms, but bid-shape signals are mixed across techs. Evidence of rising solar cannibalisation — already a serious profitability issue.
- **Open question:** did market microstructure affect grid stability? Three facts in the open: (i) REE's post-blackout response is technology-specific (CCGT doubles in restriction dispatch); (ii) realised renewable supply was similar in spring 2024 vs 2025, yet only 2025 blacked out; (iii) nuclear's auction-plus-bilateral output dropped sharply in the 40 days between ID15 and the

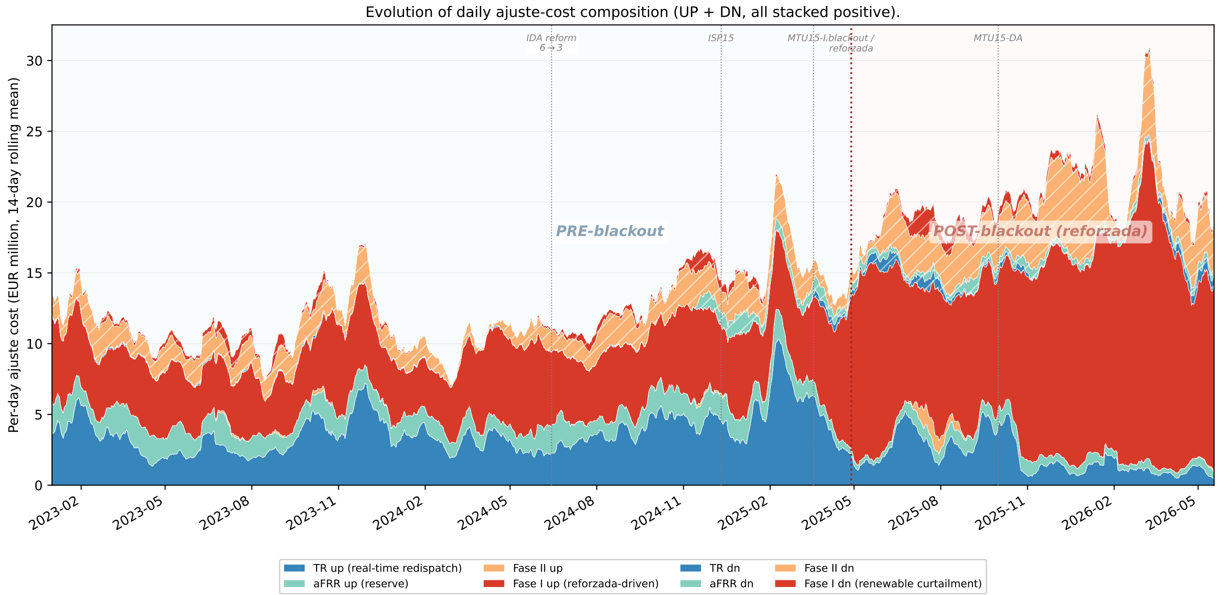


Figure 8: Daily adjustment-cost composition, 2023–2026 (EUR M / day, 14-day rolling mean). Each channel is the daily sum of the pay-as-bid system-cost series REE publishes via ESIOS: *TR up/dn* is real-time technical restrictions under PO 3.7 (cost of energy REE buys/sells to enforce grid security between gate closure and delivery); *aFRR up/dn* is the automatic Frequency Restoration Reserve energy cost; *Fase I up/dn* is the pre-IDA technical-restrictions process applied to PDBF (pay-as-bid of a separate restriction offer per PO 3.2 and PO 14.4 ap. 20.1, distinct from the day-ahead market offer); *Fase II up/dn* is the post-Fase-I rebalance that re-absorbs the up-redispatch into the schedule. All series are EUR sums collected at native 15-min granularity (raw daily totals smoothed only by the 14-day moving average), with down-direction channels taken as absolute values so the chart reads as gross costs REE pays providers in both directions. Up-direction stacked solid (TR-up, aFRR-up, Fase II up, Fase I up); down-direction stacked hatched. Blue background: pre-blackout (granularity reforms shrink the up-direction TR and aFRR channels through the ISP15-win and MTU15-IDA cutovers, visible against the stable 2023 baseline). Red background: post-blackout reforzada (Fase I up-redispatch balloons by EUR ∼5 M/day, almost entirely CCGT voltage support; this is the level shift between the 2023 baseline and 2025-Q4 totals).

blackout (Appendix D.7). The ENTSO-E Expert Panel does not list ID15 as a root cause, but CNMC flags quarter-hourly trading as contributing to abrupt voltage variations.

- **The biggest welfare losses live in the ancillary markets**, not in the wholesale auctions (Brown and Reguant, 2026; Daví-Arderius and Graf, 2026).
- **Urgent need to strengthen network security** to bring down retail prices — not finer time resolution.

CNMC report on the blackout flags quarter-hourly trading among contributors to within-hour voltage variability (CNMC, 2026, p. 44):

> *The implementation of the 15-minute trading unit in the Spanish intraday market was completed on 18 March 2025, one month before the incident. . . . The evolution to quarter-hourly trading and settlement has made the appearance of transitions between negative and positive prices increasingly frequent, leading to strong changes in production volumes in certain zones.*

References

Kay H. Brodersen, Fabian Gallusser, Jim Koehler, Nicolas Remy, and Steven L. Scott. Inferring causal impact using Bayesian structural time-series models. *Annals of Applied Statistics*, 9(1): 247–274, 2015.

David P. Brown and Mar Reguant. The price impacts of renewable power: A tale of two sources. University of Alberta Working Paper 2026-04, 2026.

CNMC. Resolución de 25 de abril de 2024, por la que se modifican las condiciones relativas al balance y los procedimientos de operación para la participación del sistema eléctrico peninsular español en las plataformas europeas de balance MARI y PICASSO. BOE-A-2024-11535, 2024a.

CNMC. Resolución de 23 de mayo de 2024, por la que se modifican procedimientos de operación para su adaptación a las subastas intradiarias europeas. BOE-A-2024-11959, 2024b.

CNMC. Resolución de 28 de febrero de 2025, por la que se publican las reglas de funcionamiento de los mercados diario e intradiario de electricidad para su adaptación a la negociación cuarto-horaria y a la nueva tipología de ofertas del mercado diario. BOE-A-2025-4908, 2025.

CNMC. Informe de recomendaciones y medidas propuestas a raíz del incidente eléctrico del 28 de abril de 2025. Expediente pro/cnmc/001/26, Comisión Nacional de los Mercados y la Competencia, March 2026.

Javier Colón. NeuroDATE II: El mercado cuartohorario y el modo re-reforzado. Industry webinar, NeuroEnergía / ACENEL, https://www.youtube.com/watch?v=zk8Q6IfPHYw, December 2025.

Daniel Daví-Arderius and Christoph Graf. Iberian peninsula blackout: A trade-off between operational risk and consumer cost. Working paper, University of Barcelona and New York University, March 2026.

Clément de Chaisemartin and Xavier D'Haultfœ uille. *Causal Inference with Differences-in-Differences: Credible Answers to Hard Questions*. Princeton University Press, 2026. Forthcoming.

European Commission. Commission regulation (EU) 2017/2195 of 23 november 2017 establishing a guideline on electricity balancing. Official Journal of the European Union, L 312, 2017.

European Union. Regulation (EU) 2019/943 of the european parliament and of the council of 5 june 2019 on the internal market for electricity (recast). Official Journal of the European Union, L 158, 2019.

James J. Heckman, Hidehiko Ichimura, and Petra E. Todd. Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *The Review of Economic Studies*, 64(4):605–654, 1997.

Koichiro Ito and Mar Reguant. Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7):1921–57, 2016.

Paul D. Klemperer and Margaret A. Meyer. Supply function equilibria in oligopoly under uncertainty. *Econometrica*, 57(6):1243–1277, 1989.

Joscha Märkle-Huß, Stefan Feuerriegel, and Dirk Neumann. Contract durations in the electricity market: Causal impact of 15-min trading on the EPEX. *Energy Economics*, 88:104760, 2020.

NEMO Committee. EUPHEMIA public description: Single price coupling algorithm. https://www.nemo-committee.eu/assets/files/euphemia-public-description.pdf, 2025.

OMIE. Modelo de ficheros para la distribución pública de información del mercado de electricidad. Versión 1.37, September 2025a. URL https://www.omeldiversificacion.es/sites/default/files/2025-09/formato_ficheros_inf_pub_137.pdf.

OMIE. Mercados intradiario y continuo: descripción del funcionamiento, 2025b. URL https://www.omie.es/sites/default/files/inline-files/mercados_intradiario_y_continuo.pdf.

Red Eléctrica de España. Guía para ser proveedor de servicios de ajuste del sistema eléctrico. Version 5.0, December 2024.

Redeia. Shareholders structure. https://www.redeia.com/en/shareholders-and-investors/share/shareholders-structure, 2025. Accessed June 2026.

A Theoretical derivations

This appendix proves the results of §3 in the order the body uses them: the setup and the arrival of information, the aggregator's hedging gain, the strategic bidder's optimal ladder, the three equilibrium states $(1,1) \to (1,Q) \to (Q,Q)$, the comparative statics behind Table 8, the balancing ledger, and the welfare account. I take the reader to have §3 in hand and reference its objects without redefining them.

A.1 Setup details and information arrival

This subsection records the timing of information arrival, the Ito and Reguant residual-demand specification used throughout, and how that specification specialises when one or both markets are hourly.

Timing. Stage 1 (forward gate): $\bar{A}$, the profile $\{\zeta_t\}$, and all distributions are common knowledge; each forward product clears against its realised clearing state v_{1t}, screened by the stage-1 ladder. Between the gates the within-hour update ψ_t realises: $\psi_t = \zeta_t + \mu_t$ when the forward is hourly, $\psi_t = \mu_t$ when it is per-quarter. Stage 2 (spot gate): each spot product clears against ψ_t and its own clearing state v_{2t}, screened by the stage-2 ladder. Stage 3 (balancing): the unpriceable noise ε_t realises and the settlement prices residual deviations at the penalty γ.

Residual demand. The strategic residual monopolist with constant marginal cost c serves what the operational aggregator does not cover, net of any arbitrage position s_t carried from forward to spot. Per delivery quarter t, with all quantities in MWh:

$$q_{1t} \;=\; \bar{A} + \zeta_t\, \mathbb{1}\{Q^F = Q\} + v_{1t} - b_{1t}\, p_{1t} - s_t - S^{op}_{1t}, \qquad q_{2t} \;=\; b_{2t}\left(p_{1(t)} - p_{2t}\right) + \psi_t + s_t - S^{op}_{2t}, \tag{7}$$

where $p_{1(t)}$ is the forward price covering quarter t (one number per clock-hour when $Q^F = 1$), S^{op}_{mt} is the aggregator's offered MWh in market m at quarter t (uniform S^{op}_m/Q when market m is hourly), and b_{mt} is the market's residual-demand slope, with per-quarter asymmetry allowed (when market m is hourly, $b_{mt} \equiv b_m$ and $p_{mt} \equiv p_m$).

Hourly residual demand by specialisation. When market m is hourly, its product clears against the within-hour *average* of the per-quarter objects in (7): the profile and every redistributive component drop out by $\sum_t \zeta_t = \sum_t \mu_{mt} = 0$, the screened state is the hour-level $\theta_m = Q^{-1} \sum_t v_{mt}$, and the spot's traded update is the hour-level $\bar{\psi} = Q^{-1} \sum_t \psi_t$. The within-hour variation that this averaging suppresses does not vanish from the model: it loads onto the balancing residual (§A.8) and shows up in the system imbalance cost C^I_{sys} in the welfare derivations (§A.9).

Tiling lemma: uniform amplification under aggregation. §3.1 states that the uniform family is closed under the aggregation $v_{mt} = \theta_m + \mu_{mt}$ — that one stochastic news process delivers both $\theta_m \sim U[-\Delta_m, \Delta_m]$ and $v_{mt} \sim U[-\rho\Delta_m, +\rho\Delta_m]$ at once. The tiling lemma is the construction behind that claim: it exhibits the μ_{mt} that closes the family and pins ρ to the support of the news.

Lemma 1 (Tiling)**.** *Let $\theta_m \sim U[-\Delta_m, \Delta_m]$ and let μ_{mt} be supported on $\{-\Delta_m, +\Delta_m\}$, taking each value with probability $1/2$ in each quarter (with $\sum_t \mu_{mt} = 0$ enforced across quarters). Then $v_{mt} = \theta_m + \mu_{mt} \sim U[-2\Delta_m, +2\Delta_m]$ exactly: $\rho = 2$. More generally, supporting μ_{mt} on the integer multiples $\{-(2j-1)\Delta_m, +(2j-1)\Delta_m\}$ for $j = 1, \ldots, J$ delivers $v_{mt} \sim U[-(2J)\Delta_m, +(2J)\Delta_m]$ exactly, i.e. $\rho = 2J$, an arbitrary even integer.*

Proof. For the $J = 1$ case: $\theta_m + \Delta_m \sim U[0, 2\Delta_m]$ when $\mu_{mt} = +\Delta_m$, and $\theta_m - \Delta_m \sim U[-2\Delta_m, 0]$ when $\mu_{mt} = -\Delta_m$. Each sub-distribution has density $1/(2\Delta_m)$ on its support; the equiprobable mixture has density $1/(4\Delta_m)$ on $[-2\Delta_m, 2\Delta_m]$, which is $U[-2\Delta_m, 2\Delta_m]$. The two sub-supports tile $[-2\Delta_m, 2\Delta_m]$ without overlap or gap. The $J > 1$ case is the same tiling argument with J pairs of sub-supports. □

The body carries a general ρ rather than the even-integer values delivered by the tiling; I treat ρ as a primitive of the hour, pinned by the support of the redistributive component the market's per-quarter object carries. The same construction with a smaller support delivers the news-only value $\rho^\mu \leq \rho$ used in §3.5.

Model abstractions: residual monopolist, supply functions, and the passive balancing stage. Three simplifications deserve flagging. First, the strategic bidder is a single residual monopolist per product, even though Spanish wholesale concentration is moderate (firm-level Herfindahl ∼1 100) and several firms compete on the bid stack. The residual monopolist is a stylised device that delivers clean comparative statics on conduct near the clearing price. Second, within a product the bid *is* a supply function, and with a single strategic bidder facing one-dimensional residual-demand uncertainty the optimal schedule is exact (§3.2); what I do not model is strategic interaction in supply functions across several large firms — the supply-function-equilibrium concept of Klemperer and Meyer (1989) — which gets intractable in a sequential-market setting. Third, the balancing stage in the model is a passive settlement of the residual imbalance, *not* an active auction with bidding. The aggregator internalises its own deviation through the penalty γ; the strategic bidder produces with certainty and so carries no individual imbalance; the unpriced within-hour residual collects in C^I_{sys} for the welfare account (Appendix A.8, A.9). In reality the system operator runs active balancing auctions; I abstract from them to keep the focus on how forward and spot granularity reshape the auction stages. What the model keeps from this layer is its prediction: finer auctions absorb the within-hour state upstream and shrink the balancing residual.

Renewable fringe and our relation to Ito and Reguant. Ito and Reguant's residual-demand interpretation already absorbs a renewable (or thermal) competitive fringe into the slope b_m: the strategic bidder serves $A - b_m p_m$, the implicit fringe serves the complement. Their setup delivers the strategic-bidder, arbitrage, and price-wedge mechanisms but is silent on two objects our data record directly: *why* the fringe's willingness to supply changes with granularity, and *what shape* the strategic bidder's offer takes. I make the fringe explicit (the operational aggregator with forecast errors and an imbalance penalty γ), so the hedging gain from per-period scheduling has the closed form $G(\Delta_\eta) = \gamma Q \Delta_\eta / 2$ (Appendix A.2); and I let the strategic bidder choose the step offer itself, so the geometry of the competing tier — depth, dispersion, gradient — comes out in closed form (§3.2).

A.2 Operational aggregator: closed-form gain from per-period scheduling

This subsection derives the aggregator's expected hedging gain $G(\Delta_\eta)$ from per-period (rather than hourly) spot scheduling, in closed form under uniform shocks. It then explains why the aggregator's *individual* forward-vs-spot allocation is not pinned down by an interior first-order condition, so that the migration onto the granular leg is an equilibrium statement about G being positive rather than an individual allocation result.

A.2.1 *Problem statement and primitives*

The aggregator allocates total hourly expected production $\bar{Y}_h$ between forward and spot. Forecast-error shock $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$, i.i.d. across t, with variance $\sigma_\eta^2 = \Delta_\eta^2/3$. We set the structural per-subperiod expected production constant ($\bar{y}_t = \bar{Y}_h/Q$) for the closed-form derivation; the structural variation $\sigma_a^2 = \text{Var}(\bar{y}_t - \bar{Y}_h/Q)$ adds linearly to G and is handled in the closing remark of §A.2.2. Imbalance penalty is $\gamma|\mathcal{I}_t|$ per period (linear in the magnitude of the gap).

What the reform changes in the aggregator's choice set. The reform changes what the aggregator can choose at Stage 2:

	Hourly spot ($Q^S = 1$)	**Granular spot** ($Q^S = Q$)
At Stage 2 the aggregator chooses	ONE number S_2 for the whole hour	Q numbers $\{S_{2,t}\}$ per period
Per-period schedule	uniform $(S_1 + S_2)/Q$	$S_1/Q + S^*_{2,t}$
Can respond to η_t per period?	no	yes, exactly
Per-period imbalance	η_t	0

A.2.2 *Expected penalty under each regime*

Granular spot ($Q^S = Q$)**.** At Stage 2 the aggregator observes η_t and sets the per-subperiod allocation $S_{2t}^{op*} = \bar{y}_t + \eta_t - S_1^{op}/Q$. The realised per-period imbalance is 0 (modulo post-Stage-2 noise ε_t, which is regime-invariant and we drop for clarity):

$$\mathbb{E}[\mathcal{C}_{\text{gran}} \mid \mathcal{H}_S] \;=\; 0.$$

Hourly spot ($Q^S = 1$)**.** The aggregator commits $S_1^{op} + S_2^{op} = \bar{Y}_h$ before η_t resolves; the per-subperiod schedule is therefore uniform at $\bar{Y}_h/Q$. With $\bar{y}_t = \bar{Y}_h/Q$ the per-period imbalance is exactly η_t. For $\eta_t \sim \text{Uniform}[-\Delta_\eta, \Delta_\eta]$:

$$\mathbb{E}|\eta_t| \;=\; \frac{\Delta_\eta}{2}, \qquad \mathbb{E}[\mathcal{C}_{\text{hour}} \mid \mathcal{H}_S] \;=\; \gamma Q \cdot \frac{\Delta_\eta}{2}.$$

Closed-form hedging gain. The aggregator's expected hourly saving from per-period (granular) spot scheduling relative to hourly spot scheduling is

$$\boxed{G(\Delta_\eta) \;=\; \mathbb{E}[\mathcal{C}_{\text{hour}} \mid \mathcal{H}_S] - \mathbb{E}[\mathcal{C}_{\text{gran}} \mid \mathcal{H}_S] \;=\; \frac{\gamma Q\, \Delta_\eta}{2}.} \tag{8}$$

Comparative statics are immediate:

$$\frac{\partial G}{\partial \Delta_\eta} \;=\; \frac{\gamma Q}{2} \;>\; 0, \qquad \frac{\partial G}{\partial \sigma_\eta^2} \;=\; \frac{\gamma Q}{2\sqrt{3}\,\sigma_\eta} \;>\; 0. \tag{9}$$

The gain is strictly positive and strictly increasing in the forecast-error variance. *Closing remark (structural variation).* When $\bar{y}_t \neq \bar{Y}_h/Q$, the hourly regime's uniform schedule additionally misses the deterministic profile, adding $\gamma \sum_t |\bar{y}_t - \bar{Y}_h/Q|$ to its bill; granular scheduling removes this term too, so it enters G additively and we omit it from the display.

A.2.3 *Why Δ^* is not pinned down by an individual FOC*

A natural follow-up is to ask whether G pins down the aggregator's allocation S_1^{op} via a first-order condition. It does not. The aggregator's expected profit under either regime is

$$\Pi(S_1^{op}) \;=\; (p_1 - p_2)\, S_1^{op} \;+\; p_2\, \bar{Y}_h \;-\; \mathbb{E}[\mathcal{C} \mid \mathcal{H}_F],$$

where the expected imbalance cost $\mathbb{E}[\mathcal{C} \mid \mathcal{H}_F]$ is independent of S_1^{op} once we impose the no-average-bias condition $S_1^{op} + S_2^{op} = \bar{Y}_h$. The FOC reduces to $p_1 - p_2 = 0$, a corner solution rather than an interior Δ^*. This is a structural feature of the price-taker setup: the imbalance cost depends on the resolution of information (whether the aggregator can react to η_t), not on how forward and spot allocations split.

The migration onto the granular leg is therefore an *equilibrium* statement: across aggregators, $G(\Delta_\eta) > 0$ generates a positive premium for participation on the granular spot; equilibrium prices adjust so that the wedge $p_1 - p_2$ partially incorporates this premium; the equilibrium counterpart is an aggregate shift of supply onto the granular leg. The claim is directional, through $G > 0$; the model does not deliver a level prediction for Δ^* at the individual aggregator's optimum.

A.3 Strategic bidder: the optimal n-tranche ladder along the monopoly line

This subsection proves the bid-ladder results of §3.2 in turn: the continuous benchmark (Lemma 2), the optimal n-tranche ladder (Proposition 1), the competing-tier geometry (Corollary 1), level neutrality (Remark 1), the optimal tranche depth under a complexity cost (Proposition 2), the granularity–variation interaction (Corollary 2), and the stage-1 pass-through (Corollary 3).

Fix one delivery product. Residual demand is $q = A - b\,p$ with $A = \bar{a} + v$ and $v \sim U[-\Delta, \Delta]$; the bidder has constant marginal cost c. The auction is uniform-price; the bidder submits a non-decreasing step function with corner points $\{(p_k, q_k)\}$ — cumulative quantity q_k offered once the price reaches p_k — and the clearing pair $(p^\star, q^\star)$ solves $S(p^\star) = A - b\,p^\star$ on the relevant segment.

Assumption 1. $q^m(\bar{a} - \Delta) \geq \Delta/(2(2n + 1))$: the monopoly quantity is interior in every state, by a margin of a quarter cell.

A.3.1 Continuous benchmark

Lemma 2 (Continuous benchmark)**.** *The supply schedule $S^*(p) = b\,(p - c)$ clears at the full-information monopoly point $p^m(A) = (A + bc)/(2b)$, $q^m(A) = (A - bc)/2$ in every state A, and is the unique non-decreasing schedule that does so on the contested range. Expected profit equals $\mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m]/(4b)$, the monopoly value.*

Proof. In state A, the full-information monopoly point maximises $(p - c)(A - b\,p)$ over p; the FOC gives $p^m(A) = (A + b\,c)/(2b)$ and $q^m(A) = (A - b\,c)/2$. Eliminating A, the monopoly line is $\{(q, p) : q = b(p - c)\}$, a line of slope b through $(p, q) = (c, 0)$. Define $S^\star(p) = b(p - c)$. For any state A,

$$S^\star(p^m(A)) \;=\; b\Big(\frac{A + b\,c}{2b} - c\Big) \;=\; \frac{A - b\,c}{2} \;=\; q^m(A),$$

so $(p^m(A), q^m(A))$ is on $S^\star$ in every state. *Uniqueness on the contested range*: a non-decreasing schedule that clears at the monopoly point in every state must contain the line, which is strictly increasing; a monotone schedule containing a strictly increasing curve coincides with it on its range. Expected profit at $S^\star$ is $\mathbb{E}[(p^m(A) - c)q^m(A) \mid \mathcal{H}_m] = \mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m]/(4b)$, the monopoly value. □

A.3.2 Optimal n-tranche ladder (proof of Proposition 1)

Clearing geometry under a step offer. Index the n competing tranches $1, \dots, n$ by ascending price $p_1 < \cdots < p_n$, let q_0 be the inframarginal base priced at the floor, and let Q_k be the cumulative quantity through tranche k ($Q_0 = q_0$). Three clearing regimes are possible per state A, alternating riser–horizontal–riser:

(V_0) *Base riser.* For $A \in [\bar{a} - \Delta,\ Q_0 + b\,p_1]$: quantity is Q_0 and the realised demand sets the price, $p = (A - Q_0)/b$ between the floor and p_1.

(H_k) *Horizontal segment at p_k.* For $A \in [Q_{k-1} + b\,p_k,\ Q_k + b\,p_k]$: clearing is $(p_k,\ A - b\,p_k)$ — the marginal tranche is partially filled.

(V_k) *Riser at Q_k.* For $A \in [Q_k + b\,p_k,\ Q_k + b\,p_{k+1}]$ (the top riser V_n closing at $\bar{a} + \Delta$): clearing is $\big((A - Q_k)/b,\ Q_k\big)$ — the realised demand sets the price.

This partitions the state space into $2n + 1$ full cells: one base riser, n horizontals of length $Q_k - Q_{k-1}$ (in A-units), and n risers of length $b(p_{k+1} - p_k)$. Profit is continuous at every switch: both adjacent regimes deliver the same price–quantity pair at the boundary state.

Each instrument serves an interval of states and loses a squared distance. Profit in state A is a quadratic hill around the monopoly point. On a horizontal segment at price p, completing the square gives $\pi(p;\,A) = \pi^m(A) - b\,(p - p^m(A))^2$; on a riser at quantity q (the base included), $\pi(q;\,A) = \pi^m(A) - (q - q^m(A))^2/b$. Each instrument is therefore exactly right in one state — the step at p in the state whose monopoly price is p, the riser at q in the state whose monopoly quantity is q; call that state the instrument's *target* $\hat{A}$. Because p^m and q^m are linear in A with slopes $1/(2b)$ and $1/2$, both losses reduce to the same expression,

$$\pi^m(A) - \pi \;=\; \frac{(A - \hat{A})^2}{4\,b},$$

so the bidder's problem is to place $2n + 1$ targets so that whichever state realises, it lands close to a target.

Equal intervals are optimal. An instrument serving an interval of length L does best with its target at the interval's midpoint, since the loss is symmetric in the distance; the mean squared distance of a uniform draw from the midpoint is $L^2/12$ (the variance of a uniform), and the interval is reached with probability $L/(2\Delta)$. Its contribution to the expected loss is therefore at least

$$\frac{L^2}{12} \cdot \frac{1}{4b} \cdot \frac{L}{2\Delta} \;=\; \frac{L^3}{96\,b\,\Delta},$$

and the total expected loss is at least $\sum_j L_j^3/(96\,b\,\Delta)$ over the $2n+1$ intervals with $\sum_j L_j = 2\Delta$. Since x^3 is convex, moving length from a longer interval to a shorter one always lowers the sum, so the minimum splits the state range into *equal* intervals of width $w \equiv 2\Delta/(2n+1)$.

The clearing rule places the boundaries halfway by itself. The bidder chooses only the targets; the interval boundaries then follow from the clearing geometry, and they land exactly halfway between adjacent targets. The switch from the step at p_k to the riser at Q_k occurs at the state $A^\star = Q_k + b\,p_k$; substituting $p_k = p^m(\hat{A}_{H_k})$ and $Q_k = q^m(\hat{A}_{V_k})$,

$$A^\star \;=\; \frac{\hat{A}_{H_k} + bc}{2} + \frac{\hat{A}_{V_k} - bc}{2} \;=\; \frac{\hat{A}_{H_k} + \hat{A}_{V_k}}{2},$$

and likewise at every other switch. Placing the targets at the equal-grid midpoints $m_j = \bar{a} - \Delta + (j - \tfrac{1}{2})\,w$, $j = 1, \ldots, 2n+1$, therefore reproduces exactly the equal-interval partition with every target at its interval's midpoint: the lower bound is attained.

Closed forms and feasibility. The alternation assigns V_0 to interval 1, H_k to interval $2k$, V_k to interval $2k{+}1$, so

$$q_0 = q^m\big(\bar{a} - \Delta + \tfrac{w}{2}\big), \qquad p_k = p^m(m_{2k}) = p^m\big(\bar{a} - \Delta + (2k - \tfrac{1}{2})\,w\big), \qquad Q_k = q^m(m_{2k+1}):$$

every competing tranche carries $Q_k - Q_{k-1} = (m_{2k+1} - m_{2k-1})/2 = w$ MW (the first measured from the base) and the price grid is even with step $p_{k+1} - p_k = (m_{2k+2} - m_{2k})/(2b) = w/b$. Feasibility under Assumption 1: the base $q_0 = q^m(\bar{a} - \Delta) + w/4 \geq w/4 > 0$, and the lowest realised price (in V_0 at $A = \bar{a} - \Delta$) is $p^m(\bar{a} - \Delta - w/2) \geq c$ iff $q^m(\bar{a} - \Delta) \geq w/4$ — the stated condition.

Expected profit. With equal intervals the mean squared distance to the target is $w^2/12$ everywhere, so the total expected loss is $w^2/(48\,b) = \Delta^2/(12\,b\,(2n+1)^2)$ and

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] \;=\; \frac{\mathbb{E}[(A - b\,c)^2 \mid \mathcal{H}_m]}{4\,b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2}. \qquad \square$$

A.3.3 Competing-tier geometry (proof of Corollary 1)

The competing tranches carry equal MW w, so each tranche's share of the tier is $w_k \equiv (Q_k - Q_{k-1})/\sum_j (Q_j - Q_{j-1}) = 1/n$, and the Herfindahl of the tier is

$$\sum_{k=1}^{n} w_k^2 \;=\; \sum_{k=1}^{n} \frac{1}{n^2} \;=\; \frac{1}{n}.$$

The tranche prices form an arithmetic progression $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$ centred on $p^m(\bar{a})$. Shares are equal, so MW-weighted and unweighted moments coincide. The mean tranche price is $\bar{p} = (1/n)\sum_k p_k = p^m(\bar{a})$ by symmetry, and the variance of tranche prices is

$$\sigma(\Delta;\, b)^2 \;=\; \frac{1}{n}\sum_{k=1}^{n}(p_k - \bar{p})^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \sum_{k=1}^{n}\left(k - \tfrac{n+1}{2}\right)^2 \;=\; \frac{1}{n}\left(\frac{w}{b}\right)^2 \cdot \frac{n(n^2-1)}{12} \;=\; \left(\frac{w}{b}\right)^2 \frac{n^2-1}{12}.$$

Substituting $w = 2\Delta/(2n+1)$,

$$\sigma(\Delta;\, b) \;=\; \frac{2\Delta}{(2n+1)\,b}\sqrt{\frac{n^2-1}{12}}.$$

As $n \to \infty$, $\sigma(\Delta;\, b) \to \Delta/(2b\sqrt{3}) = \sigma_v/(2b)$ since $\sigma_v = \mathrm{sd}_v(v) = \Delta/\sqrt{3}$ for $v \sim U[-\Delta, \Delta]$.

For the gradient, the cumulative quantity at tranche k is $Q_k = q_0 + k\,w$ and the price is $p_k = p^m(\bar{a}) + (k - (n+1)/2)\,(w/b)$. Both advance linearly in k, so the slope of tranche price on cumulative quantity is the ratio of the two increments:

$$\frac{p_{k+1} - p_k}{Q_{k+1} - Q_k} \;=\; \frac{w/b}{w} \;=\; \frac{1}{b}.$$

And since p_k is an exact affine function of Q_k, a quadratic fit adds nothing: the curvature coefficient is zero. $\square$

A.3.4 Level neutrality

Remark 1 (Level neutrality). $\mathbb{E}[\text{clearing price} \mid \mathcal{H}_m] = p^m(\bar{a})$ for every n: ladder geometry reshapes dispersion and surplus, but never the mean price.

Proof. Take expectation of the clearing price over states. Each cell contributes its midpoint-monopoly price weighted by cell length: horizontal cells contribute $p_k = p^m(m_{2k})$ over length w; riser cells contribute the average realised price over the riser, $\mathbb{E}[(A - Q_k)/b \mid \mathcal{H}_m, V_k] = (m_{2k+1} - q^m(m_{2k+1}))/b = p^m(m_{2k+1})$, using $A - b\,p^m(A) - q^m(A) \equiv 0$; the base riser contributes $p^m(m_1)$ by the same demand-sets-price computation. The resulting weighted average over all $2n+1$ cells (each of length w) is the simple average of the $2n+1$ midpoint-monopoly prices, which — because p^m is affine and the midpoints m_j are symmetric around $\bar{a}$ — equals $p^m(\bar{a})$. So $\mathbb{E}[\text{clearing price} \mid \mathcal{H}_m] = p^m(\bar{a})$, independent of n. $\square$

Distributional consequence. The step structure reshapes the variance and the per-state surplus split, but never the mean clearing price. Bid-shape changes (regrading the ladder) are therefore *level-neutral* on price; mean price changes come only from changes in residual demand (the slope b or the market supply S^{op}), which is the channel underpinning the markup identity (1) in the body.

A.3.5 Optimal tranche depth under a complexity cost

Proposition 2 (Tranche budget)**.** *With a linear cost $\kappa > 0$ per competing tranche (treating n as continuous; the integer optimum is one of the two adjacent integers), the optimal depth satisfies*

$$2n^* + 1 \;=\; \left(\frac{\Delta^2}{3\,b\,\kappa}\right)^{1/3}:$$

ladders deepen with the screened range, $n^* + \frac{1}{2} \;=\; \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$, and flatten with market thickness b and complexity cost κ.

Proof. With a complexity cost $\kappa > 0$ per competing tranche, the objective is

$$\mathbb{E}[\pi_n \mid \mathcal{H}_m] - \kappa\, n \;=\; \frac{\mathbb{E}[(A - bc)^2 \mid \mathcal{H}_m]}{4b} \;-\; \frac{\Delta^2}{12\,b\,(2n+1)^2} \;-\; \kappa\, n.$$

Treating n as continuous, the FOC in n is

$$\frac{d}{dn}\left[-\,\frac{\Delta^2}{12b\,(2n+1)^2}\right] \;=\; \kappa \quad\Longleftrightarrow\quad \frac{\Delta^2}{3b\,(2n+1)^3} \;=\; \kappa \quad\Longleftrightarrow\quad 2n^* + 1 \;=\; \left(\frac{\Delta^2}{3b\,\kappa}\right)^{1/3}.$$

The integer optimum is one of the two adjacent integers; either way $n^* + \frac{1}{2} = \frac{1}{2}\,\Delta^{2/3}\,(3\,b\,\kappa)^{-1/3}$ at the continuous optimum. Joint signature with Corollary 1: a wider screened range Δ raises the tier dispersion and lowers the Herfindahl $1/n^*$ — they move together. □

A.3.6 Granularity $\times$ within-hour variation (proof of Corollary 2)

Each part substitutes the granularity map $(\Delta, b) \to (\rho\Delta, b')$ into the closed forms already in hand — the dispersion and span of §A.3.3 and the depth of Proposition 2.

(i) Under the continuous benchmark the clearing price is $p^m(A) = (A + bc)/(2b)$, affine in A with slope $1/(2b)$; over $A \in [\bar{a} - \Delta, \bar{a} + \Delta]$ its range is $2\Delta/(2b) = \Delta/b$, and substituting $(\rho\Delta, b')$ gives $\rho\Delta/b'$. The span $\frac{2(n-1)\Delta}{(2n+1)b}$ and the dispersion $\sigma(\Delta;\, b) = \frac{2\Delta}{(2n+1)b}\sqrt{(n^2-1)/12}$ of §A.3.3 are linear in Δ at fixed n, so the same substitution multiplies both by $\rho\, b/b'$; the tier Herfindahl $1/n$ involves neither Δ nor b.

(ii) Substituting $(\rho\Delta, b')$ into Proposition 2: $2n^*+1 = \big((\rho\Delta)^2/(3b'\kappa)\big)^{1/3} = \rho^{2/3}\,\big(\Delta^2/(3b'\kappa)\big)^{1/3}$.

(iii) The change in an internally flat hour ($\rho = 1$) is $\sigma(\Delta; b') - \sigma(\Delta; b)$ — the pure slope shift, since the screen stays at Δ. Subtracting it from the change in an hour with amplification ρ, $\rho\,\sigma(\Delta; b') - \sigma(\Delta; b)$, leaves

$$(\rho - 1)\,\sigma(\Delta; b') \;=\; (\rho - 1)\,\frac{2\Delta}{(2n+1)\,b'}\,\sqrt{\frac{n^2-1}{12}},$$

zero iff $\rho = 1$ and strictly increasing in ρ. □

Assumption 2 (Stage-1 second-order condition)**.** *$B_2 \equiv \sum_t b_{2t} < 2b_1$: the total continuation slope of the spot products hanging off the forward product is strictly less than twice the forward slope.*

A.3.7 Continuation slope and the forward monopoly line (proof of Corollary 3)

Apply backward induction at the spot stage. With Q spot products $t = 1, \ldots, Q$ each of slope b_{2t}, the per-quarter strategic best response is $p^*_{2t}(p_1) = (p_1 + c)/2$ at the symmetric solution (omitting arbitrage and aggregator terms which are p_1-independent for the slope calculation). The per-quarter cleared quantity is $q^*_{2t}(p_1) = b_{2t}(p_1 - p^*_{2t}) = b_{2t}(p_1 - c)/2$. Spot profit at the best response is

$$\pi^*_{2t}(p_1) \;=\; (p^*_{2t} - c)\, q^*_{2t} \;=\; \frac{p_1 - c}{2} \cdot \frac{b_{2t}\,(p_1 - c)}{2} \;=\; \frac{b_{2t}\,(p_1 - c)^2}{4}.$$

Summing across spot products, the stage-1 total profit in state v at forward intercept $A_1 \equiv \bar{A} + v$ is

$$\Pi_1(p_1, v) \;=\; (p_1 - c)(A_1 - b_1\,p_1) \;+\; \frac{(p_1 - c)^2}{4} \sum_t b_{2t} \;=\; (p_1 - c)(A_1 - b_1\,p_1) \;+\; \frac{B_2\,(p_1 - c)^2}{4},$$

with $B_2 \equiv \sum_t b_{2t}$. The stage-1 FOC:

$$\frac{\partial \Pi_1}{\partial p_1} \;=\; (A_1 - b_1\,p_1) + (p_1 - c)(-b_1) + \frac{B_2(p_1 - c)}{2} \;=\; 0 \quad\Longleftrightarrow\quad A_1 - 2b_1p_1 + b_1c + \frac{B_2}{2}(p_1 - c) \;=\; 0.$$

Solving for p^*_1:

$$p^*_1(v) \;=\; \frac{A_1 + c(b_1 - B_2/2)}{2b_1 - B_2/2} \;=\; \frac{\bar{A} + v + c(b_1 - B_2/2)}{2b_1 - B_2/2}.$$

The stage-1 monopoly-line slope in v is therefore

$$\boxed{\frac{dp^*_1}{dv} \;=\; \frac{1}{2b_1 - B_2/2}}$$

exactly as claimed in the body (5). Assumption 2 ($B_2 < 2b_1$) ensures the denominator is positive, so the line is increasing in v and the stage-1 problem is concave at the interior optimum.

Price range. Over the screened states $v \in [-\Delta_1, \Delta_1]$ the stage-1 clearing-price range under the continuous benchmark is exactly

$$2\Delta_1 \cdot \frac{dp^*_1}{dv} \;=\; \frac{2\Delta_1}{2b_1 - B_2/2},$$

so the forward tier's price span widens one-for-one with the line's slope.

Ladder gradient. The cleared quantity at the monopoly point in state v is $q^*_1(v) = A_1 - b_1\,p^*_1(v)$. Substituting,

$$q^*_1(v) \;=\; \frac{(b_1 - B_2/2)(\bar{A} + v - b_1c)}{2b_1 - B_2/2}, \qquad \frac{dq^*_1}{dv} \;=\; \frac{b_1 - B_2/2}{2b_1 - B_2/2}.$$

Assumption 2 also makes $dq^*_1/dv > 0$: the line is increasing in *both* coordinates, hence implementable by a non-decreasing ladder. Building the step offer along this line as in §A.3.2, tranche prices and cumulative quantities both advance linearly with the state — at the rates dp^*_1/dv and dq^*_1/dv — so both are evenly spaced sequences and the ladder's gradient is the ratio of the two rates, independent of the grid width:

$$\frac{dp^*_1/dv}{dq^*_1/dv} \;=\; \frac{1/(2b_1 - B_2/2)}{(b_1 - B_2/2)/(2b_1 - B_2/2)} \;=\; \frac{1}{b_1 - B_2/2}.$$

Spot-reform jump. At the asymmetric state $(1, Q)$, the spot market becomes per-quarter, so B_2 jumps from a single hourly slope b_2 to the sum $\sum_t b_{2t}$ of Q per-quarter slopes. When the per-quarter slopes are of the same order as the hourly one, the continuation coefficient B_2 jumps from b_2 to roughly $Q \cdot b_2$ — a large factor. The denominators $2b_1 - B_2/2$ and $b_1 - B_2/2$ shrink correspondingly (within the second-order condition): the stage-1 price range $2\Delta_1/(2b_1 - B_2/2)$ widens and the ladder gradient $1/(b_1 - B_2/2)$ steepens, both in exact closed form. This is the cross-market anticipation channel of §3.4. □

Remark 2 (Price-takers post blocks). A price-taking unit optimally offers its marginal-cost schedule; for a zero-marginal-cost renewable that is a single block at the floor — a one-tranche tier with zero price dispersion and Herfindahl one, below the contested range at any granularity. The bid-shape margin is therefore specific to dispatchable technologies; the renewable margin of the reform is the quantity re-scheduling of §3.4.

A.4 $(1, 1)$ equilibrium

We solve the pre-reform game by backward induction: the Stage 2 best response (§A.4.1), then the four arbitrage regimes at Stage 1 (§A.4.2). The scarcity wedge $p_1 > p_2 > c$ survives in regimes (1), (3), and (4); only competitive full arbitrage closes it. *Convention:* throughout this subsection we report the stage-1 problem in the Ito and Reguant form, suppressing the stage-1 continuation term; Corollary 3 restores it and shows it tilts the stage-1 line (slope $1/(2b_1 - b_2/2)$ here, since $B_2 = b_2$ at this state) without altering the regime classification or the wedge formulas, which depend on s and the spot best response only.

A.4.1 Stage 2 best response

The strategic bidder's Stage 2 problem, taking p_1, q_1 and s as given:

$$\max_{p_2}(p_2 - c)q_2 \quad \text{s.t.} \quad q_2 = b_2(p_1 - p_2) + s - S_2^{op}.$$

FOC: $b_2(p_1 - p_2) + s - S_2^{op} + (p_2 - c)(-b_2) = 0$, giving

$$p_2^*(p_1, s) \;=\; \frac{p_1 + c}{2} + \frac{s - S_2^{op}}{2\, b_2}. \tag{10}$$

A.4.2 Stage 1 best response and equilibrium in (p_1, s)

Substituting (10) into the strategic bidder's total profit and using $q_1 = \bar{A} - b_1 p_1 - s - S_1^{op}$,

$$\Pi(p_1, s) \;=\; (p_1 - c)(\bar{A} - b_1 p_1 - s - S_1^{op}) \;+\; (p_2^*(p_1, s) - c)q_2^*(p_1, s).$$

The four arbitrage cases pin s differently.

No arbitrage ($s = 0$). The strategic bidder solves a Cournot problem in each market separately. The forward FOC gives $p_1^* = (\bar{A} - S_1^{op} + b_1 c)/(2b_1)$. Combining with (10) yields $p_1^* > p_2^* > c$ and $q_1, q_2 > 0$ (Ito and Reguant Result 1).

Full arbitrage ($p_1 = p_2$). Setting $p_1 = p_2$ in (10) gives $s = b_2(p_1 - c) + S_2^{op}$. Substituting back, total cleared MWh $q_1 + q_2 = \bar{A} - b_1 p_1 - S_1^{op} - S_2^{op} = \bar{A} - b_1 p_1 - \bar{Y}_h$ (using $S_1^{op} + S_2^{op} = \bar{Y}_h$). The strategic bidder maximises $(p_1 - c)(\bar{A} - b_1 p_1 - \bar{Y}_h)$, giving

$$p_1^{*,F} \;=\; \frac{\bar{A} - \bar{Y}_h + b_1 c}{2b_1}, \qquad p_2^{*,F} \;=\; p_1^{*,F}. \tag{11}$$

Note that $s > 0$ *reduces* the strategic bidder's total output relative to no arbitrage (Ito and Reguant Result 2): even competitive arbitrage does not restore competitive prices, because the bidder withholds more in response.

Limited arbitrage ($s \le K_a$ binding). If K_a is below the full-arbitrage level, $s = K_a$ binds and $p_1 - p_2 > 0$. From (10), $p_1 - p_2 = (p_1 - c)/2 - (K_a - S_2^{op})/(2b_2)$; the wedge depends on K_a and on S_2^{op} (Ito and Reguant Result 3). The smaller K_a, the larger the residual wedge.

Strategic arbitrage. The arbitrageur maximises $\pi^A(s) = s(p_1 - p_2)$ subject to (10):

$$\pi^A(s) \;=\; s\left[\frac{p_1 - c}{2} - \frac{s - S_2^{op}}{2\,b_2}\right].$$

FOC in s:

$$\frac{p_1 - c}{2} - \frac{2s - S_2^{op}}{2\,b_2} \;=\; 0 \quad\Longleftrightarrow\quad s^{*,S}(p_1) \;=\; \frac{b_2(p_1 - c)}{2} + \frac{S_2^{op}}{2}.$$

Substituting into (10),

$$p_2^{*,S} \;=\; \frac{p_1 + c}{2} + \frac{p_1 - c}{4} - \frac{S_2^{op}}{4\,b_2} \;=\; \frac{3p_1 + c}{4} - \frac{S_2^{op}}{4\,b_2}, \qquad p_1 - p_2^{*,S} \;=\; \frac{p_1 - c}{4} + \frac{S_2^{op}}{4\,b_2}.$$

With the aggregator's spot leg netted into the spot intercept ($S_2^{op} = 0$ in the display — the $S_2^{op}/(4b_2)$ term is a level shift common across regimes), this is the boxed body benchmark (2): $p_1 - p_2 = (p_1 - c)/4$ (Ito and Reguant Result 4).

A.5 $(1, Q)$ equilibrium

Here the forward stays hourly and the spot becomes per-period. I derive the per-period spot best response, then the within-block price dispersion under full and strategic arbitrage, and finally the within-hour wedge SD reported in the body.

The strategic bidder's Stage 2 problem now has Q spot products. Directly from the spot residual demand in (7) (suppressing the update term, restored in §A.5.3) and the FOC $q_{2t} = b_{2t}(p_{2t} - c)$, the strategic per-period spot best response is

$$p_{2t}^{*}(p_1, s_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{s_t - S_{2t}^{op}}{2\,b_{2t}}. \tag{12}$$

A.5.1 Full arbitrage: block parity and within-block dispersion

Block parity requires $p_1 = (1/Q)\sum_t p_{2t}$. Substituting (12),

$$p_1 \;=\; \frac{p_1 + c}{2} + \frac{1}{2Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}} \quad\Longleftrightarrow\quad p_1 - c \;=\; \frac{1}{Q}\sum_t \frac{s_t - S_{2t}^{op}}{b_{2t}}.$$

Setting $S_{2t}^{op} \equiv 0$ and uniform arbitrage $s_t \equiv \bar{s}$ to isolate the slope-asymmetry mechanism, and $Q = 2$ for concreteness,

$$p_1 - c \;=\; \frac{\bar{s}}{2}\left(\frac{1}{b_{21}} + \frac{1}{b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{21} + b_{22}}{b_{21}b_{22}} \quad\Longleftrightarrow\quad \bar{s} \;=\; \frac{2(p_1 - c)\,b_{21}b_{22}}{b_{21} + b_{22}}.$$

The within-block dispersion follows from differencing p_{2t}^{*} across t:

$$p_{21} - p_{22} \;=\; \left(\frac{\bar{s}}{2b_{21}} - \frac{\bar{s}}{2b_{22}}\right) \;=\; \frac{\bar{s}}{2}\cdot\frac{b_{22} - b_{21}}{b_{21}\,b_{22}}.$$

Substituting $\bar{s}$ from above,

$$\boxed{p_{21} - p_{22} \;=\; (p_1 - c)\cdot\frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(full arbitrage).} \tag{13}$$

A.5.2 *Strategic arbitrage: block wedge and attenuated dispersion*

The strategic arbitrageur internalises price impact and stops at

$$p_1 - (1/Q)\sum_t p_{2t} \;=\; \frac{p_1 - c}{4}.$$

The block-wedge condition is the average of (2) across the Q matched spot products. Repeating the algebra of §A.5.1 with this modified parity, the arbitrage position $\bar{s}$ falls to exactly half of the full-arbitrage value:

$$\bar{s}^S \;=\; \frac{(p_1 - c)\, b_{21} b_{22}}{b_{21} + b_{22}} \;=\; \frac{\bar{s}^F}{2}.$$

The within-block dispersion correspondingly becomes

$$\boxed{p_{21} - p_{22} \;=\; \frac{p_1 - c}{2} \cdot \frac{b_{22} - b_{21}}{b_{21} + b_{22}}} \qquad \text{(strategic arbitrage).} \tag{14}$$

The dispersion is attenuated by a factor of two relative to the full-arbitrage case, reflecting the smaller arbitrage volume traded under strategic incentives.

A.5.3 *Wedge SD in the asymmetric state*

Update channel: $\sigma_\psi/(2b_2)$ (derivation of body eq. 4). Decompose the spot residual demand at the per-quarter level using the within-hour state of §3.1: $A_{2t} = \bar{A} + \zeta_t + v_{2t}$. While the forward is hourly, the update $\psi_t = \zeta_t + \mu_t$ realises between the gates and becomes the spot-side traded object (plus an hour-level component common across t, which is absorbed into the mean and does not affect dispersions). The strategic spot best response (12) acquires an ψ_t term:

$$p^*_{2t}(p_1, s_t, \psi_t) \;=\; \frac{p_1 + c}{2} \;+\; \frac{\psi_t + s_t - S^{op}_{2t}}{2\, b_{2t}}.$$

With p_1 constant across t (hourly forward), the per-subperiod wedge $p_1 - p^*_{2t}$ inherits the cross-period variation in p^*_{2t}. Taking variance across t at uniform $b_{2t} \equiv b_2$, s_t, S^{op}_{2t} (so the only t-varying object is ψ_t):

$$\mathrm{sd}_t(p^*_{2t}) \;=\; \frac{\mathrm{sd}_t(\psi_t)}{2\, b_2} \;=\; \frac{\sigma_\psi}{2\, b_2}, \qquad \sigma^2_\psi \;=\; \sigma^2_\zeta + \sigma^2_\mu,$$

and the within-hour wedge SD equals $\mathrm{sd}_t(p^*_{2t})$ exactly because p_1 contributes no cross-period variation. This is the main-body (4). Largest in ramp hours where σ_ζ is large; mechanically zero before the reform because the hourly spot averages ψ_t to its hour-level component $\bar{\psi}$. (Arbitrage on the hourly forward is a block position, $s_t \equiv \bar{s}$: it disciplines the mean wedge but cannot move the dispersion of the granular leg.)

Slope-asymmetry channel (block-parity arbitrage). The $\sigma_\psi/(2b_2)$ channel above holds at uniform b_{2t}. With the per-subperiod slope b_{2t} varying across quarters, an additional dispersion term emerges through the block-parity arbitrage of §A.5.1–§A.5.2. For two matched products ($Q = 2$) the population SD contribution from slope asymmetry alone (setting $\psi_t = 0$) is

$$\mathrm{sd}^{F,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{2(b_{21} + b_{22})}, \qquad \mathrm{sd}^{S,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{(p_1 - c)\,|b_{22} - b_{21}|}{4(b_{21} + b_{22})}, \tag{15}$$

the strategic value exactly half of the full-arbitrage value. Under limited arbitrage with per-product capacity $s_t = K_a/Q$ binding, the slope-asymmetry SD becomes

$$\mathrm{sd}^{L,\mathrm{thin}}_t(p_1 - p_{2t}) \;=\; \frac{K_a\,|b_{22} - b_{21}|}{8\, b_{21}\, b_{22}} \qquad (Q = 2,\; s_t = K_a/2), \tag{16}$$

which carries K_a rather than $(p_1 - c)$. The total wedge SD in the asymmetric state is the sum (in quadrature) of the update channel $\sigma_\psi/(2b_2)$ and the slope-asymmetry channel above; the update channel dominates whenever σ_ζ is large.

A.6 (Q, Q) equilibrium

Now both markets are per-period. I derive the per-product forward best response, the three arbitrage regimes, the Cournot cleared-MWh scale-up, and the stage-1 value of holding a granular continuation.

The strategic per-period forward best response is the continuation-adjusted monopoly line of Corollary 3 with the matched single spot product ($B_2 = b_{2t}$); arbitrage and aggregator legs shift the intercept only and are omitted from the display:

$$p_{1t}^*(v) \;=\; \frac{\bar{A} + \zeta_t + v + c\,(b_{1t} - b_{2t}/2)}{2\,b_{1t} - b_{2t}/2}, \qquad \lambda_1 \;\equiv\; \frac{1}{2\,b_{1t} - b_{2t}/2}. \tag{17}$$

The ramp ζ_t and the screened state v both enter at the pass-through λ_1, so the across-quarter dispersion of expected forward prices is $\mathrm{sd}_t(\mathbb{E}[p_{1t}^* \mid \mathcal{H}_F]) = \lambda_1\,\sigma_\zeta$, and the per-product forward screen is the ρ-amplified $v \sim U[-\rho\Delta_1, \rho\Delta_1]$.

A.6.1 Full arbitrage: product parity

Per-product parity $p_{1t} = p_{2t}$ pins s_t at $s_t = b_{2t}(p_{1t} - c) + S_{2t}^{op}$ (same structure as the $(1,1)$ full-arbitrage condition, but per product). The wedge SD is zero by construction.

A.6.2 Strategic arbitrage: product wedge and SD

Per-product strategic arbitrage — the strategic-arbitrage derivation of §A.4 applied product by product — gives $p_{2t} = (3p_{1t} + c)/4$ (aggregator leg netted). The per-product wedge is $(p_{1t} - c)/4$. Cross-period SD:

$$\mathrm{sd}_t(p_{1t} - p_{2t}) \;=\; \tfrac{1}{4}\,\mathrm{sd}_t(p_{1t}^*) \;=\; \frac{\lambda_1\,\sigma_\zeta}{4},$$

using $\mathrm{sd}_t(\mathbb{E}[p_{1t}^* \mid \mathcal{H}_F]) = \lambda_1\sigma_\zeta$ from (17) — the unlimited-strategic benchmark of the body.

A.6.3 Limited arbitrage at the symmetric state

With Q matched products per clock-hour the arbitrageur's capacity K_a is spread over Q positions, so the effective per-product capacity is K_a/Q. When this capacity binds — the benchmark regime maintained throughout (§3.3) — the per-product equilibrium is Ito and Reguant Result 3 (limited arbitrage) applied product by product.

Strategic spot best response under binding per-product capacity. Setting $s_t = K_a/Q$ in the strategic Stage 2 FOC (12), with the matched per-product forward price p_{1t} in place of the hourly p_1:

$$p_{2t}^* \;=\; \frac{p_{1t}^* + c}{2} + \frac{K_a/Q - S_{2t}^{op}}{2\,b_{2t}}.$$

The per-product wedge is

$$p_{1t}^* - p_{2t}^* \;=\; \frac{p_{1t}^* - c}{2} \;-\; \frac{K_a/Q - S_{2t}^{op}}{2\,b_{2t}}.$$

Wedge SD: $\lambda_1\sigma_\zeta/2$**.** At (Q, Q) the per-quarter forward absorbs the ramp upstream: by (17) the forward price across quarters has SD $\mathrm{sd}_t(p_{1t}^*) = \lambda_1\,\sigma_\zeta$. The capacity term $(K_a/Q - S_{2t}^{op})/(2b_{2t})$ is common across quarters when b_{2t} and S_{2t}^{op} are, so only the first term disperses, feeding the wedge at half the rate:

$$\boxed{\mathrm{sd}_t(p_{1t}^* - p_{2t}^*) \;=\; \tfrac{1}{2}\,\mathrm{sd}_t(p_{1t}^*) \;=\; \frac{\lambda_1\,\sigma_\zeta}{2}.}$$

This is the main-body (6). Under unlimited strategic arbitrage the wedge SD would halve further to $\lambda_1\sigma_\zeta/4$; binding per-product capacity doubles it. If the per-period slope b_{2t} also varies across t, the wedge carries an additional slope-asymmetry component on top of $\lambda_1\sigma_\zeta/2$.

Per-product capacity binding. For the limited regime to bind, K_a/Q must be smaller than the strategic-arbitrage equilibrium position $s_t^{\mathrm{strat}} = b_{2t}(p_{1t} - c)/2$ (aggregator spot leg netted into the intercept). Binding therefore requires $K_a \le Q \cdot \bar{s}^{\mathrm{strat}}$ where $\bar{s}^{\mathrm{strat}}$ is the average per-product strategic position. In practice, arbitrage capacity is set by participation rules and balance-sheet limits that do not scale with Q when granularity reforms multiply products; the constraint binds at the granular state by construction.

A.6.4 Cournot cleared-MWh under per-period slopes

To isolate the slope channel, shut the continuation and arbitrage channels in both states ($B_2 = 0$, $s_t = 0$) — a channel isolation, not an approximation: both states are evaluated under the same convention. The per-period cleared monopoly quantity is then $q_{1t}^* = (\bar{A} + \zeta_t - b_{1t}c - S_{1t}^{op})/2$. Summing across t and using $\sum_t \zeta_t = 0$:

$$\mathbb{E}\Big[\sum_t q_{1t}^* \,\Big|\, \mathcal{H}_F\Big] \;=\; \frac{Q\bar{A} - c\sum_t b_{1t} - \sum_t S_{1t}^{op}}{2}.$$

The hourly counterpart is $Q \cdot q_1^*|_{(1,1)} = Q \cdot (\bar{A} - b_1 c - S_1^{op})/2$. Comparing at equal aggregator totals ($\sum_t S_{1t}^{op} = S_1^{op}$):

$$\mathbb{E}\Big[\sum_{t=1}^{Q} q_{1t}^* \,\Big|\, \mathcal{H}_F\Big]\Big|_{(Q,Q)} \;-\; Q \cdot q_1^*\big|_{(1,1)} \;=\; \frac{c}{2}\left(Q\,b_1 - \sum_{t=1}^{Q} b_{1t}\right). \tag{18}$$

Positive whenever $\sum_{t=1}^{Q} b_{1t} < Q\,b_1$, as expected (each 15-min product faces a less elastic residual demand than the hourly aggregate it replaces). This is the Cournot scale-up of the markup channel in §3.5.

A.6.5 Stage-1 value of the granular continuation

Maximised monopoly profit is convex in the demand intercept ($\pi^m(A) = (A - bc)^2/(4b)$), so a bidder that can re-optimise per quarter values the update as a mean-preserving spread: viewed from stage 1, where μ_t is not yet realised, expected spot profit exceeds spot profit at the mean update by exactly $\mathbb{E}[\mu_t^2 \mid \mathcal{H}_F]/(4\,b_{2t}) = \Delta_\mu^2/(12\,b_{2t})$ per product under the uniform update — an exact equality, because the profit is quadratic. The term raises the stage-1 *value* of holding a granular continuation (part of why the granular state is privately valuable to the strategic bidder), but it does not involve p_1, so it leaves the stage-1 first-order condition unchanged: the forward-*ladder* response to the spot reform operates entirely through the continuation slope of Corollary 3, not through an option-value tilt.

A.7 Comparative statics across the three states

Table 8 collects the state-by-state closed forms behind the results of §3.6.

Table 8: Comparative statics across the reform path under limited per-product arbitrage, the maintained regime. Notation: $\lambda_1(B) \equiv (2b_1 - B/2)^{-1}$ is the stage-1 pass-through at continuation coefficient B (Corollary 3; with the market's own forward slope, b_{1t} per product at (Q,Q)); $\rho^{\mu} \le \rho$ is the news-only amplification of §3.5, and $\rho^{\mu} = \rho = 1$ when the hour is internally flat. In the two ladder rows, twice the product of the entries is the product's conditional monopoly-price range (Corollary 2(i)); the tier dispersion, the tranche-price span, and the Herfindahl follow in closed form from Corollary 1 and Proposition 2 evaluated at the screen and slope shown. Imbalance row: quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$, omitting the unpriceable σ_ε^2 common to all states; for uniform shocks $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$ (§3.1, §3.6).

Outcome	$(1,1)$	$(1,Q)$	(Q,Q)
Aggregator hedging gain $G(\Delta_\eta)$	0	$\dfrac{\gamma Q \Delta_\eta}{2}$	$\dfrac{\gamma Q \Delta_\eta}{2}$
Spot tier: screen × pass-through	$\Delta_2 \times \dfrac{1}{2b_2}$	$\rho\Delta_2 \times \dfrac{1}{2b_{2t}}$	$\rho^{\mu}\Delta_2 \times \dfrac{1}{2b_{2t}}$
Forward tier: screen × pass-through	$\Delta_1 \times \lambda_1(b_2)$	$\Delta_1 \times \lambda_1\big(\sum_t b_{2t}\big)$	$\rho\Delta_1 \times \lambda_1(b_{2t})$
Within-hour wedge SD	0	$\dfrac{\sigma_\psi}{2b_2},\ \ \sigma_\psi^2 = \sigma_\zeta^2 + \sigma_\mu^2$	$\dfrac{\lambda_1\,\sigma_\zeta}{2}$ + slope asym.
Across-quarter forward price dispersion	0	0	$\lambda_1\,\sigma_\zeta$
Balancing Δ_t	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$	ε_t	ε_t
Imbalance cost $C^I/(\gamma Q)$, quadratic	$\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2$	0	0

A.8 Balancing discrepancy

Per quarter, the balancing residual collects whatever no auction carried and no agent could re-schedule:

$$\Delta_t \;=\; \big(\text{within-hour demand objects unpriced by any auction}\big) + \big(\text{unhedged production error}\big) + \varepsilon_t. \tag{19}$$

State $(1,1)$. Both auctions hourly: the spot trades only the hour-level update $\bar{\psi}$, so the ramp, the redistributive news, and the aggregator's forecast error all land in balancing: $\Delta_t = \zeta_t + \mu_t + \eta_t + \varepsilon_t$.

State $(1,Q)$. The spot becomes per-period: it absorbs the full update $\psi_t = \zeta_t + \mu_t$, and the aggregator re-schedules η_t per quarter (§A.2): $\Delta_t = \varepsilon_t$.

State (Q,Q). The per-quarter forward carries the ramp ζ_t, the spot carries the news μ_t, the aggregator hedges η_t: $\Delta_t = \varepsilon_t$.

State	ramp ζ_t	redistributive μ_t	forecast error η_t	residual Δ_t
$(1,1)$	balancing	balancing	balancing	$\zeta_t + \mu_t + \eta_t + \varepsilon_t$
$(1,Q)$	spot	spot	aggregator (hedged)	ε_t
(Q,Q)	forward	spot	aggregator (hedged)	ε_t

The rows for $(1,Q)$ and (Q,Q) are identical in the residual column: the forward reform relocates rows between the two auctions but produces no further imbalance gain, as stated in §3.5. These reductions are mechanical in the contracting grid and hold under any arbitrage regime.

A.9 Welfare derivations

This subsection collects closed-form welfare components by state, consistent with the balancing ledger of §A.8 and with the channels listed in the body §3.6.

A.9.1 Components

Per clock-hour:

$$W \;=\; CS \;+\; \pi^{SB} \;+\; \pi^{Agg,\mathrm{net}} \;+\; \pi^{Arb} \;-\; C^{I}_{\mathrm{sys}}. \tag{20}$$

For linear residual demand $q = \bar{A} - bp$, consumer surplus at p^* is $CS = (\bar{A} - bp^*)^2/(2b)$. Gross surplus from consuming q at state A (willingness to pay minus production cost at c; balancing premia accounted separately) is

$$S(q;\,A) \;=\; \int_0^q \frac{A-x}{b}\,dx \;-\; c\,q \;=\; \frac{q\,(A-bc)}{b} \;-\; \frac{q^2}{2b}.$$

A.9.2 Imbalance cost decomposition (closed form, quadratic penalty)

Under the quadratic penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ per quarter and the ledger of §A.8, with ζ_t deterministic ($Q^{-1}\sum_t \zeta_t^2 = \sigma_\zeta^2$) and μ_t, η_t mean-zero and independent:

	$(1,1)$	$(1,Q)$	(Q,Q)
$\gamma\,\mathbb{E}[(\mathcal{I}^{Agg})^2]$ (in $\pi^{Agg,\mathrm{net}}$)	$\gamma Q\,\sigma_\eta^2$	0	0
C^I_{sys} (system side: $\zeta + \mu$)	$\gamma Q\,(\sigma_\zeta^2 + \sigma_\mu^2)$	0	0
Total imbalance cost	$\gamma Q\,(\sigma_\zeta^2 + \sigma_\mu^2 + \sigma_\eta^2)$	0	0

(21)

omitting the unpriceable $\gamma Q\sigma_\varepsilon^2$, common to all three states. The two channels are disjoint (η_t enters only $\pi^{Agg,\mathrm{net}}$; ζ_t and μ_t only C^I_{sys}). For uniform shocks, $\sigma_\mu^2 = \Delta_\mu^2/3$ and $\sigma_\eta^2 = \Delta_\eta^2/3$. The whole imbalance saving arrives at the spot reform; the (Q,Q) column equals the $(1,Q)$ column, the model's DA15 imbalance null.

A.9.3 Allocative cost of pricing the state per quarter (discrimination term)

Pricing the within-hour state per quarter is not free: it is third-degree price discrimination by a residual monopolist. Fix one clock-hour, the within-hour object ψ_t with $\sum_t \psi_t = 0$, and the margin b at which it is priced; write $\bar{q} = q^m(\bar{A})$ and $A_t = \bar{A} + \psi_t$, so $A_t - bc = 2\bar{q} + \psi_t$.

Regime U (one price for the hour). The monopolist sets $\bar{p} = p^m(\bar{A})$; quarter-t consumption realises on the demand curve, $q_t^U = A_t - b\bar{p} = \bar{q} + \psi_t$; the contracted profile is flat at $\bar{q}$ and the balancing layer supplies the residual ψ_t. Substituting into S:

$$S(q_t^U;\,A_t) \;=\; \frac{(\bar{q}+\psi_t)(3\bar{q}+\psi_t)}{2b}.$$

Regime D (per-quarter prices). The monopolist prices each quarter at $p^m(A_t)$; consumption is $q_t^D = q^m(A_t) = \bar{q} + \psi_t/2$ — monopoly pass-through one half — and there is no balancing residual:

$$S(q_t^D;\,A_t) \;=\; \frac{3\,(\bar{q}+\psi_t/2)^2}{2b}.$$

Difference. Summing over t and using $\sum_t \psi_t = 0$,

$$\sum_t S^D - \sum_t S^U \;=\; \frac{1}{2b}\sum_t \Big[3\bar{q}^2 + 3\bar{q}\,\psi_t + \tfrac{3}{4}\psi_t^2 \;-\; 3\bar{q}^2 - 4\bar{q}\,\psi_t - \psi_t^2\Big] \;=\; -\,\frac{\sum_t \psi_t^2}{8\,b}\;:$$

total output is unchanged and output is misallocated across quarters, at expected cost $Q\,\mathbb{E}[\psi^2]/(8b)$. Against it stands the avoided quadratic balancing penalty $\gamma\,Q\,\mathbb{E}[\psi^2]$, so the net system-side gain from pricing ψ at margin b is

$$Q\,\mathbb{E}[\psi^2]\,\Big(\gamma - \frac{1}{8b}\Big), \qquad \text{positive iff } \gamma > \frac{1}{8b},$$

a threshold independent of the distribution of ψ. At the spot reform the relevant margin is b_2; balancing premia in Spanish electricity markets exceed wholesale margins by a factor of 2–10, far above $1/(8b_2)$.

A.9.4 Markup channel

From the Cournot identity (1), $p_t - c = q_m/b_{mt}$: at given cleared quantity, $\partial(p_t - c)/\partial b_{mt} = -q_m/b_{mt}^2 < 0$, so a rise in the per-firm residual-demand slope compresses the margin — and, by level neutrality (Remark 1), this is the *only* channel through which the reform moves mean prices; ladder geometry contributes nothing to the mean.

A.9.5 Discreteness channel

The optimal-ladder expected profit shortfall relative to the continuous benchmark is $\Delta^2/(12\, b\,(2n^*+1)^2)$ (Proposition 1); the deeper endogenous depth at amplified screens (Corollary 2(ii)) reduces it. By Remark 1 the mean clearing price is unchanged by the ladder, so within a product this channel is mostly a CS$\leftrightarrow \pi^{SB}$ transfer, net of tranche costs $\kappa\, n^*$; the efficiency content is the discreteness loss itself.

A.9.6 Less strategic withholding under limited per-product arbitrage

The spot deadweight-loss triangle at markup $M = p_2 - c$ and slope b_2 is $b_2 M^2/2$. Under unlimited strategic arbitrage, $p_2 - c = 3(p_1 - c)/4$ (§A.4, R4): DWL $= 9\, b_2(p_1 - c)^2/32$. Under binding per-product capacity, the wedge is $(p_1 - c)/2$, so $p_2 - c = (p_1 - c)/2$ and DWL $= b_2(p_1 - c)^2/8 = 4\, b_2(p_1 - c)^2/32$. The per-product saving from the regime change is

$$\frac{5\, b_2\,(p_1 - c)^2}{32},$$

per Ito and Reguant Result 2 logic — a welfare gain that accompanies the wedge-SD non-collapse of §3.5, not a cost of it.

A.9.7 Net welfare deltas (closed form)

Combining the contractibility channels (writing $a_1 \equiv \bar{A} - b_1 c$, aggregator and arbitrage legs netted into $\bar{A}$):

$$\Delta W^{\text{spot}} \;=\; \gamma\, Q\, \sigma_\eta^2 \;+\; Q\,(\sigma_\zeta^2 + \sigma_\mu^2)\Big(\gamma - \frac{1}{8\, b_2}\Big), \tag{22}$$

$$\Delta W^{\text{forward}} \;=\; \frac{Q\,\sigma_\zeta^2}{8}\Big(\frac{1}{b_2} - \frac{1}{b_1}\Big) \;+\; \frac{c\, a_1}{4\, b_1}\Big(Qb_1 - \sum_t b_{1t}\Big). \tag{23}$$

The spot-reform delta is the aggregator's hedging saving plus the net value of pricing the update at the spot margin: positive whenever $\gamma > 1/(8b_2)$, and dominated by the imbalance saving at observed balancing premia. The forward-reform delta is a *relocation*: the ramp's discrimination cost moves from the spot margin b_2 to the (thicker) forward margin b_1 — the first term, positive when $b_1 > b_2$ and zero when the margins coincide — plus the cleared-quantity scale-up of (18) valued at the pre-existing forward markup, carrying the sign of $Qb_1 - \sum_t b_{1t}$. Both forward-reform terms are second-order relative to (22): no imbalance term appears at the forward reform (the (Q, Q) column of (21) equals the $(1, Q)$ column), so the model predicts a small aggregate footprint for the forward reform. The markup-compression and withholding channels add on top wherever b_f rises or the arbitrage constraint binds harder, with the signs derived above; the bid-ladder rent channel is mostly distributional.

B Robustness

Each subsection stress-tests one ingredient of the design: parallel trends, the bandwidth, the hour-class partition, the renewable controls, the estimation window, and bid composition.

B.1 Pre-trend placebo

Split the pre-window at its midpoint and run the same DiD on the two halves; genuine reform effects should leave that placebo near zero. ID15 own-market cells pass cleanly (sign flips between real and placebo). The DA15 own-market CCGT placebo runs same-sign as the real, so the DA15 CCGT widening is continuation-plus-augmentation rather than a clean jump. Figure 9 is the visual analogue.

	ID15 IDA (own)		**ID15 DA (cross)**		**DA15 DA (own)**		**DA15 IDA (cross)**	
Tech	Hdln	Plac	Hdln	Plac	Hdln	Plac	Hdln	Plac
Intercept α (EUR/MWh; positive = higher curve)								
CCGT	11.8*	4.9	−40.2***	1.8	28.3***	40.2***	6.0**	3.8**
Hydro	87.3	10.2***	0.2	−0.4	19.7	53.3*	9.3***	6.3***
Pump-storage	−3.0	−12.7***	1.1	1.3	−1.8	−4.3	16.1	−4.9
Slope β (EUR/MWh per MW; more negative = steeper)								
CCGT	−0.881*	0.146	0.269***	0.004	−0.150***	−0.206***	0.200	−0.101
Hydro	−1.498	−0.156**	0.154	−1.703***	−1.785***	1.176***	−0.012	−0.022
Pump-storage	0.071	0.248	−0.010	0.015*	0.008	0.046***	−0.254	0.007
Curvature γ (EUR/MWh per MW^2; positive = convex)								
CCGT	0.0100	0.0334*	−0.0040***	0.0015***	0.0010***	0.0022***	0.0076	0.0002
Hydro	−0.1088	0.2455	−0.3055	−1.3947	−7.4938***	1.6198	0.0016	−0.0105
Pump-storage	0.0014	0.0005	0.0001	0.0002	0.0001	0.0000	−0.0002	−0.0002

Table 9: Pre-only midpoint placebo for the bid-shape DiD. "Hdln" = headline DiD (§6.4); "Plac" = same DiD on the pre-window split at its midpoint, which should be null under parallel trends.

Parallel-trends diagnostics, metric by metric. Figures 9 and 10 plot critical against flat hours around each cutover for every dispatchable technology. The split follows the headline-vs-supporting logic of §4.2: σ_p and $\mathrm{HHI}_{\mathrm{tr}}$ — first and second moments of the in-band price distribution — show flat pre-differentials on the headline cells, while $\hat{\beta}$ and $\hat{\gamma}$ — within-curve OLS objects, mechanically sensitive to tranche counts and price-grid density — drift in most cells. (σ_p also needs only price-side variation and is well-defined on single-tranche curves, where $\hat{\beta}$ requires joint p–q variation; for near-linear curves $\sigma_p \approx |\hat{\beta}|\,\sigma_Q$, so the two track each other up to estimator noise.)

Figure 9: Parallel trends for σ_p and $\mathrm{HHI}_{\mathrm{tr}}$: clean on the headline cells. Critical (red) vs flat (blue), 14-day rolling means; dashed line = cutover. Panels top to bottom: ID15 DA, ID15 IDA, DA15 DA, DA15 IDA.

Figure 10: Parallel trends for $\hat{\beta}$ and $\hat{\gamma}$: drift. Same construction as Figure 9. These are within-curve OLS objects, mechanically sensitive to tranche counts and grid density — hence supporting, not headline.

B.2 Bandwidth δ

I rerun the DiD at $\delta = 100, 140, 200$ EUR/MWh on the headline metrics σ_p and $1/\mathrm{HHI}_{\mathrm{tr}}$ (Table 10). The CCGT deconcentration at DA15 is the most bandwidth-robust cell: the $1/\mathrm{HHI}_{\mathrm{tr}}$ DiD is positive and significant at every bandwidth. On the ID15 IDA leg both CCGT metrics strengthen as the band widens beyond the contested p_{90} margin (σ_p insignificant at $\delta = 100$, significant at 140 and 200), consistent with the re-spaced tranches extending past the tight headline band. Hydro's σ_p widening is significant throughout, while its DA15 $1/\mathrm{HHI}_{\mathrm{tr}}$ changes sign at wide bands; pump-storage moves are concentrated at ID15 on $1/\mathrm{HHI}_{\mathrm{tr}}$.

	ID15 IDA			**DA15 DA**		
δ	CCGT	Hydro	Pump	CCGT	Hydro	Pump
Dispersion σ_p (EUR/MWh)						
$\delta = 100$	0.65	1.10^{***}	0.03	−0.68	1.31^{***}	−0.31
$\delta = 140$	2.45^{***}	1.17^{***}	0.25	1.30^{*}	1.69^{***}	-0.63^{*}
$\delta = 200$	2.09^{***}	1.06^{**}	0.24	0.50	1.82^{***}	-0.68^{**}
Effective tranche count $1/HHI_{tr}$						
$\delta = 100$	0.66	0.20^{***}	0.16^{**}	1.11^{***}	0.22^{***}	−0.01
$\delta = 140$	0.79^{***}	0.21^{***}	0.18^{**}	0.44^{***}	−0.04	−0.05
$\delta = 200$	0.84^{***}	0.24^{***}	0.19^{**}	0.42^{***}	-0.13^{***}	−0.04

Table 10: Bandwidth robustness, σ_p and $1/\mathrm{HHI}_{\mathrm{tr}}$. DiD θ at $\delta = 100, 140, 200$ EUR/MWh; the headline p_{90} estimates are in Table 6. Positive $1/\mathrm{HHI}_{\mathrm{tr}}$ = deconcentration. SEs clustered by date.

Per-IDA-session BSTS estimates with the long pre-window cluster around EUR −45/MWh across IDA1, IDA2, IDA3, confirming session symmetry. A winter-only pre-window inflates estimates to ∼−80/MWh because the renewable coefficient is identified off too narrow a range. This motivates the long pre with year-by-year renewable interactions in BSTS.

B.3 Critical/flat definition: midday falsification and cross-border control

Midday falsification: substituting midday for critical hours, the ID15 IDA CCGT effect flips sign (no engagement in solar-saturated hours) while DA15 DA carries through (the day-ahead reform engaged the whole non-flat zone). *Cross-border control:* adding hourly Spain-France net flow to the ID15 IDA DiD leaves the coefficients essentially unchanged, ruling out the contemporaneous XBID/SIDC reform.

B.4 Convexity and non-stationarity of the renewable response: year-by-year solar coefficient

The BSTS uses long pre-windows with year-by-renewable interactions because Spanish wind+solar capacity grew ∼6× over 2018–2025, so the marginal price-suppression of an additional GWh is non-stationary; a flat-coefficient spec mis-fits the cross-year merit-order shift.

Table 11 reports the cumulative solar coefficient (base + year-dummy) by calendar year, extracted from the BSTS posterior of the price spec under both windows (ID15 pre ends 2025-03-18, DA15 pre ends 2025-09-30). **Conditional posterior means** (the coefficient when the spike-and-slab includes the regressor). The wind base coefficient is stable around −0.50 EUR/MWh per GWh; the wind×year interactions all have inclusion probability below 1.5%. The solar coefficient is the one that varies.

Year	ID15 IDA spec	DA15 IDA spec	DA15 DA spec
2022 (base)	−0.22	−0.23	−0.22
2023	−0.02	−0.23	−0.17
2024	**−0.46**	**−0.47**	**−0.40**
2025 †	**−0.29** *(Jan-Mar only on ID15)*	**−0.50**	**−0.49**
Implied weighted average for the post-window prediction			
2024+2025 weighted avg ‡	−0.43	−0.45	−0.44

Table 11: Cumulative solar coefficient (EUR/MWh per GWh of solar) by year, extracted from the BSTS posterior of the corresponding price spec. Cumulative coefficient = base + year-dummy interaction, conditional posterior mean among MCMC draws that include the regressor. †The ID15 spec pre-window ends 2025-03-18, so its 2025-by-solar coefficient is fitted on Jan-Mar 2025 only, when Spanish solar production is at its lowest seasonal level; the DA15 spec pre-window extends through 2025-09-30 and includes summer, giving the more representative estimate. ‡Months-weighted average of the 2024 and 2025 cumulative coefficients (12 months of 2024 + available months of 2025): this is the coefficient one would extrapolate forward if 2024 and 2025 were pooled into a single dummy for the post-window prediction; the per-year BSTS instead extrapolates using the 2025-row coefficient alone.

Reading the table: the DA15 spec (which sees Jan-Sep 2025) tells the cleaner story. Solar's marginal price-suppressing effect is roughly constant from 2022 to 2023, then nearly doubles from 2023 to 2024 (Spain added ∼5 GW of solar capacity in 2024 alone), and continues to deepen in 2025 as solar saturates more clock-hours and negative-price events become routine. The ID15 spec's 2025 coefficient looks deceptively small (Jan-Mar winter sun is weak) but the underlying renewable response in 2025 is the most negative on record. The non-stationarity is exactly what the year interaction is designed to absorb: a flat-coefficient spec would average −0.22 (2022) and −0.50 (2025) into a single value around −0.35, mis-fitting both ends of the window.

Calibrated-solar robustness: how much of the price drop is solar vs the reform? A natural concern about the headline BSTS counterfactual is that Spanish solar capacity grew rapidly through 2024 and 2025, so the marginal price-suppressing effect of solar is itself non-stationary. The BSTS per-year spike-and-slab attempts to capture this via year-by-solar interactions; the conditional posterior means (Table 11) are −0.46 for 2024 and −0.29 for the Jan-Mar-2025 sample on the ID15 spec, suggesting a strong solar effect that may not be fully credited in the post-window prediction. If the post-window counterfactual under-credits solar, the residual "price drop" that we attribute to the reform is partly mechanical (it is the solar effect we failed to attribute correctly). We therefore implement a **calibrated-solar counterfactual**: pre-subtract a known solar contribution $\beta^{\text{cal}}_{\text{solar}} \cdot \text{solar}_t \cdot \mathbb{1}\{t \geq 2024\}$ from observed prices, fit the BSTS state-space model on the residual (with the usual base regressors but no year interactions), and add the calibrated solar contribution back to the post-window prediction before computing the treatment effect.

	ID15 IDA			DA15 DA		
Calibrated solar coefficient $\beta^{\text{cal}}_{\text{solar}}$ (EUR/MWh per GWh)	**Real**	**Placebo**	**Net**	**Real**	**Placebo**	**Net**
−0.21 (BSTS unconditional default)	−41	32	**−73**	30	43	−13
−0.29 (per-year cond. 2025, ID15 spec)	−38	36	**−74**	21	32	−12
−0.40 (per-year cond. 2024, DA15 DA spec)	−30	36	**−66**	18	27	−10
−0.46 (per-year cond. 2024, ID15 spec)	−26	37	**−63**	9	23	−13
−0.50 (per-year cond. 2025, DA15 spec)	−23	37	**−60**	9	19	−10

Table 12: Calibrated-solar BSTS treatment effects on clearing price. EUR/MWh; the −0.46 row uses the 2024 conditional-posterior solar coefficient. The calibrated approach explicitly subtracts $\beta^{\text{cal}}_{\text{solar}} \cdot \text{solar} \cdot \mathbf{1}\{t \geq 2024\}$ from prices, fits BSTS on the residual (no year interactions), and adds the calibrated contribution back to the post-window prediction. Placebo column uses the same calibration but with the cutover shifted one year earlier (real reform absent in the placebo window). Real, placebo, and net all measured on the price level in EUR/MWh.

The price drop at ID15 IDA survives every solar calibration. Under the most aggressive solar coefficient (−0.50, the DA15-spec conditional 2025 mean) the real-effect point estimate is −23 EUR/MWh; under the 2024 conditional value of −0.46 it is −26; under the BSTS unconditional default, −41. After 2024-same-calendar placebo-netting the net effects are uniformly larger in magnitude (−60 to −74 EUR/MWh), because the placebo windows carry their own positive "effect" under the calibrated specification. The DA15 DA real effect is essentially null across all calibrations (9 to 30 EUR/MWh); its net effect is mildly negative (−10 to −13) but far smaller than at ID15, with credible intervals that always bracket zero. So solar's strong 2024 marginal effect does not explain the ID15 drop — the reform moved the granular leg's clearing price on top of what solar alone would predict — while the DA15 effect is not separately identified. The cleanly-identified pricing-chain input remains the per-firm slope increase of Appendix C.2, which this discussion leaves untouched.

OLS cross-check. The wide credible intervals in the calibrated-solar BSTS reflect the state-space level component's flexibility, which absorbs persistent low-frequency variation that the regression then has no way to pin down. We complement the BSTS with a standard OLS regression on the same daily price panel, including calendar-month and day-of-week fixed effects (the seasonal structure that BSTS encoded via local-level + 7-day seasonal), a time trend (linear or quadratic), gas price, and per-year wind and solar interactions. Standard errors are Newey-West HAC with 7-day bandwidth to account for daily price autocorrelation. The model:

$$p_t \;=\; \alpha+\theta\,\text{POST}_t+\beta_{\text{wind}}\text{wind}_t+\beta_{\text{solar}}\text{solar}_t+\beta_{\text{gas}}\text{gas}_t+\sum_y \gamma_y\,\text{renew}_t\cdot\mathbb{1}\{\text{year}=y\}+\phi_m+\delta_{\text{dow}}+g(t)+\varepsilon_t.$$

	ID15 IDA price		DA15 DA price	
Specification (POST coef θ)	$\hat{\theta}$	HAC SE	$\hat{\theta}$	HAC SE
Spec 1: base + month FE + DOW FE	**−31.7*****	8.2	−8.3	5.3
Spec 2: + linear time trend	**−23.7*****	8.6	8.3	7.3
Spec 3: + year-by-renewable (per year)	**−45.8*****	9.6	−8.6	8.1
Spec 4: + year-by-renewable (2024+ pooled)	**−38.7*****	7.3	−4.6	7.7
Spec 5: + quadratic time trend	**−53.1*****	6.8	**−18.9****	9.2

Table 13: OLS price-effect estimates with full controls. Daily clearing-price regressions with calendar-month and day-of-week fixed effects, gas price, time trend (linear or quadratic), and year-by-renewable interactions. Newey-West HAC standard errors with 7-day bandwidth. POST = 1 on/after the reform cutover. Spec 4 (year-by-renewable, 2024+2025 pooled) addresses the Jan-Mar-2025-only fit of the per-year spec. $^{**}\,p < 0.05$, $^{***}\,p < 0.01$. ID15 IDA pre-window 2022-01-01 to 2025-03-18, post 2025-03-19 to 2025-04-27. DA15 DA pre 2022-01-01 to 2025-09-30, post 2025-10-01 to 2025-12-31.

OLS verdict matches the calibrated-solar BSTS verdict. The ID15 IDA price drop is significant at every standard significance level across all five specifications (−24 to −53 EUR/MWh, $p < 0.01$ throughout). The DA15 DA effect ranges from −4.6 (null, $p = 0.55$) to −18.9 (significant under quadratic trend, $p = 0.04$); the bulk of specifications return small null or marginally-significant negative coefficients. OLS with full controls is more precise than the BSTS (smaller standard errors, narrower confidence intervals) but qualitatively delivers the same conclusion: **ID15 has a robust own-market price drop; DA15 does not**.

This diagnostic also clarifies why several level findings (§6.1) lose significance under the year-interaction spec: forecast uncertainty about what solar's marginal price impact will be in the post-cutover window is large precisely because the coefficient has been shifting fast in the years immediately preceding the cutover.

Note on bid-shape outcomes in this appendix. The bid-shape DiD robustness tables below report on the headline quintuple $(\sigma_p, \hat{\alpha}, \hat{\beta}, \hat{\gamma}, \mathrm{HHI}_{\mathrm{tr}})$ used in §4.2. The cross-border and offer-type checks lead with σ_p (the robust price-side anchor); for a near-linear curve $\sigma_p \approx |\hat{\beta}| \cdot \sigma_Q$, so the signs on those checks track $\hat{\beta}$ up to estimator noise (and via curvature track $\hat{\gamma}$).

B.5 Convex-renewable robustness on the price OLS

Hourly OLS on the ID15 IDA price, no linear trend, Newey-West HAC lag 24×7. Pre window 2022-01-01 onwards; post = 40 days for ID15 (pre-blackout), 92 days for DA15. **D**: quadratic spec adds wind^2, solar^2 with year-by-renewable interactions on both linear and quadratic terms. **E**: cubic spec adds wind^3, solar^3 with year-by-renewable interactions on all three orders. The quadratic spec absorbs merit-order curvature in renewable price-suppression; the cubic adds essentially nothing (diminishing returns).

	ID15 (IDA reform)		**DA15 (DA reform)**	
	IDA (own)	DA (cross)	DA (own)	IDA (cross)
OLS hourly — + year-by-renewable (2024+25 pooled)	-30^{***}	-29^{***}	3	2
D + quadratic renew × year	$\mathbf{-24}^{***}$	$\mathbf{-24}^{***}$	-2	-3
E + cubic renew × year	-24^{***}	-23^{***}	-2	-2
Newey-West HAC SE (D)	7.00	7.10	5.36	5.31
Newey-West HAC SE (E)	6.85	6.95	5.37	5.32

Table 14: Convex-renewable robustness. Quadratic (D) and cubic (E) renewable terms with year interactions, no trend. ID15 holds at ≈ -24; DA15 stays null.

B.6 Cross-border flow control on the price effect

A natural worry about the ID15 price drop is that it reflects cross-border conditions rather than granularity — a coincidental import surge would depress the Spanish price over the post-window. I add the daily net flow into Spain (ENTSO-E physical flows from France plus Portugal, imports minus exports) to the headline pooled-renewable daily specification. The ID15 drop is essentially unmoved: $-38.7 \to -35.5$ EUR/MWh on the intraday leg and $-38.7 \to -35.0$ on the cross-market day-ahead, both significant at the 1% level, while DA15 stays null ($-4.6 \to -5.5$ and $-6.0 \to -6.8$, both insignificant). The flow itself enters with a small *positive* coefficient (about +0.34 EUR/MWh per GWh of net import) — the wrong sign for a purely exogenous import, and a reminder that net flows are partly demand-driven, since Spain imports more precisely when its own price is high, so the control is imperfect. But that endogeneity cuts against finding robustness, not for it: even letting a price-responsive flow variable absorb what it can, the granularity-attributable drop survives almost intact. Cross-border flow also enters the within-hour wedge test of §6.3 as a control, where it is likewise small and insignificant.

B.7 Renewable capacity vs realised production, 2024 vs 2025

(a) Installed capacity at year-start (ENTSO-E, MW). Solar PV expanded 43%; wind essentially flat.

	Jan 2024	**Jan 2025**	**Δ MW**	**Δ %**
Solar PV (B16)	29,047	**41,433**	12,386	42.6%
Wind (B18+B19)	30,932	32,451	1,519	4.9%
Solar + Wind	59,979	73,885	13,906	23.2%
Coal (B05)	1,820	1,258	−562	−30.9%
Nuclear (B14), Gas (B04)	no change	no change	—	—

(b) Realised production in the matched ID15 window (Mar 19 – Apr 27).

	Spring 2024	**Spring 2025**	**Δ**	**Δ %**
Wind (GWh/d)	167.5	161.4	−6.1	−4%
Solar (GWh/d)	122.2	138.4	16.2	13%
Gas (EUR/MWh)	431	746	315	**73**%
IDA price (EUR/MWh, raw mean)	10.9	25.1	14.2	—

B.8 Window length: long-pre controls and conditional parallel trends

I extend the pre-window to 2022-01-01, add year FE, calendar-month FE, and crit×wind/crit×solar interactions; and as a separate spec run the regression-adjustment DiD of Heckman et al. (1997) under conditional parallel trends. ID15 DA CCGT (anticipation) survives both checks on σ_p and HHI_{tr}; DA15 DA CCGT survives on σ_p, the HHI_{tr} sign flips under RA-DiD so the σ_p side is the rugged headline. Side technologies omitted: only cells that survive at least the long-pre spec are reported (Table 15).

Cell	**Outcome**	**Tight (1)**	**Long-pre + controls (2)**	**RA-DiD (3)**
ID15 DA CCGT (anticipation)	σ_p	19.42***	10.78***	12.71**
	HHI_{tr}	−0.076***	−0.098***	−0.000
DA15 DA CCGT	σ_p	1.25**	5.47***	1.72
	HHI_{tr}	−0.059***	−0.028***	0.099**

Table 15: Bid-shape DiD: tight (1), long-pre with controls (2), regression-adjustment DiD (3). Spec (2) extends the pre-window to 2022 with year, month, and crit×renewable controls; spec (3) is the Heckman et al. (1997) RA-DiD under conditional parallel trends. Wider band $H = 140$, so magnitudes exceed the headline table; signs are the comparable object.

Per-firm b_{mt} under the same long-pre control. I run the same long-pre + year + month FE on the per-firm b_{mt} increase: magnitude shrinks from the tight-window range 13%–28% to 8%–15% across the four dominant CCGT-owning firms, all firm cells still significant. Direction robust, magnitude roughly halved.

B.9 Bid types: CCGT offer-type composition

I split CCGT bids by structural type (simple / MIC / block) and re-run the DA15 DiD within each type. The composition shifts toward more block offers post-reform (block offers are mechanically single-tranche, pushing the pooled $\mathrm{HHI}_{\mathrm{tr}}$ UP), but the within-type DiD is still negative in MIC (-0.046^{***}) and block (-0.043^{***}), and pooled stays -0.059^{***}. The bid-stack deconcentration is real within-tranche behaviour, not a composition artefact.

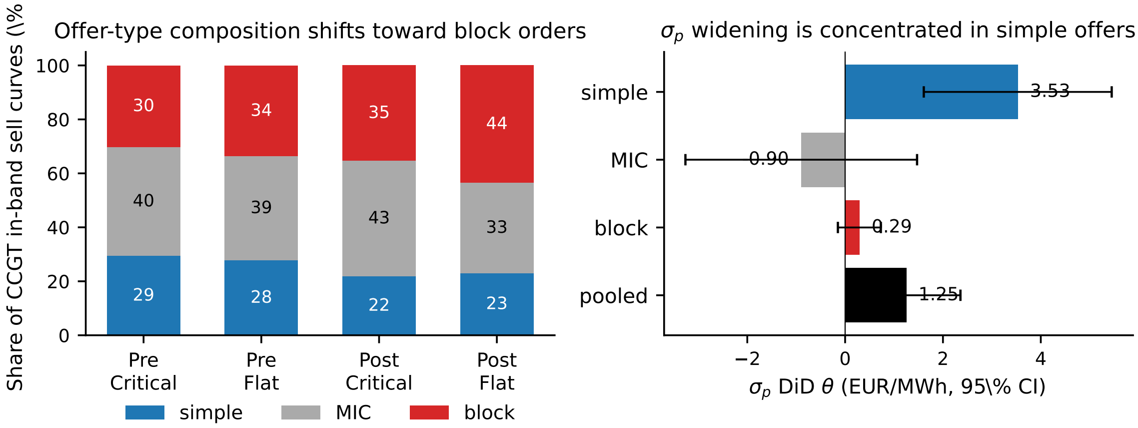


Figure 11: CCGT offer-type composition and within-type bid-shape DiD at DA15. *Left*: CCGT in-band sell-curve composition, pre vs post. *Right*: σ_p DiD by offer type, 95% CI. The $\mathrm{HHI}_{\mathrm{tr}}$ DiD is negative within every type, so the deconcentration is not a composition artefact.

B.10 Demand-side bid shape

I re-run bid-shape DiD symmetrically on buy offers (tranches sorted descending in price) per demand class. At DA15 the buy-side α rises in step with sell-side α in critical: equilibrium re-coordination at a higher reservation level, not a one-sided supply shift.

Demand class	ID15 IDA	ID15 DA	DA15 DA	DA15 IDA
Intercept α (EUR/MWh)				
Retailer	-10.6^{**}	5.4	15.0^{***}	33.1^{***}
DirectCons	0.2	—	-3.1	19.5^{***}
PumpBuy	0.3	-4.9	-2.1^{*}	0.8
Slope β (EUR/MWh per buy MW; on a descending price-ordered curve)				
Retailer	-1.688	-1.624	2.607^{***}	-2.272^{**}
DirectCons	-5.308^{***}	—	-3.746^{***}	1.555^{***}
PumpBuy	-0.012	0.133	0.039^{***}	0.017^{***}
Curvature γ (EUR/MWh per buy MW^2)				
Retailer	-6.1746	-0.7050	2.7389	4.9286
DirectCons	-25.4^{***}	—	-5.7976^{*}	-0.0206
PumpBuy	-0.0002	0.0002	0.0008^{**}	0.0005

Table 16: Demand-side bid-shape DiD, critical vs. flat. Buy curves sorted descending in price. Retailer = competitive supplier; DirectCons = industrial buyer; PumpBuy = pump-storage charging.

Retailer DA15 IDA α at 33^{***} tracks the sell-side widening; ID15 IDA retailer runs the opposite direction, corroborating the ID15 IDA price compression in §6.1.

B.11 Per-CCGT-firm decomposition

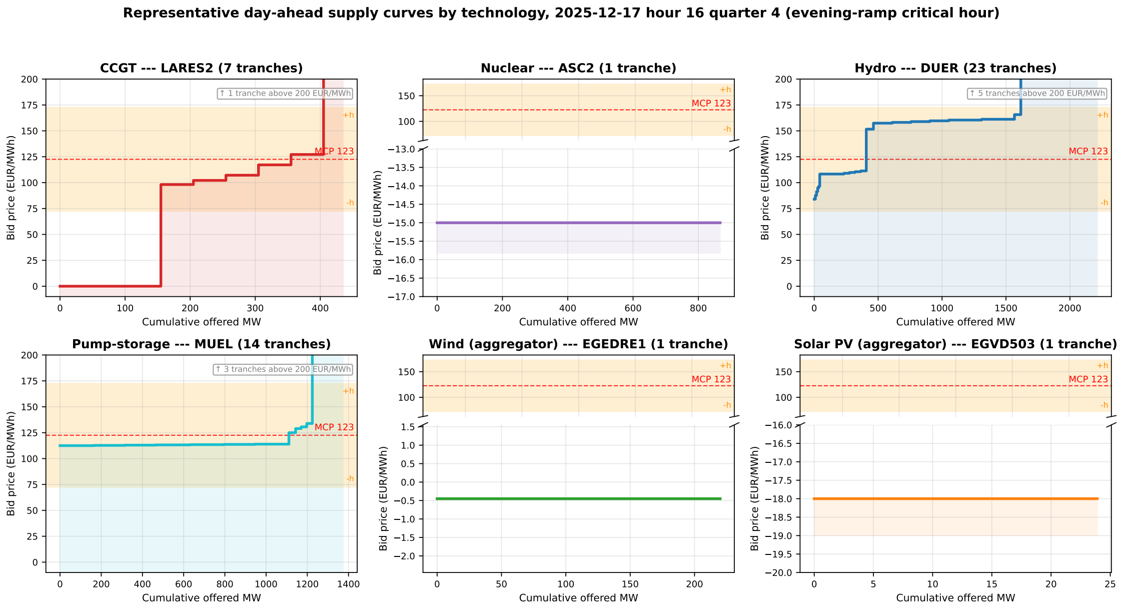
I run the DiD firm by firm on each firm's own CCGT fleet (unit FE within-firm). Naturgy is the dominant scale-up at DA15 DA (Table 17); Iberdrola flat on level; EDP-HC small same-sign. ID15 IDA spreads more evenly across firms. Sign and firm-ranking are interpretable; magnitudes are not (firm intercepts not netted out).

Firm	α	β	γ	Units
DA15 DA (own-market)				
Naturgy	69.9^{***}	-0.402^{***}	0.0022^{**}	17
Iberdrola	0.0	0.001	0.0000	10
EDP-HC	5.2^{*}	-0.021^{*}	-0.0002	7
ID15 IDA (own-market)				
Naturgy	10.3	-1.197^{*}	0.0109	17
Iberdrola	32.2	-0.253	—	9
Endesa	-2.9	0.109	0.0044^{*}	7
EDP-HC	1.7	-0.211^{**}	-0.0029	6

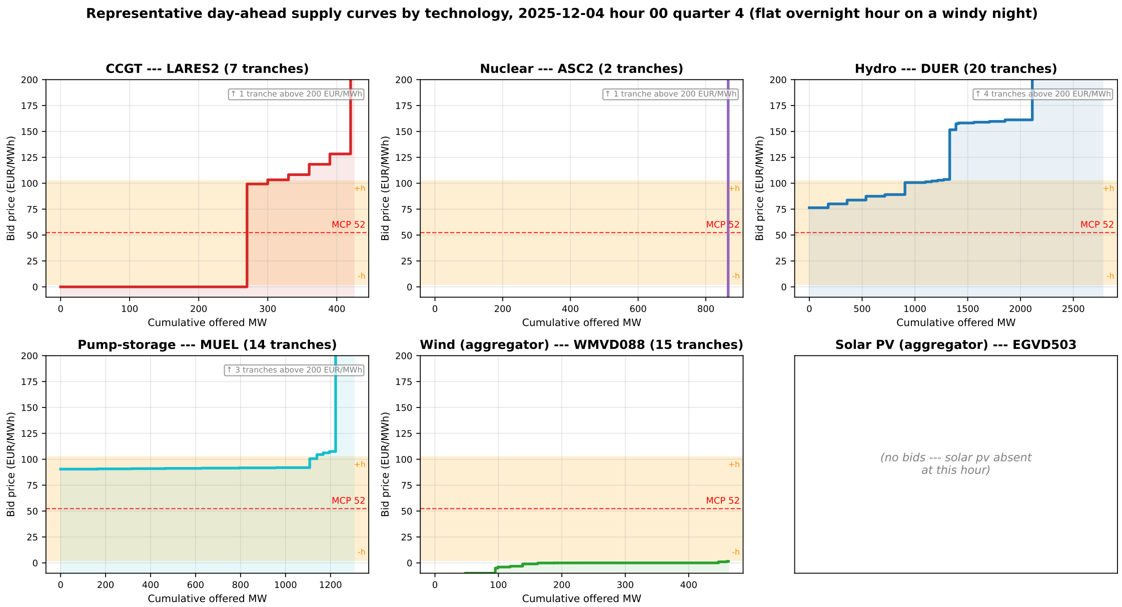
Table 17: Per-firm CCGT bid-shape DiD, own-market cells. One regression per firm on its own CCGT fleet (unit FE). Sign and ranking are interpretable; magnitudes are not comparable to the pooled headlines.

C Measurement details: bid curves, residual demand, and the margin proxy

C.1 Representative bid curves by technology



(a) Evening-ramp critical quarter: 2025-12-17, hour 17, quarter 4.



(b) Flat overnight quarter on a windy night: 2025-12-04, hour 01, quarter 4.

Figure 12: Representative day-ahead supply curves by technology (post-MTU15-DA). Red dashed: MCP; orange band: in-band region ($\delta = 50$). Dispatchables structure their competing tier inside the band; renewables submit single blocks far below it — the basis for excluding renewables from the bid-shape DiD.

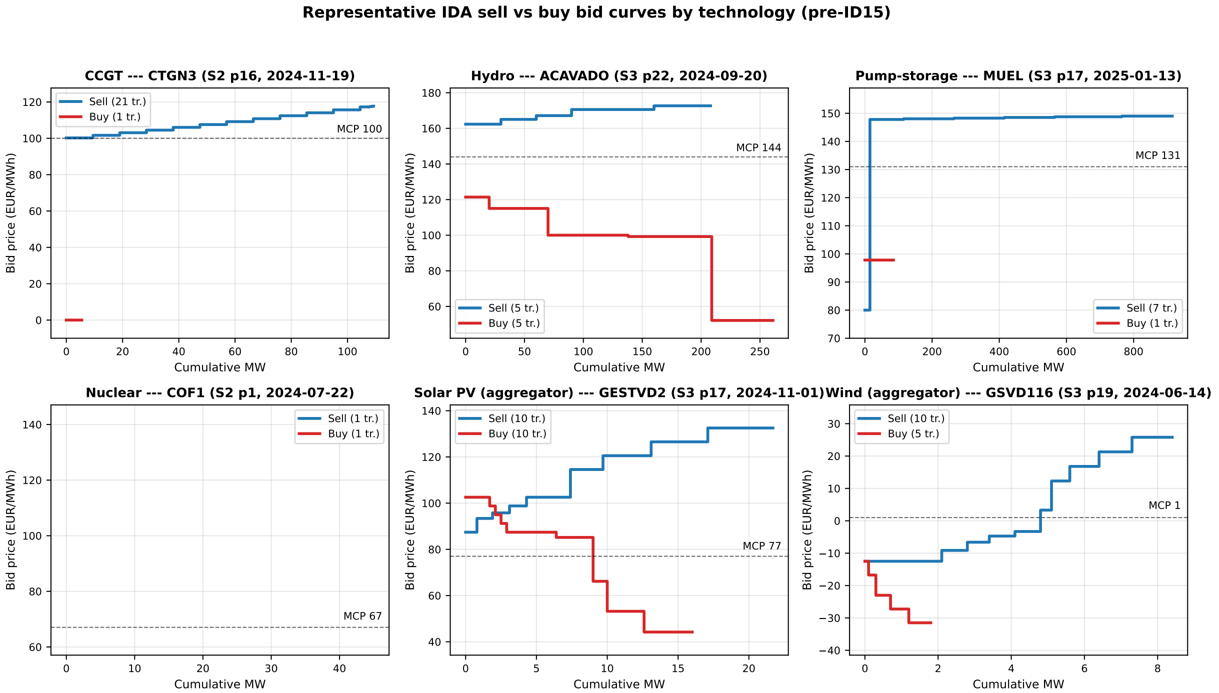


Figure 13: Representative IDA sell vs buy bid curves by technology, pre-ID15. One representative (unit, date, session, period) per tech. Blue: sell curve (supply, ascending in price). Red: buy curve (demand, descending in price). Dashed: MCP. Pre-ID15 window 2024-06-14 to 2025-03-18.

C.2 Residual-demand slope inside the bandwidth: empirical b_{mt}

I replicate Ito and Reguant (2016) Section III.A Method 1 per firm. For each (date, period, market, focal firm) I build the residual demand the firm faces as aggregate demand minus all-other-firms' supply, sample it on a price grid ±50 EUR/MWh around MCP, fit a quadratic, and take the slope at MCP. Per-firm means in Table 18.

	ID15 IDA				**DA15 DA**				**DA15 IDA**			
Firm	pre	post	Δ	ratio	pre	post	Δ	ratio	pre	post	Δ	ratio
Iberdrola	156	180	24	1.15	230	285	55	1.24	149	168	19	1.13
Endesa	180	214	34	1.19	255	321	66	1.26	171	202	31	1.18
Naturgy	173	212	39	1.23	266	330	64	1.24	159	187	28	1.17
EDP-HC	167	193	26	1.16	231	295	64	1.28	157	181	24	1.15

Table 18: Per-firm residual-demand slope b_{mt} at MCP (MW per EUR/MWh; larger = more elastic). Recovered as in Ito and Reguant (2016) III.A.

The residual-demand slope b_f rises for every firm in every (reform, market) cell, by 13–28% (tight) and 8–15% (long-pre + year + month FE). The Cournot first-order condition then implies markup compression. Per-firm b is the model's pricing primitive and it moves as predicted.

Per-IDA-session decomposition. I rebuild b_{mt} per IDA session (S1, S2, S3) by joining each session's MCP to its own bids. Pre-reform the model predicts b thins toward delivery (S1 ¿ S2 ¿ S3); the data confirm it. Post-MTU15-IDA the S2/S3 gap collapses (Table 19).

Grid bandwidth	Pre-MTU15-IDA S1	S2	S3 (closest to delivery)	Post-MTU15-IDA S1	S2	S3
±50 EUR/MWh (baseline)	80.2	68.3	59.4	**99.9**	73.0	72.3
±70 EUR/MWh	71.7	62.6	55.0	**84.0**	61.8	60.5
±90 EUR/MWh	64.2	57.1	50.8	**73.4**	54.5	53.0

Table 19: Mean per-firm b_{mt} by IDA session and bandwidth. MW per EUR/MWh, mean across the four dominant firms.

At every bandwidth: pre-reform residual demand becomes less elastic toward delivery ($b_{S1} > b_{S2} > b_{S3}$, the sequential-markets ordering); post-reform the session-1 slope rises 14–24% and the S2-vs-S3 gap collapses, making the two later sessions look like the same product. Industry reporting confirms the liquidity ordering (IDA1, IDA2 ∼37% each; IDA3 12%).

C.3 Strategic markup proxy: q_f^{strat}/b_f per firm

Cournot first-order condition: $(p - MC)/p = q_f^{\text{strat}}/(b_f \cdot p)$, where q_f^{strat} is the focal firm's strategic cleared MW (total cleared minus own renewable cleared MW: wind, solar PV, run-of-river, CSP, biomass-renewable, RE Tarifa CUR). Renewables clear at marginal cost ≈ 0 regardless of bidding choices, so they are infra-marginal and do not enter the markup calculation. Reported as the period mean of q_f^{strat}/b_f per (firm, window). Bigger = more pricing power per strategic MW.

	%Δ b_f (residual demand) IB	GE	GN	HC	%Δ markup$_f$ IB	GE	GN	HC
ID15 IDA	15	19	23	16	−12	152‡	−13	62‡
ID15 DA	**116**	**93**	**100**	**100**	−19	29‡	**−38**	23‡
DA15 DA	**24**	**26**	**24**	**28**	**−27**	**−26**	−12	−18
DA15 IDA	13	18	17	15	7	10	−15	40

Table 20: Per-firm residual-demand slope and strategic margin proxy, pre vs post. Margin proxy $= q_f^{\text{strat}}/b_f$, q_f^{strat} net of own renewables. ‡ ID15 margin signs are compositional (small 40-day post base), not behavioural. DA15 DA is the clean cell: b_f up 24–28%, margin proxy down 12–27% on all four firms.

C.4 Imbalance settlement data and what its prices mean

The imbalance evidence of §6.5 uses REE's settlement, retrieved through ESIOS. The public series are easy to misread, so this subsection states what they are, distinguishes them from the neighbouring balancing markets they are often confused with, and flags a measurement subtlety specific to the ISP15 window.

Settlement, not procurement. Two different objects in the Spanish system are both called "deviations." One is the *gestión de desvíos*, a procurement market REE runs in the hour before delivery to buy upward or downward energy when the forecast system imbalance is large (in excess of roughly 300 MWh); this is the "deviation market" that Brown and Reguant (2026) place inside the ancillary-service stack alongside the tertiary reserve. The other — the one §6.5 uses — is the ex-post *settlement* of each balance-responsible party against its final schedule: residual imbalance equals metered delivery minus the final scheduled programme, and REE charges or pays for it at a per-period settlement price. I use the settlement prices, not the procurement-market prices, because the model's penalty $\gamma\,\mathbb{E}[\Delta_t^2]$ is precisely the cost a party bears for the imbalance the auctions could not contract away.

Single price, with a dual-price exception. Since the European imbalance-settlement harmonisation (applied in Spain from December 2021), the default is a *single* imbalance price per period: one price, the marginal cost of the balancing energy activated, that charges deviations aggravating the system imbalance and rewards — at the same price — those that relieve it. A *dual* price, with separate up- and down-deviation prices, applies only in periods with significant simultaneous upward and downward activation. The two series I report, the up- and down-deviation settlement prices, therefore coincide in single-priced periods and diverge in dual-priced ones; their gap is the asymmetric penalty a systematically forecast-wrong party feels, which the industry puts at roughly 29 against 13 EUR/MWh for the costly versus the cheap side. The system-average price can fall, as it does at ID15, while that asymmetric gap widens a little — the two are not in tension.

The ISP15 measurement subtlety. The settlement period went from 60 to 15 minutes on 2024-12-01 (the CNMC ISP15 reform). But the many small and domestic meters are still read hourly, and their quarter-hourly settlement is a cubic-spline interpolation of the hourly reading (Operating Procedure 10.5), so through the ISP15 window the settlement *reference* was quarter-hourly while part of the underlying *measurement* was synthetic. This does not bias the prices I use — they are formed from activated-reserve costs, not from interpolated meters — but it is why the relevant clock is best described as the settlement reference bidders respond to rather than a fully metered quarter-hour.

How the price is formed. The deviation price is not a black box: it is fully defined by regulation. Under the European Electricity Balancing Guideline (Regulation (EU) 2017/2195, Articles 52 and 55) and the CNMC-approved Spanish balancing terms (Resolution of 11 December 2019, BOE-A-2019-18423; Operating Procedure 7.x), the imbalance price in each settlement period is built from the prices of the balancing energy actually activated to resolve the net system imbalance — the secondary regulation (aFRR), the tertiary regulation (mFRR), and the replacement reserve (RR) — each of which is itself cleared at a *marginal* price from the providers' balancing-energy bids, plus, where applicable, the value of avoided activation. So the cost of a deviation is, by construction, the marginal cost of the secondary, tertiary, and replacement energy REE had to call to cover the residual; it depends on those reserve prices and on the activated bids, exactly as one would expect. The single price collapses the upward and downward components into one in periods without heavy two-directional activation, and splits them otherwise.

This pins down what I am measuring. The two directional prices and the net penalty are settlement outputs of that rule; I lead with the directional prices because they are the cleanest read of it, and report the net penalty as a one-number summary and the volume alongside. That this layer is worth the care is clear from Brown and Reguant (2026): ancillary-service costs, the deviation market among them, have grown to roughly a third of the total cost of delivered energy in Spain. These prices are the empirical face of the model's balancing cost $\gamma\,\mathbb{E}[\Delta_t^2]$ — the cheaper the secondary, tertiary, and replacement energy needed to settle a deviation, the smaller the penalty the system pays for the residual the auctions could not contract — and the model predicts that cost falls once the spot becomes granular and not again at the forward reform, which is what both estimators in Table 7 show.

D Additional descriptive findings

Brief descriptive evidence adjacent to the main results. None of it carries identification weight; it is recorded because it informs the reading of the headline estimates.

D.1 Continuous-market response at the three reforms

BSTS on ES-leg continuous-market series under headline BSTS. ID15 is the only structural cutover of the three: trade count jumps 24,269/day placebo-net ($p < 0.001$) as participants take up the four 15-min contracts per clock-hour, and energy turnover drops -12.36 GWh/day placebo-net ($p < 0.001$) as smaller contracts and partial migration toward the now-finer IDA auctions thin total volume. The continuous-market volume-weighted price is not separately identifiable from the renewable-driven price drop once year-interactions control the wind/solar coefficient (real -43.6^*, placebo-net -18.3 ns). ISP15 is a methodological null (settlement-only reform, contract grid unchanged); DA15 leaves the continuous-market contract grid alone (already 15-min post-ID15) and no outcome survives the year-interaction control. The continuous-market reading is consistent with the auction-side reading: granularity reforms shift the contract grid and the participation pattern, but the price-level component cannot be separated from the renewable-coefficient drift.

D.2 Reforzada and the ancillary layer: the cost increase concentrates on CCGT

The post-blackout intervention is technology-specific. Decomposing each technology's weekly dispatch chronologically (day-ahead, bilaterals, intraday auctions, continuous, then post-IDA redispatch), the gas panel of Figure 14 shows the day-ahead-cleared base collapsing post-blackout while the REE-routed Fase I layer balloons; other technologies are flat (Figure 15). Fase I is the pre-IDA technical-restriction redispatch, paid pay-as-bid through a separate restriction offer (PO 3.2; PO 14.4 ap. 20.1).

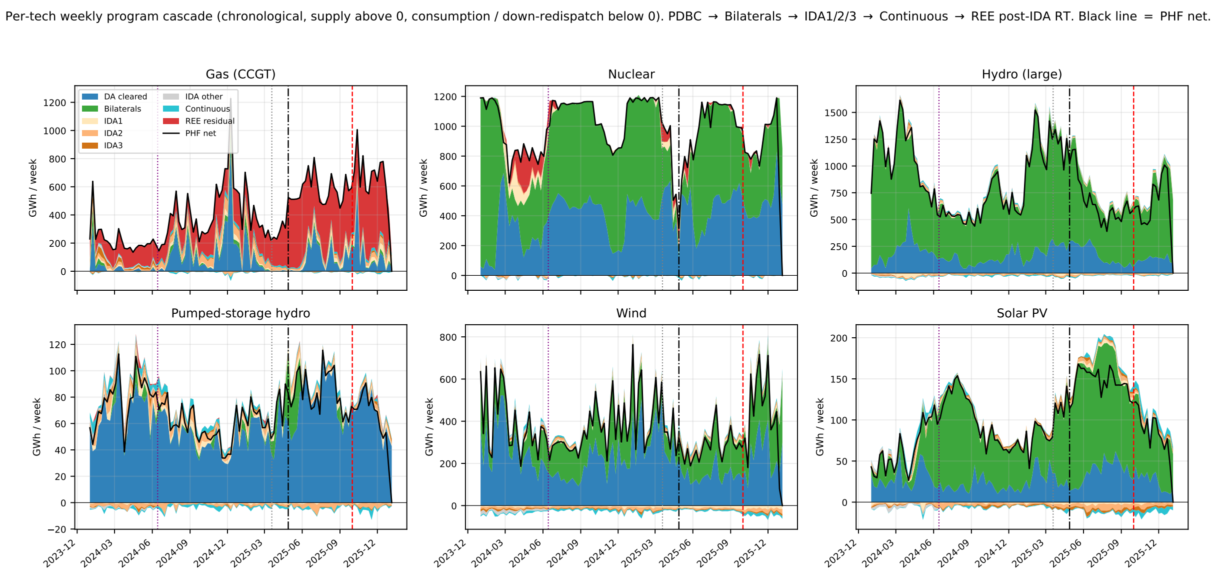


Figure 14: Weekly program cascade by technology. Chronological stack: day-ahead + bilaterals + IDA1–3 + continuous + post-IDA redispatch; supply above zero, consumption and down-redispatch below; black line = PHF net. Vertical guides: ID15, blackout, DA15.

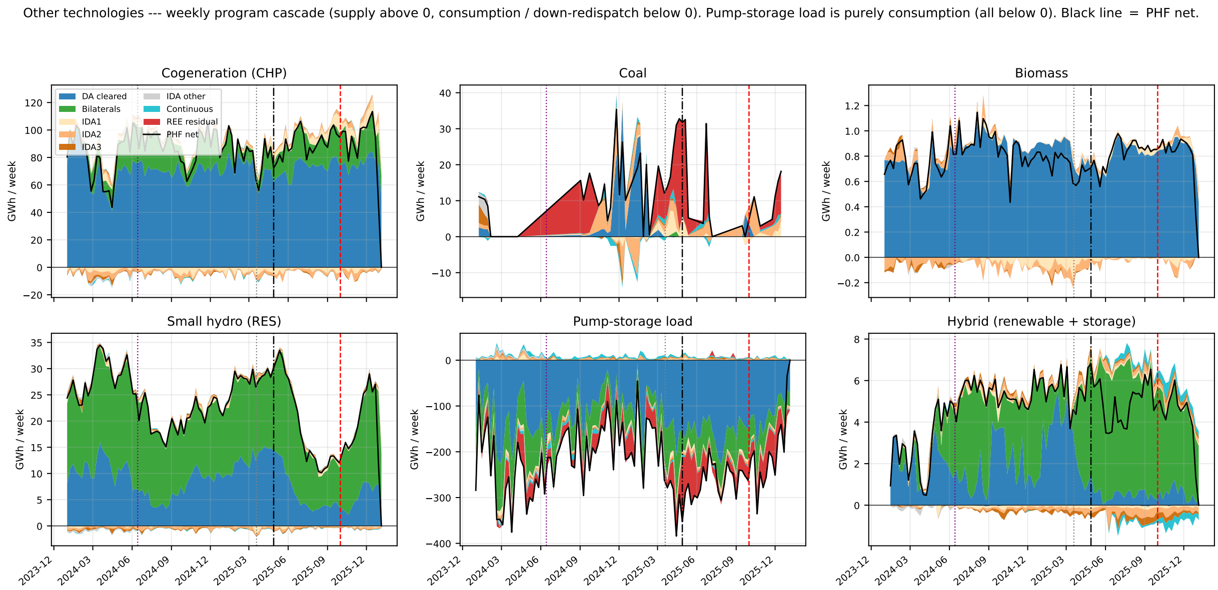


Figure 15: Weekly program cascade, other technologies. Same convention as Figure 14: supply above zero, consumption and down-redispatch below zero, black line = PHF net. Pump-storage load is purely consumption and plots entirely below zero.

The CCGT-only pattern is the cleanest evidence that the reforzada cost increase is separate from the granularity reforms: a reform touching all firms equally would not produce a single-technology intervention.

D.3 Per-channel BSTS on REE ajuste-cost decisions

BSTS on the eight daily ajuste-cost channels (raw posterior mean). MTU15-IDA shrinks real-time technical restrictions and aFRR-up cleanly (−EUR 3.9M/day total); MTU15-DA is dominated by the reforzada-driven Fase I-up rise (+EUR 5.1M/day). No same-calendar placebo: these are operator procurement decisions, not market outcomes, so a year-ago window is not a counterfactual.

Channel	**MTU15-IDA (pre-blackout post)**	**MTU15-DA (post-window straddles reforzada)**
Fase I up	1,988	4,554
Fase I dn	-197^{**}	−123
Fase II up	−24	-143^{***}
Fase II dn	-963^{*}	$1{,}959^{**}$
TR up	$\mathbf{-4{,}182}^{**}$	−709
TR dn	390^{***}	−29
aFRR up	$\mathbf{-842}^{***}$	−325
aFRR dn	−44	−132
Sum	**≈ −3,874**	**≈ 5,102**

Table 21: BSTS on daily REE ajuste-cost channels (kEUR/day). Stars: 95% credible interval excludes zero.

Daví-Arderius and Graf (2026) put the reforzada cost rise at ∼EUR 1.24B/year, the same order as the raw MTU15-DA Fase I-up posterior here (≈EUR 1.7B/year) — directionally reassuring, not an identification claim.

D.4 Post-DA15 IDA activity rise and the Fase I displacement channel

Post-DA15 vs pre-DA15 IDA activity (reforzada-constant window): CCGT supply in IDA3 falls 15% while demand-side volume rises 15/10/29% across IDA1/IDA2/IDA3 — consistent with re-reinforced Fase I displacement (CCGT dispatched pre-IDA, leaving residual imbalance for the demand side in late sessions).

	IDA1		**IDA2**		**IDA3**	
Side / outcome	MW/day	tranches/curve	MW/day	tranches/curve	MW/day	tranches/curve
CCGT supply	11.6%	−3.8%	0.5%	−1.9%	**−15.0**%	−18.3%
All supply	−5.7%	16.7%	−10.2%	6.9%	−17.9%	3.8%
Demand	**14.7**%	14.9%	**10.0**%	12.0%	**29.4**%	9.7%

Table 22: IDA bidding activity, post-DA15 vs. pre-DA15. % change in in-band MW/day and tranches per curve. Block orders excluded.

This composes with Table 21: the TR-up shrink at MTU15-DA is procurement-timing substitution into Fase I, not a market-mechanism cleanup.

D.5 Per-session IDA bid-shape decomposition: CCGT bidding intensity is IDA2-concentrated pre-reform and homogenizes post-reform

Pre-reform CCGT bidding intensity concentrates in IDA2 (σ_p and $\hat{\beta}$ roughly 2.5× IDA1); post-reform the three sessions converge. Descriptive across three calendar windows, not an identified effect.

	Pre-MTU15-IDA			**Post-MTU15-IDA (pre-blackout)**			**Post-DA15 (incl. re-reinforce**		
Outcome	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3	IDA1	IDA2	IDA3
σ_p (EUR/MWh)	1.74	4.08	0.34	4.03	5.77	2.47	3.64	4.79	2.07
$\hat{\beta}$ (EUR/MWh per MW)	0.125	0.335	0.161	0.122	0.135	0.123	0.099	0.125	0.098
$\mathrm{HHI}_{\mathrm{tr}}$	0.25	0.50	0.60	0.21	0.33	0.33	0.22	0.25	0.33
n_{curves}	7,647	11,398	10,515	8,962	10,254	7,208	56,484	60,147	43,998

Table 23: Per-session IDA CCGT bid-shape, critical evening hours 16–22. Medians at $\delta = 150$ EUR/MWh across three calendar windows; descriptive, not an identified effect.

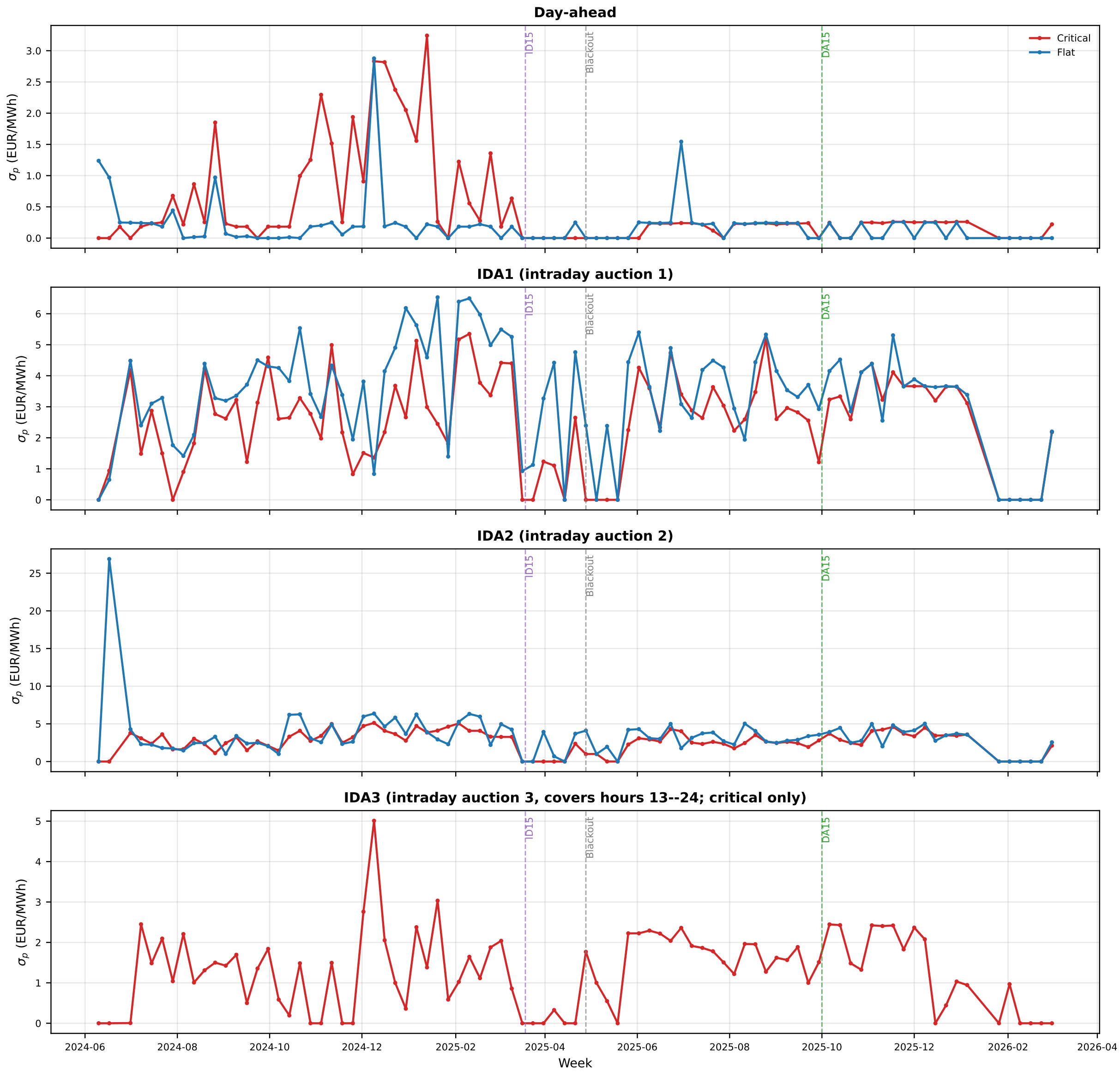


Figure 16: Weekly CCGT σ_p by hour-class: day-ahead and the three IDA sessions. Red: critical hours; blue: flat. Dashes mark ID15, blackout, DA15. The IDA sessions converge in level post-reform.

D.6 Battery bid behavior in the IDA — descriptive

Battery curves are single-tranche throughout, so the bid-shape DiD does not apply; what changes is participation and bid placement relative to the clearing price (eleven buy-side and three sell-side OMIE storage units, across three regimes).

Regime	Side	Curves	Median p_{bid} − MCP	In-band share (±50)	Avg MW/bid
Pre-MTU15-IDA	Charge	8	38.0	63%	4.4
	Sell	4	−47.6	50%	4.2
Post-MTU15-IDA	Charge	1,115	0.2	96%	3.3
	Sell	1,017	0.2	98%	3.9
Post-DA15	Charge	2,184	27.4	74%	3.3
	Sell	1,598	−10.0	72%	3.1

Table 24: Battery IDA bidding by regime (descriptive). Positive $p_{\mathrm{bid}} - \mathrm{MCP}$ on the charge side = bidding above MCP to guarantee clearing.

Two shifts: participation rises from a dozen curves pre-MTU15-IDA to thousands post-DA15 (a build-out alongside the reforms), and the strategy moves from at-MCP price-taking at the MTU15-IDA cutover to spread-around-MCP bidding (charge above, sell below) at DA15 — the signature of active storage arbitrage.

Post-DA15 one battery (Batalha) starts placing bids on both sides of the same period in 18% of cases — a bracketing position that clears in either direction, consistent with two-sided arbitrage. Sample is tiny; descriptive only.

D.7 Nuclear PDBF shortfall before the blackout

Headline BSTS on nuclear auction-plus-bilateral dispatch (PDBF, PHF) at both reforms. Nuclear output drops sharply in the 40 days between ID15 and the blackout; the DA15 window is null.

Reform window	**Outcome**	**Real**	**Placebo 2024**	**Placebo-net**
ID15 (Mar 19 – Apr 27, 2025, pre-blackout)	PDBF (DA + bilateral)	$\mathbf{-72.8}^{***}$	−18.9	−53.9
	PHF (post-IDA total)	$\mathbf{-54.1}^{***}$	17.3	−71.4
DA15 (Oct 1 – Dec 31, 2025, reforzada-constant)	PDBF	−6.1	−29.4	23.3
	PHF	−7.8	−31.2	23.4

Table 25: Nuclear PDBF and PHF BSTS (GWh/day). Placebos: same-calendar 2024 windows.